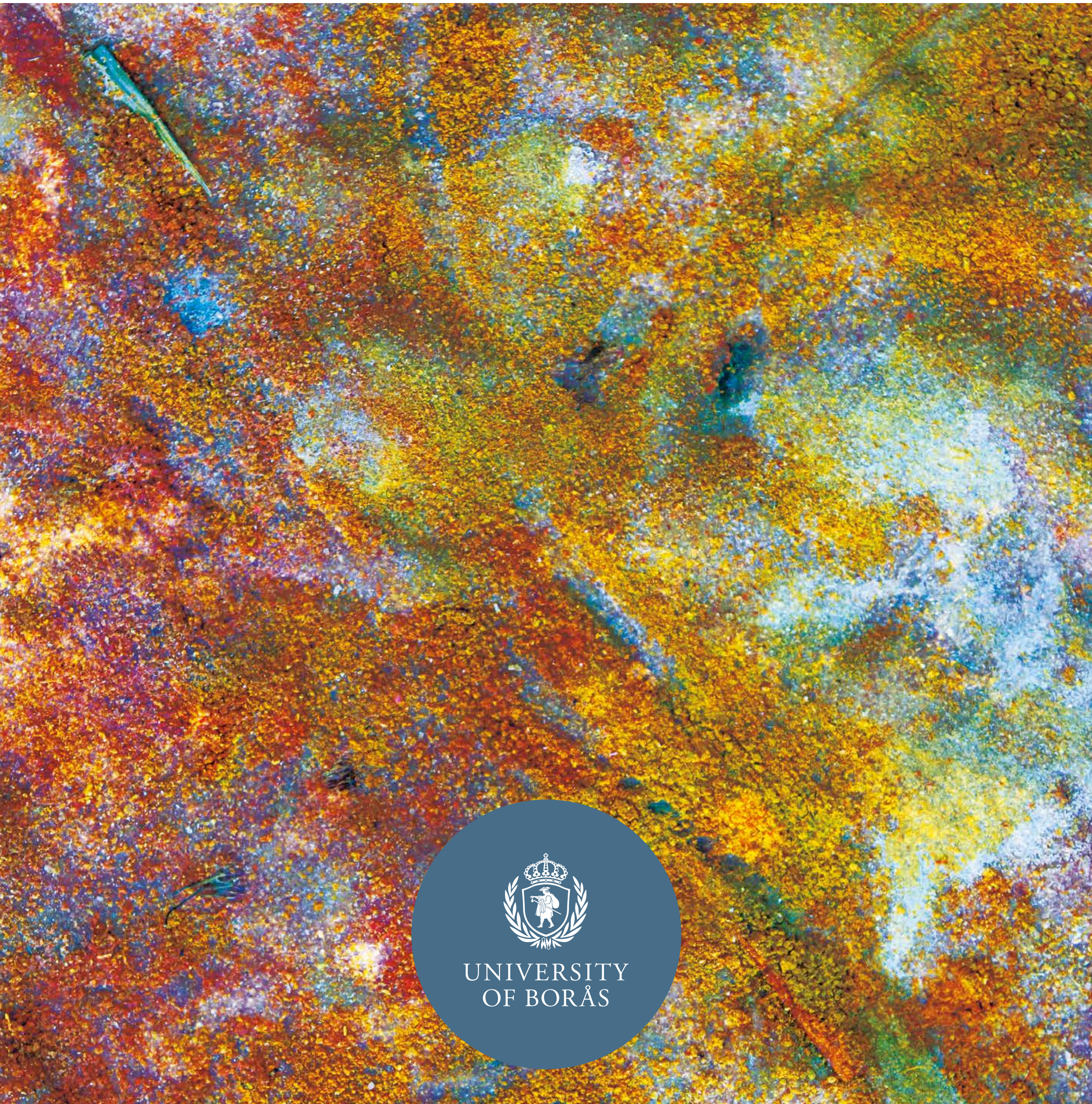
DOCTORAL THESIS
Resource Recovery

Prevention and valorisation of surplus bread at the supplier–retailer interface

PEDRO BRANCOLI



THESIS FOR THE DEGREE DOCTOR OF PHILOSOPHY

Prevention and valorisation of surplus bread at the supplier–retailer interface

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Prevention and valorisation of surplus bread at the supplier–retailer interface

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ABSTRACT

The global food system is a major driver of many environmental impacts, particularly those related to climate change, biodiversity loss, and depletion of freshwater resources. These problems are aggravated by a substantial waste of food throughout the supply chain, where retailers are responsible for large quantities of waste. Although other parts of the supply chain account for relatively higher waste generation, retailers are particularly important because of their influence both downstream and upstream in the supply chain.

This thesis aims to design and evaluate strategies for food waste prevention and valorisation, particularly for bread products, by analysing food waste quantities, identifying the causes and risk factors, and proposing and evaluating measures for preventing and valorising food waste.

This aim was achieved through a variety of approaches. First, food waste was quantified for one year in a typical mid-sized urban supermarket in Sweden. This information was used to identify hotspots at the product-level in relation to mass, environmental impacts, and cost. Bread was identified as a hotspot and also as a product with a high potential for waste prevention and valorisation measures. A second quantification was performed with the goal of estimating the quantity of surplus bread throughout the Swedish supply chain and to identify the risk factors for waste generation, particularly at the supplier–retailer interface. Finally, this thesis investigated current and future circular economy strategies for the prevention, valorisation, and management of bread surplus by evaluating the environmental performance of multiple strategies and comparing them with current waste management practices.

The results from the first quantification indicated that bread was a category with significant contribution in all environmental impact categories analysed, with the greatest contribution in terms of the total mass of waste and the economic costs incurred by the supermarket. The second quantification estimated 80 500 tonnes of bread waste/year in Sweden, equivalent to 8 kg per person/year, which was mainly concentrated at household and retail levels, specifically at the supplier–retailer interface. The results provided evidence that the take-back agreement between suppliers and retailers is a risk factor for high waste generation. Therefore, current business models may need to be changed to achieve a more sustainable bread supply chain with lower waste generation. However, the currently established return system between bakeries and retailers enables a segregated flow of bread waste that is not contaminated with other food waste products. This provides an opportunity for alternative valorisation and waste management options that are not viable for mixed waste streams.

The results from the environmental assessment for the prevention, valorisation and waste management pathways supported a waste hierarchy, where prevention has the highest environmental savings, followed by donation, the use of surplus bread as animal feed, and for beer and ethanol production. Anaerobic digestion and incineration offer the lowest environmental savings, particularly in low impact energy systems. The results suggest that Sweden can make use of the established return system to implement environmentally preferred options for the management of surplus bread.

Keywords: Food waste; Life cycle assessment; Bread; Prevention; Valorisation

Dedicated to vó Laíde

LIST OF PUBLICATIONS

This doctoral thesis is based upon the following peer-reviewed articles, which will be referred to in the text by their Roman numerals. The articles are appended at the end of the thesis

PAPER I: BRANCOLI, P., ROUSTA, K. & BOLTON, K. 2017. Life cycle assessment of supermarket food waste. Resources, Conservation and Recycling, 118, 39-46.

PAPER II: BRANCOLI, P., LUNDIN, M., BOLTON, K. & ERIKSSON, M. 2019. Bread loss rates at the supplier–retailer interface–Analysis of risk factors to support waste prevention measures. Resources, Conservation and Recycling, 147, 128-136.

PAPER III: BRANCOLI, P., BOLTON, K. & ERIKSSON, M. 2020. Environmental impacts of waste management and valorisation pathways for surplus bread in Sweden. Waste Management, 117, 136-145.

PAPER IV: BRANCOLI, P., GMOSER, R., TAHERZADEH, M. J. & BOLTON, K. 2021. The Use of Life Cycle Assessment in the Support of the Development of Fungal Food Products from Surplus Bread. Fermentation, 7, 173**.**

STATEMENT OF CONTRIBUTION

My contributions to the above publications were:

PAPER I: Planned the paper in cooperation with the co-authors. Performed the data collection, observation in the stores and the analysis of the data. Responsible for the life cycle assessment. Interpreted the data and wrote the manuscript with the co-authors.

PAPER II: Planned the paper in cooperation with the co-authors. Together with the co-authors, I was responsible for the data collection, processing of the data and statistical analysis. Interpreted the data and wrote the manuscript with the co-authors.

PAPER III: Planed the study. Responsible for the data collection and life cycle assessment. Interpreted the data and wrote the manuscript with the co-authors.

PAPER IV: Responsible for the design of the scenarios with the co-authors, and for the life cycle assessment. Interpreted the data and wrote the manuscript with the co-authors.

LIST OF PUBLICATIONS NOT INCLUDED IN THE THESIS

Book chapters

BRANCOLI, P. & BOLTON, K. 2019. Life Cycle Assessment of Waste Management Systems. Sustainable Resource Recovery and Zero Waste Approaches. Elsevier.

BRANCOLI, P., BOLTON, K., ROUSTA, K. & ERIKSSON, M. 2020. Life cycle assessment and sustainability aspects of food waste. In: WONG, J., KAUR, G., TAHERZADEH, M., PANDEY, A. & LASARIDI, K. (eds.) Current Developments in Biotechnology and Bioengineering: Sustainable Food Waste Management: Resource Recovery and Treatment. Elsevier.

Articles

BRANCOLI, P., FERREIRA, J. A., BOLTON, K. & TAHERZADEH, M. 2018. Changes in carbon footprint when integrating production of filamentous fungi in 1st generation ethanol plants. Bioresource technology, 249, 1069-1073.

FERREIRA, J. A., BRANCOLI, P., AGNIHOTRI, S., BOLTON, K. & TAHERZADEH, M. J. 2018. A review of integration strategies of lignocelluloses and other wastes in 1st generation bioethanol processes. Process Biochemistry, 75, 173-186.

MATZEMBACHER, D. E., BRANCOLI, P., MAIA, L. M. & ERIKSSON, M. 2020. Consumer's food waste in different restaurants configuration: A comparison between different levels of incentive and interaction. Waste Management, 114, 263-273.

SOUZA FILHO, P. F., BRANCOLI, P., BOLTON, K., ZAMANI, A. & TAHERZADEH, M. J. 2017. Techno-economic and life cycle assessment of wastewater management from potato starch production: Present status and alternative biotreatments. Fermentation, 3, 56.

WEIDEMA, B. P., SIMAS, M. S., SCHMIDT, J., PIZZOL, M., LØKKE, S. & BRANCOLI, P. L. 2019. Relevance of attributional and consequential information for environmental product labelling. The International Journal of Life Cycle Assessment, 1-5.

NOMENCLATURE

EAN	Individual code number
EU	European union
FAO	Food and Agriculture Organization of the United Nations
FFV	Fresh fruits and vegetables
FLI	Food loss index
FLW	Food loss and waste
FWI	Food Waste Index
ILCD	International Reference Life Cycle Data System
LCA	Life cycle assessment
POS	Point of sales data
SDG	Sustainable development goals
SSF	Solid-state fermentation
TBA	Take-back agreement
USFDA	U.S. Food and Drug Administration
VAT	Value-added tax
WHO	World Health Organization

TABLE OF CONTENTS

CHAPTER 1

INTRODUCTION

The current linear economic model, based on 'take–make–consume–dispose', assumes that economic growth can be achieved based on the production and consumption from an abundance of resources while neglecting the environmental, economic, and social externalities associated with the extraction of raw materials, the generation of waste and its inadequate management, and the emission of pollutants (Pandey et al., 2021).

Food production is an essential activity that requires an extensive amount of resources, such as land, nutrients, water, and energy. This leads to significant environmental impacts, such as global warming, depletion of freshwater resources, loss of biodiversity, changes in land use, and the pollution of terrestrial and marine ecosystems (Springmann et al., 2018). Moreover, agricultural production is responsible for the destabilisation of large environmental processes and the transgression of several planetary boundaries, which are defined as the 'safe operating space for humanity' (Campbell et al., 2017, Rockström et al., 2009). Campbell et al. (2017) reported that agriculture has a major role in the two planetary boundaries that have been transgressed (biosphere integrity, and nitrogen and phosphorous biogeochemical flows), meaning that it could generate unacceptable environmental changes with consequences that are potentially disastrous for human existence. Moreover, of the five planetary boundaries that have not been transgressed but are in a zone of increasing risk, three, namely land system change, freshwater use, and climate change, are significantly driven by agriculture.

It has been forecasted that by 2050, the demand for agricultural products will increase by 35–50% because of a growing population and rising incomes, leading to an even higher pressure on the environment (FAO, 2019). The environmental problem is aggravated by the substantial waste of food throughout the supply chain. The waste of food represents not only the loss of the product itself, but also all the resources used and emissions released during food production, transportation, packaging and waste management. A report from the Food and Agriculture Organization (FAO, 2014) based on older figures for waste generation, estimated that annual food loss and waste globally generates more than 4.4 gigatonnes of carbon dioxide equivalent (CO_2-eq). In another report, the FAO (2017) estimated that 10% of the world's total energy consumption is used to produce food that is lost or wasted.

There has been increasing political and public debate on the need to address food loss and waste, motivated by the increased interest in the environmental, economic, and social impacts caused by food waste. This is exemplified in the UN's Sustainable Development Goals (SDGs) on its target 12.3[1] and the inclusion of food loss and waste in the Farm to Fork Strategy, part of the European Green deal that aims for a fairer, healthier, and more environmentally friendly food system (European Commission, 2020).

Although research on food waste prevention and valorisation has increased in recent years, a gap remains in the understanding of the quantities of food waste, its causes, and the efficacy of food waste reduction and valorisation measures. The Food Waste Index (UNEP, 2021) states that research on the quantification of food waste, particularly those that cover the environmental, economic, and social aspects, is necessary to understand the causes behind food waste and to spur action for food waste prevention. The need for data on the extent of food lost and wasted throughout the supply chain and the expected benefits or impacts that may occur when food waste is reduced or valorised is further emphasised in other publications (e.g., Rutten et al., 2013, Goossens et al., 2019, De Laurentiis et al., 2020). Thus, based on the knowledge gaps presented, this thesis aims to contribute to food waste prevention and valorisation by improving waste quantification data, identifying risk factors, and evaluating prevention and valorisation solutions.

[1] By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses.

1.1. Structure of the thesis

The research was structured in two main phases. Phase 1 aimed at understanding and describing the current situation of food waste generation in Swedish retailers and bakeries in terms of indicators, such as mass, economic cost, and environmental impacts, as well as the risk factors for waste generation. The goal was to improve the understanding of the existing situation and propose key factors to address when developing strategies for food waste prevention and valorisation. Phase 2 aimed at proposing and evaluating different strategies for bread surplus prevention, management and valorisation (Figure 1).

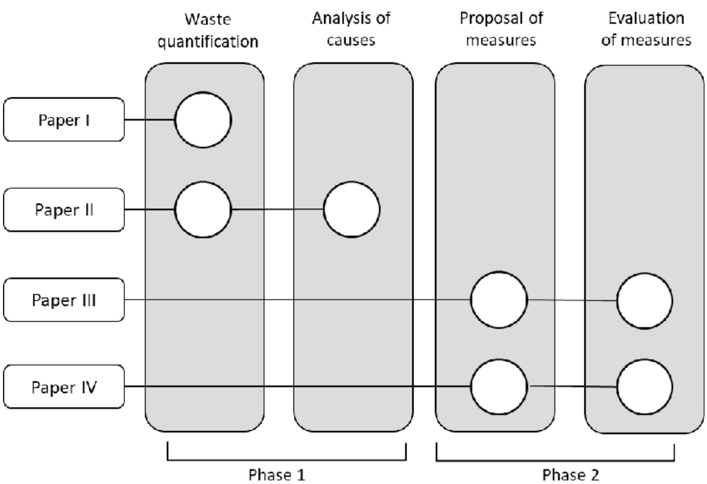

Figure 1 - Research framework.

The thesis is structured in five chapters:

- Chapter 1 introduces the research topic and explains the motivation and objectives.
- Chapter 2 provides an overview of food waste and describes the framework for food waste reduction and valorisation.
- Chapter 3 describes the methodology used in this study, including the methodology used for waste quantification, identification of risk factors for waste generation, and life cycle assessment.
- Chapter 4 presents and discusses the results of Phase 1, consisting of the quantification of food waste at the retail level, and bread surplus and waste at the national level,

covering the entire supply chain. Moreover, it presents the results of the identification of risk factors for bread surplus at the supplier–retailer interface.

- Chapter 5 presents and discusses the results of Phase 2 consisting of the evaluation of different measures for the prevention, valorisation, and management of surplus bread.
- Chapter 6 provides the main conclusions of the present research and offers insights for future work.

CHAPTER 2

BACKGROUND

Food waste has emerged as a critical global problem in the last decade because of its economic, environmental, and social impacts. Research has been conducted to understand the quantities and causes of food waste. Nevertheless, due to the complexity of modern supply chains, there is a need to further investigate the quantities, identify the reasons, and develop prevention and valorisation strategies for surplus or wasted food. This chapter presents some of the existing knowledge on food waste, including definitions and the framework for food waste reduction.

2.1. Food waste

Agricultural production is responsible for a substantial use of resources and emissions, and is a major driver to the surpassing of different planetary boundaries, including biosphere integrity and biogeochemical flows, which have already been transgressed to a high-risk zone (Campbell et al., 2017). The food system is also responsible for significant impacts on freshwater resources and on the pollution of terrestrial and aquatic ecosystems, due to the excessive use of nitrogen and phosphorus fertilisers (Springmann et al., 2018). All of these environmental impacts are exacerbated when a large amount of food is wasted, and all the resources and emissions associated with production are to no avail.

Food waste relates to multiple externalities, including food security, economic losses, and environmental impacts. FAO et al. (2020) estimated that the number of hungry people is 690 million, equivalent to 8.9% of the world population. Moreover, the same report estimates that more than 3 billion people are unable to afford a healthy diet. Reducing food loss and waste has the potential to contribute to better global food security by increasing the availability and access to food.

In addition to the environmental impacts incurred during agricultural production, food waste increases the environmental impacts at the end of its lifecycle. A significant amount of the organic waste generated in the world is landfilled or is disposed in open dumps, generating methane and other pollutants. A recent estimate indicated that 8–10% of total anthropogenic greenhouse gas (GHG) emissions are due to the loss and waste of food (Mbow et al., 2019). Springmann et al. (2018) studied various measures to keep the food system within the safe zone of the planetary boundaries, including technology development, changes in diet, and food waste reduction. The authors concluded that in order to maintain the food system within the planetary boundaries a combination of the three parameters is necessary, including changes towards plant-based diets, technological improvement leading to lower environmental impacts per unit produced and a significant reduction on food loss and waste.

The most recent studies that have attempted to estimate the amount of food waste globally are the Food Loss Index (FLI) (FAO, 2019), and the Food Waste Index (FWI) (UNEP, 2021). The former quantified food losses from post-harvest up to, but not including the retail level. The FWI covers the retail sector and consumption. According to the FLI, 14% of food produced is lost from post-harvest up to, but not including the retail level. The FWI reports the wastage of 17% of the food available at the retail and consumer levels. These two separate indexes are related to the SDG 12, more specifically they are indicators for target 12.3.

The Swedish government estimates over 1.3 million tonnes of food waste annually (Figure 2), corresponding to 133 kg of food waste per person (Naturvårdsverket, 2020). This number does not include food waste data for wholesale trade and e-commerce, which are currently missing. In addition to adopting Agenda 2030 and the SDGs, the Swedish government decided on a national target to be achieved by 2025, to reduce food waste by at least 20% and to increase the proportion of food that reaches the retail and consumer levels (Naturvårdsverket, 2016).

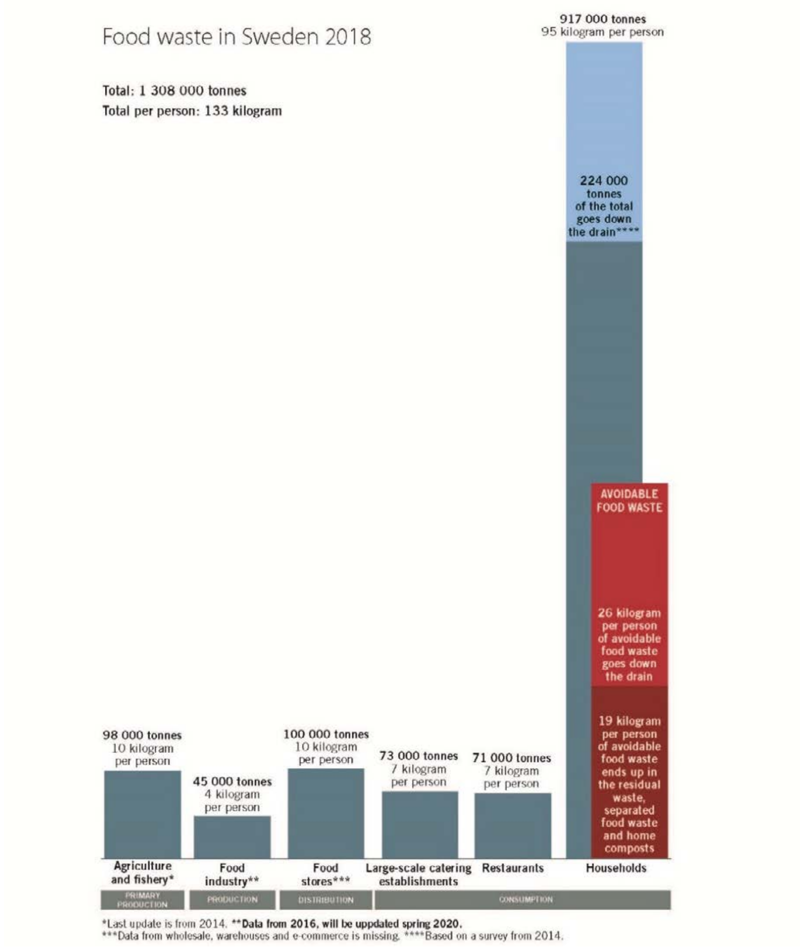

Figure 2 - Generated food waste in Sweden for the year of 2018, shown for various sectors of the food chain (Naturvårdsverket, 2020). Reprinted with permission.

The reduction of food loss and waste has the potential to contribute to the achievement of other SDGs, including SDG 2, which aims to '*end hunger, achieve food security and improved nutrition, and promote sustainable agriculture*'. Furthermore, since food waste reduction is generally connected with a decrease in the environmental burdens associated with the resources consumed and emissions that occur in the food supply chain, other SDGs related to the environment are also positively impacted, such as SDG 6 (clean water and sanitation), SDG 13 (climate action), SDG 14 (life below water), and SDG 15 (life on land).

The reduction of food waste has the further potential to provide societal gains, including an increase in productivity, economic growth, and business gains arising from, for example, an increase in the supplier's profit and monetary savings on the consumer's side (FAO, 2019). This is exemplified in a North American study, which analysed measures for food waste prevention and valorisation in the USA and estimated that businesses could potentially increase their profits by 1.9 billion dollars per year by the implementation of some of the food waste reduction measures proposed (ReFED, 2016).

Retailers are relevant actors in the supply chain and can contribute to address the issue of food loss and waste, even though their relative contribution to total food waste is smaller than that of other parts of the supply chain. In Sweden, food waste from the retail sector is estimated at 100 000 tonnes per year, corresponding to approximately 8% of the total waste (Naturvårdsverket, 2020). The importance of retailers is because the environmental impacts and economic losses related to food waste accumulate throughout the supply chain, and later stages, such as retailers and households, represent a larger amount of embedded impacts. Retailers and supermarkets fulfil a central function in the supply chain and exert influences both upstream and downstream in the supply chain through their purchasing policies, marketing, and discounts. Furthermore, retailers concentrate large quantities of waste in a few physical locations, facilitating the implementation of food waste prevention and valorisation measures. When these aspects are combined with an increased marketing concentration for retailers in Nordic countries, retailers are a relevant part of the supply chain to implement food waste reduction measures.

2.2. Definitions

2.2.1. Food waste

According to Article 2 of Regulation (EC) No 178/2002 of the European Parliament and of the Council (European Commission, 2002) food is defined as '*(...) any substance or product, whether processed, partially processed or unprocessed, intended to be, or reasonably expected to be ingested by humans*'.

Food waste is often referred to as a quantitative decrease in mass or a qualitative decrease in the nutritional value of food intended for human consumption in different stages of the supply chain. Nevertheless, different research fields and authors conceptualise food waste in various ways, varying in scope, definitions, and system boundaries. Variations in the definition of food waste have led to the development of datasets that are not always directly comparable, hindering the evaluation and monitoring of food waste prevention initiatives (Östergren et al., 2014).

There have been several attempts to harmonise the definitions of food waste, and a comprehensive review of the topic was conducted by Teigiserova et al. (2020). According to the FUSIONS definitional framework for food waste (Östergren et al., 2014), food waste is '*food and inedible parts of food removed from the food supply chain to be recovered or disposed (including composted, crops ploughed in/not harvested, anaerobic digestion, bioenergy production, co-generation, incineration, disposal to sewer, landfill, or discarded to sea)*'. Within this classification, food that is redistributed, diverted to animal feed or for materials and biochemical processing is not classified as waste. Under this framework, food waste is defined depending on its final destination.

Food waste has been further differentiated into food waste and food loss, and this differentiation has been used as indicators to the Target 12.3 of the SDGs in Agenda 2030 (United Nations, 2015). According to this definition, food loss is '*the decrease in the quantity or quality of food resulting from decisions and actions by food suppliers in the chain, excluding retail, food service providers and consumers*'. Food waste is defined as '*the decrease in the quantity or quality of food resulting from decisions and actions by retailers, food services, and consumers*' (FAO, 2019).

Food surplus is defined by Garrone et al. (2014) as '*the edible food that is produced, manufactured, retailed, or served but for various reasons is not sold to or consumed by the intended customer*'. In this classification, the definition is made at a point in time before the food reaches its final destination, in contrast to the definition of the FUSIONS framework. Therefore, surplus food is not necessarily waste, and this definition has been used in this thesis for food for which the destination is not defined or unknown.

2.2.2. Bread

Bread is defined as the product that results from the baking of dough obtained from a mixture of flour (wheat or other cereal) and water fermented with baker's yeast or a starter. In the context of this thesis, it is relevant to further define the different bread products sold in retail. The different categories are shown in Figure 3.

Prepacked bread

Prepacked bread refers to products produced by bakeries sold in packages at retailers. There are several categories of prepacked bread in supermarkets, such as white, wholegrain, and dark bread. Products within these categories can also differ in other aspects, such as the type of flour used (such as wheat, maize, barley, wholegrain, and gluten-free), type of leavening, and presence of other ingredients.

There are different distribution systems for prepacked bread, with differences in transportation, responsibilities for product handling, and disposal of unsold products. In Sweden, bread products may be distributed under a take-back agreement (TBA). For products sold under TBAs, the supplier is financially responsible for the unsold products and for the collection and treatment of returned products. The majority of prepacked bread in Sweden is sold under TBAs. Non-TBA products are often branded by the retailer, i.e. private label products. An extensive description and discussion of TBAs are provided in Section 4.1.2.

Bake-off products

This category refers to bread baked from semi-finished doughs in supermarket stores or those supplied by a bakery in a nearby store. It also includes bread produced from scratch by the store's bakery. Often, these products have a shorter shelf life in comparison with prepacked bread.

Figure 3 - Bread categories: Prepacked bread (left) and bake-off products (right). © [Didier Marti] and [Robert Kneschke] / Adobe Stock.

2.3. Framework for food waste reduction

The framework for waste reduction used in this thesis was based on an analytical approach grounded in the Deming cycle, similar to the approach of (Eriksson, 2015). This process includes four phases: planning, doing, checking, and acting. This approach allows the identification of priority areas for the implementation of measures and the evaluation of their efficacy.

Planning is the first step and consists of understanding the context of the research by identifying critical points to be prioritised and identifying the risks and opportunities that need to be addressed in the next stages of the research. The planning process can help to identify and prioritise the allocation of resources to specific areas with a potentially high impact (ISO, 2010). Planning also includes the definition of environmental performance indicators that will be used. The planning step in this study relates to waste quantification concerning mass and environmental and economic indicators in **Paper I**.

After the identification of critical points, it is necessary to understand the risk factors and causes of waste generation. This is translated in this thesis in the identification of risk factors for selected food products (**Paper II**). The identified risks and opportunities are inputs to the next phase, where solutions addressing food waste prevention and valorisation are proposed. Later, the environmental performance of the proposed measures is evaluated (**Paper III and Paper IV**). Following the methodology of Eriksson (2015), the steps are as follows.

1. Quantification of waste
2. Analysis of causes
3. Introduction of measures
4. Evaluation of measures

2.3.1. Food waste quantification

The quantification of food loss and waste (FLW) is a fundamental component for the design and implementation of measures or policies dealing with food waste reduction or valorisation. Even though FLW reduction measures can be implemented without information about the amounts generated, the quantification of food waste provides essential information such as the scale and location at which food waste is generated in the food supply chain, which is necessary for the design of efficient measures. The US Food Loss and Waste Policy Action Plan reports that the most successful organic waste management policies include provisions for measuring FLW (ReFED et al., 2021). Furthermore, the amended Waste Framework Directive (2018/851/EC) (European Commission, 2018) obliges EU member states to quantify and report food waste generation annually using a common methodology, with 2020 as the reference year. The directive establishes the minimum quality requirements for the uniform quantification of food waste.

Food waste quantification provides a baseline against which the results of a prevention measure can be evaluated. Furthermore, this is relevant for calculating indicators such as the environmental burdens, social impacts and economic costs associated with food waste. These indicators can be used to identify relevant waste fractions and consequently support the prioritisation of efforts (Caldeira et al., 2019, Corrado et al., 2019). However, food waste quantification is not a simple task, and even though there has been an increasing number of studies on the quantification of FLW, data availability is still relatively poor. Xue et al. (2017) reported that from 202 reviewed publications only 20% were based on primary data, and more than half of the studies assessed were based only on secondary data. Moreover, the majority of studies quantifying food waste focused on households, and few studies quantified food waste at retailers and wholesalers.

There is a particular lack of studies on the retail sector using primary data. This can be explained by the fact that food waste data are often sensitive to both retailers and suppliers and are subject to confidentiality agreements. There are several examples in the literature that report the hindrance of comprehensive investigations due to such issues (Lebersorger and Schneider, 2014, e.g., Garrone et al., 2016, Mourad, 2016, Mena et al., 2011, Stenmarck et al., 2011). The Food Waste Index (UNEP, 2021) reports that worldwide, there is insufficient data in the majority of the countries at the retail level, and calls for more quantification, particularly in low-income, lower-middle-income, and upper-middle-income countries.

Studies on waste quantification at the retail level have reported fruits and vegetables, bread, meat, and dairy as product categories with large amounts of waste (Cicatiello et al., 2017). Scholz et al. (2015) quantified food waste and the associated carbon footprint in six supermarkets over three years in Sweden. They demonstrated that the majority of waste, corresponding to 85% in mass, comprised fresh fruits and vegetables (FFV), and half of the carbon footprint was associated with the FFV waste stream. Meat products were observed to have a relatively low waste mass (4%), but a significant share in the carbon footprint (30%).

Bread waste is a particularly underreported fraction in the retailers' waste quantification literature. For example, Eriksson et al. (2012) and Scholz et al. (2015) did not include bread waste in their quantification since, due to TBAs, bread waste was managed by the suppliers and data were not available at the stores' database. Similarly, Cicatiello et al. (2017) studied one supermarket over one year in Italy and reported that the bakery department contributed 30% to the total mass of food waste, but at the same time, it was the department with the highest quantity of unrecorded waste, with less than 1% of the bread waste being recorded in the store's database.

Lebersorger and Schneider (2014) collected data from 612 retail outlets in Austria, covering 10% of the retail market, and quantified the amount of returned bread. The authors reported returned bread as the food category with the highest loss rate[2] by monetary value, followed by fruits, vegetables, and dairy products. The average loss rate for the assortment groups included in the study was estimated in 2.6% by monetary value. The authors observed a loss rate of 2.8% by monetary value for bread and pastry in-store, while the average loss rate for returned bread was 9.7%, ranging from 0% to 53% across individual stores. In Norway, fresh bakery products is the category with the highest loss rate in relation to turnover (Hanssen and Møller, 2013).

Some studies also combine the quantification of food waste with environmental indicators, which are calculated using life cycle assessment (LCA). The goals of such studies are to estimate the environmental impacts associated with the wasted food in retail and to identify products or departments with a high contribution to the total impact. The majority of these studies use a bottom-up approach, in which the detailed composition of the food waste gathered during the quantification is used to perform the assessment. Tonini et al. (2016) estimated the carbon footprint of retail food waste in the UK to be 2000 kg CO_2-eq t^{-1}. Brancoli et al. (2017) **(Paper I)** quantified the carbon footprint of a Swedish supermarket, estimating an impact as 3500 kg CO_2-eq t^{-1} of food waste.

2.3.2. Risk factors

Food loss and waste occurs at all stages of the supply chain due to various reasons. The identification of risk factors for food waste is essential for the development of food waste

[2] Loss rate is defined as the quotient of the sum of food waste and the sum of the respective sales, by mass or monetary value.

prevention and valorisation solutions. Moreover, food waste has different compositions and characteristics depending on the stage of the supply chain. For instance, losses occurring at the initial stages of the supply chain, such as agriculture and production, are often characterised by a relatively small variety of materials, while waste that occurs in later stages comprises a larger variety of products.

The most common reason for food waste in retail is that the food becomes unsaleable, or in a more extreme note, as noted by Eriksson (2015), an imbalance between the inflows and outflows is the root cause for any type of food waste in retailers. However, in the context of food waste prevention, it is necessary to understand why the food becomes unsaleable. Several researchers have investigated the reasons for food waste in retail stores. Mena et al. (2011) proposed a framework that categorises the causes of food waste into three groups, namely, megatrends, natural constraints, and management. Megatrends refer to trends in the industry, where the companies' strategies have limited influence, including an increase in the demand for fresh products and a decrease in the use of preservatives. Natural constraints relate to the characteristics of the products, such as short shelf-life and seasonality of supply and demand. Finally, management practices are factors that companies have a direct influence on, such as information sharing, forecasting and ordering, and performance measurements and management.

Eriksson et al. (2020) reviewed existing legislation in the European Union linked to food waste and identified the liability for food donation, zero tolerance for food contaminants, and date marking, as policies that affect the risk of food waste. Fiscal instruments were also identified as both positively and negatively affecting food waste. For example, the value-added tax (VAT) that donors must pay on donated food might prevent them from doing so, as it might be cheaper and easier to simply discard the food rather than donating it. This issue has been addressed in some EU countries by the reduction or removal of the payment of VAT on donated food. Some countries such as Portugal allow donors to deduct up to 140% of the value of the donated food, while in France and Spain, part of the value of donated food can be claimed as corporate tax credit (European Commission, 2021).

Other risk factors identified in the literature relate to managerial aspects, including inadequate ordering practices, poor inventory management, and stock rotation (Gruber et al., 2016, Canali et al., 2017, Derqui et al., 2016, de Moraes et al., 2020). These are associated with overestimated orders and a managerial lack of control over the inflows and outflows of products, particularly for those with a short shelf-life. Mena et al. (2011) reported that supermarket managers admit to having excess products in stock to guarantee a high level of service, preferring the wastage of the product rather than being out-of-stock. The absence of practices related to food waste quantification (Derqui et al., 2016, Ghosh and Eriksson, 2019), and a lack of clarity regarding the responsibilities for work procedures (Mena et al., 2011, Gruber et al., 2016) have also been identified as risk factors.

The lack of cooperation, coordination, and information sharing among actors in the supply chain is another relevant risk factor (Derqui et al., 2016, Aiello et al., 2015, Balaji and Arshinder, 2016). One phenomenon that relates to such risk factor is the 'bullwhip effect'. The phenomenon is defined as an increase in the demand variability in the supply chain from downstream actors, such as retailers, to upstream actors, such as manufacturing (Cachon et al., 2007). This can happen due to a sudden change in the consumers' purchasing behaviour, e.g. due to the promotion of a specific product. Such a change in the demand can be amplified and propagated upstream in the supply chain leading to the risk of overproduction and consequently waste when the producers can not react fast enough to such changes (Taylor, 2006). Similarly, retailers' strict aesthetic requirements can increase waste in the primary production (Mena et al., 2011).

2.3.3. Measures

The next step in the framework, after the quantification and identification of risk factors, is the development of measures that aim to solve the root causes of food waste. Overall, measures dealing with food waste can be classified into prevention and valorisation measures.

According to the Waste Framework Directive (European Commission, 2008), prevention is defined as '*(...) measures taken before a substance, material, or product has become waste*'. Prevention measures are those that '*reduce the quantity of waste, including through the reuse of products or the extension of the life span of products, the adverse impacts of the generated waste on the environment and human health, or the content of harmful substances in materials and products*'. Overall, measures that reduce food production can be categorised as prevention measures.

Valorisation measures are activities that recover materials, nutrients, or energy from waste materials. These relate to the conversion of food waste into value-added products and activities that reduce the negative environmental, economic, or social impacts (Eriksson, 2015). These includes industrial applications that use waste as a raw material, for example, for animal feed, chemicals, fuels, and materials. Valorisation also might include waste management pathways, such as anaerobic digestion, composting, and incineration with energy recovery.

Nevertheless, such definitions may vary depending on the framework used. For example, the FUSIONS definitional framework for food waste (Östergren et al., 2014) considers that materials sent to some valorisation pathways, including those for animal feed, bio-based materials, and bio-chemicals, are not defined as food waste. Therefore, these activities can be defined as prevention activities under the framework.

Stenmarck et al. (2011) summarised and classified actions aimed at waste prevention in retail in five categories: unsaleable food, control of orders, handling of food, personnel, and technical improvements. Practices of unsaleable food are related to increasing the opportunity to sell food products that are close to becoming unsaleable, such as the increased exposure of products with short shelf life, discounts of products close to their marking date, and donation or selling to restaurants. Control of orders refers to the optimisation of purchases, assortments, and amounts. Handling food practices are related to the proper handling of food (e.g., temperature and storage), packaging, and procedures for easier display and marketing of products. The training of personal on ordering and on the environmental, social, and economic impacts of food waste fall under the personnel category. Finally, technical improvements are related to the maintenance of proper temperatures in freezers, packaging, and appropriate use of cages and shelves when displaying and storing. Furthermore, Liljestrand (2016) proposed coordination mechanisms, such as information sharing, joint decisions, and rules for regulating producers, wholesalers, and retailers as efficient instruments for the prevention of food waste.

Several studies have proposed opportunities for the valorisation of food waste into value-added products. Enrica (2017) argued that the valorisation of food waste is an essential part of the circular economy model, providing opportunities to valorise waste in addition to promoting job creation and sustainable local development. Multiple pathways have been proposed for food waste valorisation, for example, the production of bulk and platform chemicals, such as succinic acid and lactic acid, bio-plasticizers, phytochemicals, animal feed, and fuels (Pleissner et al., 2016, Leung et al., 2012, Mirabella et al., 2014, Alonso et al., 2010). Nevertheless, Lin et al. (2013) reported that the use of mixed food waste as a raw material for large-scale applications in biorefineries is challenging due to the heterogeneous composition of the waste, high water content, and low calorific value.

2.3.4. Evaluation

The debate over food waste has resulted in an increasing number of initiatives dealing with food waste prevention and valorisation. However, there is a lack of studies that evaluate such measures in relation to their efficacy in preventing waste and the economic, environmental, and social benefits or burdens related to them. Clarifying the benefits associated with a measure is essential for creating an incentive for businesses and consumers to apply them.

According to De Laurentiis et al. (2020), only a few studies have assessed the efficacy of waste prevention measures in relation to the amount of waste prevented and the authors call for more evidence-based assessments of these initiatives. This is further stressed in a publication by Goossens et al. (2019), who reported that the lack of evaluation of the interventions hinders the prioritisation, comparison, and identification of trade-offs among measures.

Recent studies have evaluated prevention measures. For example, Aschemann-Witzel et al. (2017a) assessed success factors for measures aimed at consumer food waste prevention. The authors identified '*collaboration between stakeholders, timing and sequence of initiatives, competencies that the initiative is built on, and a large scale of operations*' as key success factors (Aschemann-Witzel et al., 2017a). Nevertheless, the authors clarify that '*success was not defined as an actual reduction of food waste, given it was expected that few initiatives can actually measure this. At the outset, success was defined as either having achieved media/societal awareness and popularity or having stable or growing operations of the initiative*'.

To support a common approach for the assessment of food waste prevention measures, De Laurentiis et al. (2020) developed a framework based on six criteria, namely, quality of the action design, effectiveness, efficiency, sustainability of the action over time, transferability and scalability, and intersectoral cooperation. Reynolds et al. (2019) reported on the challenges of measuring the outcomes of interventions and highlighted that the use of waste composition analysis is recommended, as it allows accurate measurements, although waste composition analysis is often time-consuming and expensive.

Several studies in the literature have focused on the use of LCAs to assess the environmental impacts and benefits of different food waste prevention, management, and valorisation alternatives. Laurent et al. (2014) reviewed 222 studies that conducted LCAs on solid waste management and concluded that for organic materials, landfilling had the worst performance in comparison with anaerobic digestion, thermal treatment, and composting. The results also indicated that most studies favoured anaerobic digestion over other options of waste management, even though the trend was less evident than the one observed for landfills.

The environmental assessment of prevention measures has been addressed by Eriksson and Spångberg (2017), who assessed the carbon footprint of food surplus redistribution and found greater environmental benefits in comparison with anaerobic digestion, conversion, and incineration. However, discrepancies in the ranking of the alternatives were found depending on the food product assessed. Martinez-Sanchez et al. (2016) compared prevention with other management options, such as incineration, use as animal feed, and anaerobic digestion. The results demonstrated that prevention provides the highest environmental savings as long as income effects are minimised. Income effects, also known as rebound effects, are related to changes in consumption due to cost reductions. For example, food waste prevention in households generates economic savings by reducing food purchases, and these savings may be used to purchase other goods or services that might have higher environmental impacts (Reynolds et al., 2019, Martinez-Sanchez et al., 2016).

The majority of the literature on LCAs of waste management and valorisation pathways focus on the lower and higher extremes of the waste hierarchy, while few studies have addressed food waste valorisation pathways. An example of LCA of valorisation pathways is the work of Albizzati et al. (2021), which evaluated the environmental and socio-economic impacts of the production of animal feed, lactic acid, polylactic acid, and succinic acid from food waste. The main conclusions of the study were that the production of chemicals still require optimization to be competitive and sustainable, while the production of animal feed substantially decreases the environmental impacts in comparison with conventional feed products.

Few studies have assessed the environmental impacts of prevention or valorisation pathways for bread surplus or waste. This could be explained by the fact that valorisation measures for segregated bread waste might not be feasible in countries without a return system for surplus bread. Moreover, as discussed in previous sections, the lack of data regarding bread waste might have deemed it as an irrelevant flow and specific measures were not considered. Studies that evaluated pathways for bread include those by Eriksson et al. (2015) and Vandermeersch et al. (2014). Eriksson et al. (2015) compared six waste management options for different food fractions and concluded that bread waste has the highest potential for greenhouse gas emission reduction. Vandermeersch et al. (2014) compared the anaerobic digestion of food waste with a scenario where bread was diverted for animal feed production, and the results indicated that using bread as animal feed reduced the environmental impacts by 26–32% in comparison with anaerobic digestion scenario.

2.4. The waste hierarchy

The waste hierarchy defined in the EU Waste Framework Directive (European Commission, 2008) defines priorities for the efficient use of natural resources aiming for the reduction of the environmental impacts caused by the generation and management of waste. It defines waste prevention and reuse as the most preferred options, followed by recycling, energy recovery, and disposal through landfills as the least preferred option (Figure 4).

Food waste has gained increasing attention in the waste management field because of its scale and economic and environmental impacts (Papargyropoulou et al., 2014). For this reason, waste hierarchies focusing on food have been developed taking into account the characteristics of the flow. Several countries, including the Netherlands (Dutch Ministry of Economic Affairs, 2014), France (Journal Officiel, 2016), the United Kingdom (UK Department for Environment Food & Rural Affairs, 2021), and the United States (USEPA, 2018) have national legislation or provide guidance on this matter. Although there are differences in the hierarchy defined by each country (see e.g., Redlingshöfer et al., 2020), overall, the food waste hierarchy follows a similar pattern, with some differences in the priority order for some pathways.

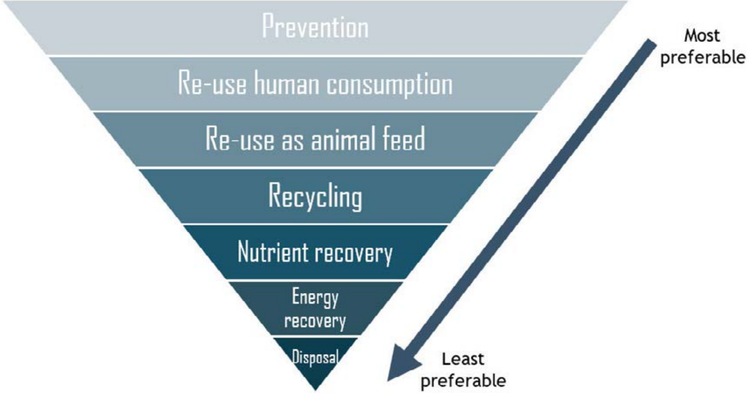

Figure 4 - Hierarchy for prioritisation of the management of food surplus, by-products, and food waste. Adapted from Sanchez Lopez et al. (2020).

The prevention of food waste has the highest priority among all food waste hierarchies. The goal is to maintain food edible, as it is its primary function. Prevention measures are varied and normally dependent on the part of the supply chain where waste occurs. It includes better logistics, improvements in storage and transportation, and consumer awareness campaigns.

Food redistribution is the second priority level, consisting of the priority use of food for humans, through food banks or other networks. In some cases, this level is categorised as prevention. Nevertheless, there are some differences when modelling the environmental impacts of donation, such as the need for transportation and the potential waste generated. Moreover, the donation of food is subject to strict legislation to ensure safe and hygienic conditions (Priefer et al., 2016).

Following the use for humans, the third level in the hierarchy is the reuse of food as animal feed. This includes the use of certain food products as feed and is regulated by strict legislation aimed at ensuring animal and human health. Recycling is the fourth level in the hierarchy, and refers to the use of food waste for the manufacturing of products with a high economic added value, such as succinic acid (Leung et al., 2012), sophorolipids (Wang et al., 2020) and polyhydroxyalkanoate for the production of bioplastics (Bassi et al., 2021). Nevertheless, many of these technologies are new and have low technology readiness levels, which are often associated with higher uncertainties in their environmental and economic performance (Albizzati et al., 2021).

Nutrient recycling and energy recovery from food waste is the fifth level in the hierarchy. This includes processes that yield low value-added products, such as biogas, bioethanol, and compost. This level includes processes such as composting and anaerobic digestion. These processes are higher in the hierarchy when comparing to incineration with energy recovery. The least preferred option in the hierarchy is disposal, which includes incineration without energy recovery, landfilling, and sewage disposal (Sanchez Lopez et al., 2020).

In order to comply with the strict legal frameworks and environmental protection targets defined by governments, waste management and valorisation systems have become progressively more complex and there are situations where the waste hierarchy might not represent the best pathway in relation to environmental outcomes (Manfredi et al., 2011). This might occur because such systems can be influenced by local conditions, such as waste composition, sorting efficiency, waste collection, the energy matrix, and the technology used in the system. Deviations from the waste hierarchy are allowed as long as they are supported in a scientifically robust manner. This support is often based upon the use of LCA methodology. It is important to note that the hierarchy is not always only based on environmental indicators, but also considers economic benefits and societal improvements. In this thesis, the main focus is on the environmental performance of the analysed pathways.

CHAPTER 3

MATERIALS AND METHODS

This chapter presents the different methodologies used over the course of this research to achieve the proposed goals. **Paper I** and **Paper II** focus on food waste quantification and identification of risk factors. **Paper III** and **Paper IV** focus on the life cycle assessment of prevention, management, and valorisation pathways for surplus bread.

3.1. Food waste quantification

Ideally, food waste quantification studies would encompass the entire supply chain and include a composition analysis to understand the weight and characteristics of products being wasted and where the losses occur in the supply chain. However, by increasing the accuracy, and consequently, the usefulness of the data, the costs for undertaking such studies also increase. Therefore, the methods used in a quantification study must reflect and be adjusted accordingly to the goals of the study.

Food waste quantification can be used for various purposes, such as to understand how much and where food waste occurs, to understand the reasons for waste generation, and to prioritise effective actions for reducing food waste (**Paper I** and **Paper II**). The goal is to understand the quantities and the composition of the products being wasted. This information can be used to identify hotspots in the supply chain for food waste prevention and valorisation measures. For this purpose, details on what (quantities and composition), why (risk factors), and where food is being wasted are required.

The quantification of food waste can also be used for the identification of food waste prevention measures and valorisation pathways (**Paper II**). To propose and prioritise prevention measures or valorisation pathways for food waste, it is necessary to account for waste generation at a macro level (Caldeira et al., 2019). It is also necessary to map and quantify the flows that are to be managed in order to identify business opportunities for the valorisation and management of the waste flows. In this thesis, two quantifications were carried out with different goals. The details of each quantification method are described below.

3.1.1. Quantification for hotspot identification

The primary goals of the first food waste quantification (**Paper I**) was to obtain a better understanding of the current situation in relation to waste generation at Swedish retailers and to identify hotspots and product categories with a high potential for waste reduction.

Data for a typical mid-sized urban supermarket located in the city of Borås, Sweden, were collected for one year. Data collection relied on the supermarket's database, and the supermarket had a routine for recording food waste before the study. The quantification is done by the scanning of food waste by the staff, a commonly used method by retailers to track the store's inventories. Bulk products, without barcodes, such as fruits and vegetables, were weighed and manually added to the database.

Eriksson (2015) classified food waste at retailers into four categories. Pre-store waste comprises products rejected by the supermarket due to non-compliance with quality standards and is often discarded at the supermarket, even though, in accounting terms, it belongs to the supplier. In-store waste is defined as waste occurring after purchase from the supplier and is divided into recorded and unrecorded. The last category is defined as 'missing quantities' and refers to the difference in the mass balance of inflows and outflows, mainly due to theft and water evaporation. Considering the framework developed by Eriksson (2015), only the flow 'recorded in-store waste' was considered in **Paper I**. Pre-store waste and unrecorded in-store waste were not included in the quantification. The data were available at the product level and included the individual code number (EAN code), name of the product, brand, amount and units, the cause of wastage with commentaries, date of the record, and price. However, as no data were available for the quantities of products sold, it was not possible to calculate the loss rate. When the waste data were not available in mass units, for example, when the data were in volume or unit, the data were converted using approximated density or packaging size. The waste products were aggregated into eight categories, namely, bread, pastry, vegetables, fruits, meat, dairy, ready meals, and others. Further, the data were analysed for its environmental impact and economic costs.

3.1.2. Quantification for understanding the quantities and risk factors

The goal of the second quantification (**Paper II**) was twofold: first, to estimate bread surplus and waste throughout the Swedish supply chain, and second, to identify risk factors for waste generation, particularly at the supplier–retailer interface. Therefore, in addition to the quantities of waste generated, it was necessary to collect data that could be used as indicators for different risk factors. Risk factors can be further investigated by the correlation between food waste and the indicators individually and in combination.

Due to the lack of available data covering all actors in the supply chain, primary data were combined with secondary data to estimate the quantity of bread waste in the entire supply chain. Primary data covered bakeries and retailers, while the remaining stages of the supply chain waste were estimated using literature data.

Primary data were collected from retailers and bakeries. Retailers' data were collected over one year from 380 retail stores, including a diverse types of shops, e.g. discount and premium stores, and a range of sizes, from 300 to 4000 m^2. For the bakery sector, data were collected from an industrial-scale bakery over the same period. The market for bread in Sweden is dominated by a few companies (Table 2), making the data collected representative. The measurements were adjusted to provide a national estimate of the remaining bakeries and retailers. The data on surplus bread were scaled using market share as a basis. This approach follows the recommendations of the FUSIONS manual for food waste quantification (Tostivint et al., 2016). A detailed description of the calculations is presented in **Paper II**. The collected primary data included sales, delivery amounts, returns, primary package size, shelf life, and type of distribution. These data were essential to enable risk factor analysis (see Section 3.2).

3.2. Identification of risk factors for waste generation

Risk factors for bread waste at the supplier–retailer interface were investigated in **Paper II.** Regression models and non-parametric correlation coefficient Kendall's tau-b were used to investigate the effect of sales, shelf life, package size, and their combination, on the loss at the product level.

Multiple linear regression analysis was used to develop a model for predicting bread loss rates based on the following product characteristics: sales, shelf life, and primary package size. The variables were defined as follows: sales are the total amount of product sold during the year, shelf life is the period between the production and the best-before date as stated on the packaging, and package size is the mass of the product sold. In total, seven models were tested, one for each independent variable, three pairwise combinations of the independent variables, and one combining the three independent variables.

Moreover, in order to investigate whether the loss rate means were different for products under different distribution systems (TBA and no-TBA), a two-sample t-test was used. In all statistical analyses, a significance level of $p < 0.05$, was used.

3.3. Life cycle assessment

Life cycle assessment (LCA) is a standardised methodology that aims to evaluate the environmental impacts associated with a product or service from cradle-to-grave (ISO, 2006a, ISO, 2006b). It consists of four phases, namely, the goal and scope definition, inventory analysis, impact assessment, and interpretation. The goal and scope definition is the first stage in an LCA and describes the purpose of an LCA study. At this stage, the functional unit, system boundaries, and methodological choices are defined. Inventory analysis comprises activities related to data collection for the processes within the system boundaries and the normalisation in relation to the functional unit. The impact assessment consists of translating the inputs and outputs collected in the inventory analysis into environmental indicators, which can then be further normalised and weighed, depending on the goal of the study. Finally, the interpretation consists of the identification of significant issues and the evaluation of the results, leading to conclusions and recommendations (Figure 5).

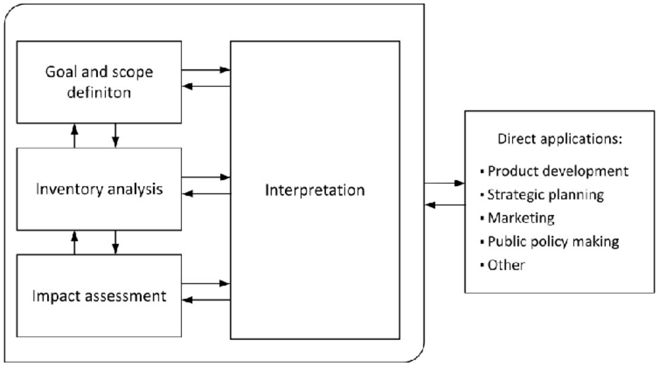

Figure 5 - LCA Framework (adapted from ISO (2006a)).

The LCA methodology was used in both phases of the research framework. In the descriptive phase (Phase 1), the methodology was used to assess the environmental impacts of different food waste fractions generated by retailers, complementing the data on other parameters, such as mass and economic cost. The goal was to support the decision of selecting a specific waste fraction for further analysis and development of prevention and valorisation measures (**Paper I**). Therefore, the functional unit used was the mass of waste generated by the supermarket over one year.

In the prescriptive phase (Phase 2), LCA was used to design, evaluate, and rank different prevention, management, and valorisation pathways for surplus bread. **Paper III** assesses the environmental impacts of technologies that can use surplus bread as a raw material. It has a waste management perspective, focusing on the service provided by the valorisation and management pathways. The functional unit used is 'the management, or prevention, of 1 kg of surplus bread in Sweden'. **Paper IV** uses an eco-design approach to support the development of a novel fungal fermentation technology. The focus is on the functionality of the product, aiming to highlight hotspots in its life cycle and to support process improvements aimed at reducing environmental burdens. Ultimately, it aims to answer how the new technology compares with technologies used for other products that provide the same functionality. The functional unit used was 1 kg of fermented fungal product after fermentation. Table 1 summarises the different perspectives, system boundaries, and functional units of the studies which used LCA.

Table 1 - Perspectives, system boundaries, and functional units of the different LCA studies.

Paper	**I**	**III**	**IV**
Perspective	Hotspot identification	Waste management	Product development
Functional unit	Amount and composition of food waste generated by a Swedish supermarket over one year	The management, or prevention, of 1 kg of surplus bread in Sweden	1 kg of fermented fungal product after solid-state fermentation
System boundaries	Cradle to grave	Collection of surplus bread to valorisation, waste management or donation	Collection of surplus bread to plant gate

Although the methodology was similar for all studies, differences in the modelling were made according to the goal and scope of the study. These details are explained in the following sections.

3.3.1. LCA of supermarket food waste

The goal of this study was to quantify food waste at a Swedish retailer in relation to its mass, economic cost, and environmental impact. Moreover, it aimed to identify hotspots at the product level. The functional unit of the LCA in **Paper I** was the mass of waste generated by the supermarket from October 2014 to September 2015 for selected products, namely beef, pork, chicken, bread, strawberries, bananas, tomatoes, lettuce, potatoes, carrots, cabbages, and apples. Data from the waste composition were complemented with biochemical and physical properties of the food products. These properties affect waste management technologies, for example, the potential methane yield of the waste flow in the anaerobic digestion scenario. Besides the inclusion of food, it is necessary to include packaging, as this affects the environmental impacts at the production stage and during waste management downstream. Data for the type of packaging and the material were retrieved from the literature and compositional analysis.

The system boundaries in the study were cradle-to-grave, including the resources and emissions from agricultural production until waste management. The geographical scope of this study was Sweden. The modelling for agricultural production used global average data. Representative production countries were selected for each product category and these were used to calculate the distances and transport modes to Sweden. The life cycle impact assessment method used was the International Reference Life Cycle Data System (ILCD) for midpoint indicators (European Commission, 2011). Details on the inventory and impact assessment data can be found in the Supplementary Material of **Paper I.**

3.3.2. LCA of management and valorisation pathways for bread waste

The goal of the LCA studies in Phase 2 of the project (**Paper III and Paper IV**) was to propose and develop alternative management and valorisation pathways for surplus bread, particularly for the returned fraction. **Paper III** focuses on the identification and ranking of prevention, valorisation, and management pathways for surplus bread.

One important methodological aspect was the handling of multifunctional processes. Waste treatments are often multifunctional since they lead to the production of secondary products, such as energy, materials, and nutrients, in addition to the main function of the management of the waste flow. In all LCA studies in this thesis, multifunctionality was handled via substitution, also known as avoided burdens, a common practice in waste management LCAs (Bakas et al., 2018). In the substitution approach, multifunctionality is solved by reducing the system to a single-function system by subtracting alternative systems for the production of the co-products functions (ISO, 2006a).

The most common functional unit for LCA studies dealing with waste management and valorisation alternatives is the ‘mass of waste generated’. As described by Ekvall et al. (2007), this implies that the management systems do not affect the amount of waste generated. This allows the simplification of the system boundaries in comparative studies to exclude upstream processes, which are identical in the compared scenarios. This approach is often referred to as the ‘zero burden assumption’, where waste streams come in with an environmental load equal to zero (Ekvall et al., 2007). However, when assessing activities that affect the amount of waste generated, such as prevention and donation, the upstream processes prior to the generation of the surplus or waste material must be accounted for, since prevention measures might reduce the production of food. Therefore, to analyse the consequences of prevention measures, it is necessary to include the benefits derived from the avoided food production.

Figure 6 illustrates the system boundaries for the prevention scenario and a waste management scenario, namely, anaerobic digestion. The system boundaries for anaerobic digestion include the biogas and digestate produced. It is assumed that the biogas is upgraded and substitutes diesel as a vehicle fuel, and the digestate is applied on land and substitutes mineral fertiliser. Moreover, the rejected material during the pre-treatment was assumed to be incinerated, and an equivalent amount of heat and electricity was substituted. For the prevention scenario, the avoided production of bread was considered and included the production of raw materials, processing, and transportation. A similar approach has been used in other publications (e.g., Albizzati et al., 2019, Sanjuan-Delmás et al., 2021, Eriksson et al., 2015).

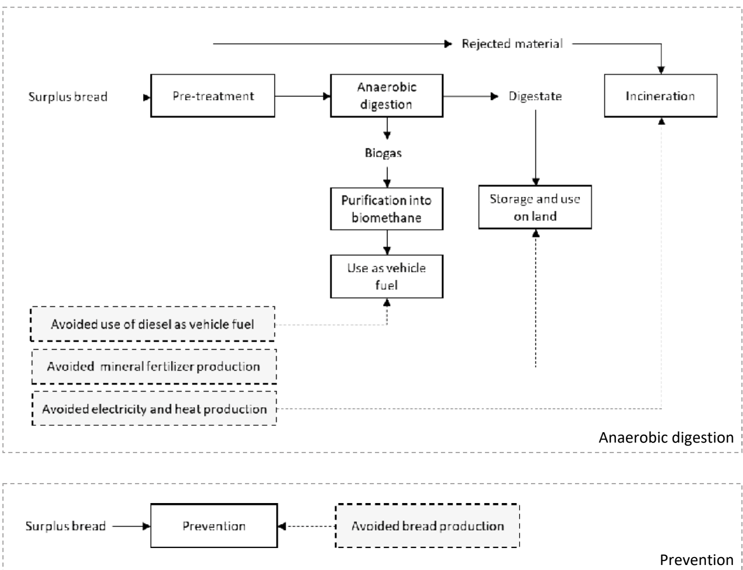

Figure 6 - System boundaries for the anaerobic digestion process in Paper III. The dashed lines and dashed boxes represent the substituted products in the market (Brancoli et al., 2020).

Paper IV has a different approach compared to the previous studies. It is a standalone LCA focused on product development. The technology under investigation in **Paper IV** is a new technology at an early stage of development and it can be considered as an emerging technology that currently exists at a lab-scale. The integration of an LCA in the early stages of technology development is useful to assess the implications of design choices on the environmental performance of the technology (Cucurachi et al., 2018). Decisions in the early stages of technology development can have a significant influence on the later environmental impacts of a product. Therefore, such studies are crucial since they can identify and prevent environmental impacts before the technology has been embedded by path dependency and lock-in (Jeswiet and Hauschild, 2005). The negative aspect of LCA studies of technologies in the early stage of development is the limited data availability and the higher level of uncertainty. This is illustrated by the Collingridge dilemma (Collingridge, 1982), which can be interpreted in LCA studies by the higher possibility of influencing technological development in the early stages accompanied by lower knowledge of the environmental burdens.

CHAPTER 4

DESCRIPTIVE STUDY

This chapter presents and discusses the results from the first phase of the research (**Paper I** and **Paper II**). The first part of the chapter describes the results of the quantification of food waste at retailers and the amount of bread waste in the Swedish supply chain. The second section describes and discusses the risk factors identified for bread surplus and waste generation. Finally, in the third section, the limitations and uncertainties of the research are discussed.

4.1. Quantities

4.1.1. Food waste at retailers

The Swedish market for retail and wholesalers is integrated within retail groups. Common to all Nordic countries, the market is dominated by a few very large retail and wholesale chains. Three actors control approximately 90% of the market share in Sweden, while the remaining 10% is controlled by two retailer groups (Table 2).

Table 2 - Swedish market share held by different bakeries and retailers (DLF et al., 2020, Company confidential, 2018).

Company	Market share
Bakery	
Pågen	39%
Fazer	25%
Polarbröd	16%
Private label	7%
Retailers	
ICA	52.3%
COOP	18.8%
Axfood	18.3%
Bergendahls	5.3%
Lidl	5.1%

The waste quantification in **Paper I** was undertaken at a medium-sized supermarket using data from October 2014 to September 2015. The waste was categorised into the following fractions: bread, meat, fruits, vegetables, pastry, ready-meals, dairy, and other. The results demonstrated that the supermarket wasted 22.5 tonnes of food during the one-year period of the study. Figure 7 shows the percentage of waste in relation to the categories assessed.

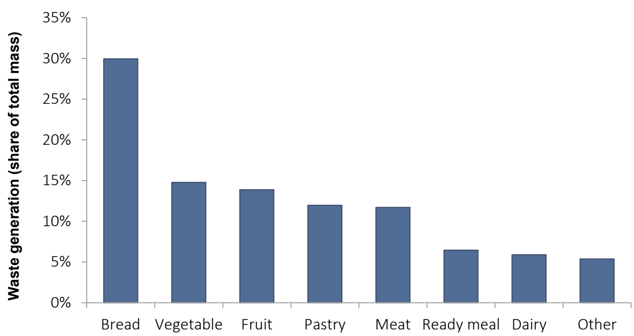

Figure 7 - Contribution of different food waste fractions to the total food waste at the supermarket (Brancoli et al., 2017).

The category with the highest mass fraction in waste was bread, with a waste mass of 6.7 tonnes, followed by vegetables and fruits, which together contributed 29% of the waste mass. This is in line with the results of previous studies, which reported a supermarket waste composition dominated by bakery products, fruits, and vegetables (e.g., Lebersorger and Schneider, 2014, Cicatiello et al., 2017, Stensgård and Hanssen, 2016, Parfitt et al., 2016b). Nevertheless, primary data on bread waste, particularly at the supplier–retailer interface, are scarce in the literature and have not been included in some studies due to the data being held by bakeries (e.g., Scholz et al., 2015). A detailed analysis of bread waste is provided in Section 4.1.2.

The analysis at the product level for fruits and vegetables show that a limited number of products contributed to larger shares of the waste. For vegetables, eight products were responsible for half of the waste. The highest amount of waste was found for salad products, tomatoes, and potatoes, similar to the results reported by Eriksson et al. (2012). For fruits, strawberries, bananas, lemons, melons, and kiwi were the products with the greatest amounts of waste. Another relevant category was pastry with a total waste of 2.7 tonnes, contributing to 12% of the total waste mass. This category comprises products baked at the store, including buns and cakes (*kaffebröd*). The detailed results for waste quantification are presented in Table 3.

Table 3 – Results for the waste quantification per product category for one supermarket.

Category	Waste mass (kg year^{-1})
Bread	6 723
Vegetable	3 317
Fruit	3 115
Pastry	2 686
Meat	2 627
Pork	*1 192*
Beef	*645*
Chicken	*360*
Other	*430*
Ready meal	1 449
Dairy	1 322
Other	1 207
Total	**22 445**

An important contribution of **Paper I** is the inclusion of bread in the quantification, which has been disregarded in many previous studies. Results from a questionnaire study conducted in the UK by the Waste & Resources Action Programme (WRAP et al., 2011) indicate that there is a consumer perception that there is not a significant amount of bread wasted at the individual level and the belief that bread waste is not an environmental issue. Both of these behaviours were experienced to some extend with retail managers. Therefore, the results of **Paper I** also fulfil the role of raising awareness, as a key part of seeking a change related to the indifference concerning bakery waste, and the misperception of the quantities generated, the true cost of the waste, and the associated environmental impacts.

There are multiple reasons for reducing food waste. For example, it can lead to a positive impact on economic outcomes, food security, and environmental sustainability. Depending on the goal of the food waste reduction campaign, different products may be prioritised. As discussed previously, the design of effective strategies aiming at food waste reduction requires information on the location, magnitude, and underlying causes. It is also critical to identify the waste fractions with a high potential for waste reduction and that it would reflect in a substantial decrease in the environmental impact and cost, as the investments in food waste reduction typically have diminishing returns, meaning that the initial investments might be low, but the costs increase for subsequent food waste reductions (FAO, 2019).

Food waste quantification can be done using different units, e.g. mass, economic cost and environmental impacts, and the choice of the analytical method has an impact on the results. According to Strid (2012), the weakness of using only mass as an indicator is that products with a high environmental impact might not be prioritised because of their low mass. The strengths of using mass are transparency and comparability since the majority of studies report food waste using a unit of mass.

The use of different indicators should be defined according to the goals of the study to ensure that actions aimed at food waste reduction are measured in relation to the intended outcomes. This is exemplified in the work of Eriksson (2015), who measured food waste in supermarkets over five years in relation to mass and carbon footprint. The results show that, in some years, the waste in mass increased but the carbon footprint decreased due to an increase in the waste of fruits and vegetables and a decrease in wasted meat products, which have a relatively higher carbon footprint.

The results from **Paper I** combined the quantification of waste with environmental assessment and estimates of the economic costs of selected products. LCA was performed for selected product categories, which covered 50%, by mass, of all food waste at the supermarket during the year of study. The economic cost was approximated by the losses in sales and costs incurred for waste treatment. Figure 8 shows the relative contribution of each waste fraction included in the LCA to the total mass, its economic cost, and selected environmental impact categories.

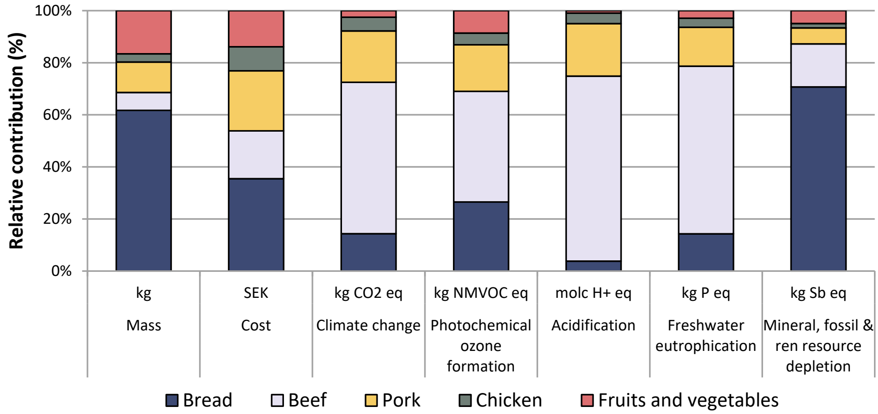

Figure 8 - Relative contribution of each waste fraction to the total mass, economic cost, and selected impact categories: climate change, photochemical ozone formation, acidification, freshwater eutrophication, and resource depletion. The data is for waste fractions that were selected for the LCA, which contribute to 49% of the total annual food waste at the supermarket (Brancoli et al., 2017).

An extensive list of the results for the environmental impact categories is presented in **Paper I.** The results estimated an average carbon footprint of 3500 kg CO_2-eq tonne^{-1} of food waste and demonstrated that, in relation to mass, bread products, and fruits and vegetables dominate the results. When the results are presented using a monetary indicator, more expensive products, such as meat, gained relevance (Figure 8). The use of environmental indicators further highlighted meat and bread products as hotspots.

Beef waste has the largest environmental impact in six of the nine environmental impact categories assessed, namely, climate change, particulate matter, photochemical ozone formation, acidification, terrestrial eutrophication, and freshwater eutrophication. Bread waste has the largest contribution in the remaining three environmental impact categories, namely, ozone depletion, freshwater ecotoxicity, and mineral, fossil, and renewable resource depletion. In the climate change category, beef waste emitted an equivalent of 22 tonnes of CO_2-eq, while bread had an impact of 5.5 tonnes of CO_2-eq.

The bread category was found to have the largest contribution in terms of the total mass of waste and the economic costs incurred by the supermarket. Beef has a relatively lower waste mass, but a higher environmental impact and cost per unit. In contrast, bread is an inexpensive product with a lower environmental impact and cost per unit. However, bread waste becomes significant when combined with the high amounts of waste.

The results from **Paper I** indicated that beef and bread are hotspots, and waste reduction in these categories would result in large economic and environmental benefits. Beef products were not prioritised due to their relatively low loss rates (waste per unit sold). Moreover, the retailer already had prevention measures in place for beef waste because of economic motivation, as this is a category with expensive products. Considering the diminishing returns to investments in food waste reduction (FAO, 2019), further reductions on beef products might be more difficult and expensive. In contrast, bread, in addition to its large waste quantities, is a product category that has been neglected in the past. Thus, considering the high potential for waste reduction, bread was selected as the focus in the next stages of research.

4.1.2. Bread waste in Sweden

The Swedish market for bread had sales of 18 625 million SEK in 2019, making it one of the highest market shares of food and beverages in the retail sector in Sweden (SCB, 2020). Bread is a staple food in the Swedish diet, with a per capita consumption of 74 kg per year (Jordbruksverket, 2021).

The business model for the majority of the largest bakeries and retailers relies on take-back agreements (TBAs), in which the supplier is financially responsible for ordering, handling, and returning unsold products. As a result, bakeries are in charge of forecasting, ordering, placement, and withdrawal of products close to the marking date from the retailer's shelves. The bakeries are also financially responsible for the unsold products and their collection and disposal, thereby operating in a reverse supply chain.

It is common in many countries for the bread supplier to be responsible for transportation to the retailer. The reason is that bread is a product with a short lead time, and the bakeries organise the distribution to ensure that the product arrives at retailers as quickly as possible. In Sweden, bread distribution is often done individually by each bakery or, more recently, some bakeries use the same reseller channel (Polfärskt Bröd AB, 2021). However, in some countries, the suppliers' responsibility is extended, and it includes additional activities such as the ordering and disposal activities mentioned previously. The occurrence of returned bread has been documented in Norway (Stensgård and Hanssen, 2016), Austria (Lebersorger and Schneider, 2014), the Netherlands (Weegels, 2010), Italy (Benozzo, 2011) and Germany (Brosowski et al., 2016), but the extension of the activities included in such agreements is unknown.

The results from **Paper II** estimated that 80 500 tonnes of bread are wasted each year in Sweden. The majority of this waste is generated at the household and retail levels (Figure 9). Bakery waste was estimated at 12 000 tonnes per year, and includes waste dough, flour, and other ingredients. The quantification of household bread waste relied on secondary data and 30 000 tonnes of bread per year was estimated to be wasted, accounting for 37% of the total waste of bread. Retailer waste was estimated at 28 000 tonnes, representing 35% of the total, similar to the value found for households. The similar volumes of bread waste at retailers and households differs from the trends in the Swedish statistics for food waste, where avoidable food waste in households is four times greater than that at retailers (Figure 2).

At the supplier–retail interface, prepacked bread was the category with the highest amount of waste, with 19 000 tonnes per year. The majority of prepacked bread is sold under TBAs, and only a small part of the product is not sold under such agreements. Products that are not sold under TBAs are often those sold under the retailers' brand. At the product level, the ten products with the highest waste in mass, which represents approximately 10% of the assortment, contributed to 60% of the total waste.

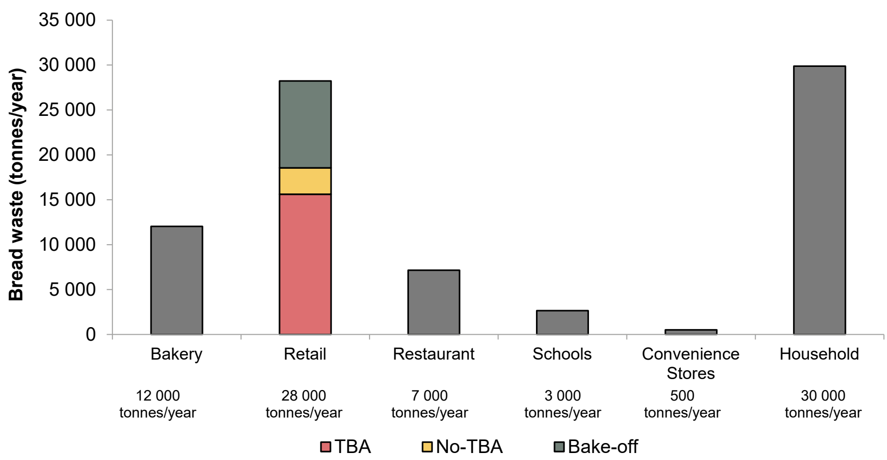

Figure 9 - Bread waste generated yearly in different parts of the value chain in Sweden (Brancoli et al., 2019).

4.2. Risk factors for bread waste

There are many reasons why bread surplus and waste occur at retailers. The identification of the causes and risk factors for bread waste is essential for designing effective measures for waste prevention. This section discusses the results from **Paper II** and the literature on the potential causes of bread waste at the supplier–retailer interface.

The demand for freshness and constant availability of a large range of products has become a common requirement in Western markets. This is particularly relevant for bread products. Stenmarck et al. (2011) report that consumers in Nordic countries expect a permanent availability of products even at closing time. Similarly, Schneider and Scherhaufer (2009) report that a consumer survey in Austria revealed that the majority of consumers consider freshness to be crucial when buying bread and that a large selection in the bakery department is important. The demand for variety and freshness is often met by full shelves and a large assortment of products, which decreases the turnover for each product and increases the complexity of forecasting (Eriksson, 2015). Products with low turnover are often wasted more in comparison with products with high turnover (Canali et al., 2017). **Paper II** shows that more than 70 different bread products from only one supplier are currently sold by Swedish retailers. Ungerth (2021) estimated that 165 types of bread are currently being sold in Swedish supermarkets. The identification of large assortments as a risk factor was further highlighted in **Paper II** by the negative correlation between sales and loss rates. This result is in line with previous findings in the literature (Lewis et al., 2017, Mena et al., 2011, van Donselaar et al., 2006).

Shelf life has been reported as a risk factor for bread waste at the category level. Lebersorger and Schneider (2014) found that the waste of 98% of bakery products was attributed to reasons related to date labelling. This can be explained by the fact that the shelf life of bread products is particularly short, and therefore, waste generation is sensitive to minor forecast errors. This is particularly relevant for bake-off products, which normally have a practical shelf life of a single day because these quickly become stale in the stores and next-day selling is very difficult. The results from **Paper II** on prepacked bread do not indicate a correlation between shelf life and loss rate at the product level. This can be explained by the low variation in shelf life between products and the fact that bread is often removed from shelves before the recommendation on the date label.

Take-back agreements as a risk factor for waste generation

Previous studies have identified TBAs as a risk factor for the generation of high volumes of bread waste (e.g., Eriksson et al., 2017, Lebersorger and Schneider, 2014). **Paper II** adds support to this claim by analysing the waste data for 380 retail stores and one industrial bread supplier. The statistical analysis indicated a higher waste level for products with TBAs in comparison with products without TBAs. Multiple perspectives need to be accounted for when attempting to explain the reasons behind the higher waste generation for TBA products.

The positive correlation between TBAs and high levels of waste can be explained by the fact that TBAs create a constant battle between suppliers and retailers, and among suppliers, over the service level on the shelves. On the one hand, retailers do not face the costs related to unsold bread and might require a high service level, while the suppliers may aim to minimise the returns. Another relevant aspect is the competition between suppliers in relation to the shelf space assigned in the stores. Although the suppliers control the amounts of products delivered to the stores, if they fail to occupy the shelf space assigned to them, this space might become available for competitors (Ghosh and Eriksson, 2019). This creates a stimulus for bakeries to occupy their assigned shelf-space, and consequently increase the amount delivered, even if it generates more waste.

The reasons for the high risk of waste generation due to TBAs are summarised below and extensively discussed in **Paper II.** Overall, the reasons for high bread waste levels in TBA products are related to information sharing and product ownership.

Information: Accurately forecasting demand is a complex task that is influenced by numerous factors, including weather, marketing campaigns, holidays, and promotions. Some bakeries believe that retailers could forecast sales more accurately because they have point of sales data (POS) and better control over events such as campaigns and stock levels. Inaccurate forecasting increases the risk that orders are bigger than the demand, and as a consequence of overstocking, products may be wasted. Another aspect related to information is the availability of data on bread waste. Returned bread is in a grey zone for waste quantification since the bakeries are financially responsible for the waste and its management, but the waste is generated at the retail level. This issue became clear when comparing studies that quantified bread waste in retail stores. Often, it is not clear if returned bread is included in these quantifications. Hence, TBAs not only increase the risk of waste but also increase the risk of underreporting the mass of waste. This can contribute to the perception that there is not a significant amount of bread waste generated, hindering the identification of bread as a relevant waste stream and the development of appropriate reduction and valorisation measures.

Ownership: Retailers have no financial stake in unsold bread and thus have no incentive to minimise retail losses. Moreover, bread stock is usually purchased by consumers in a last-in, first-out pattern, and TBAs hinder the sale of products close to their best-before date at a lower price. Ownership also influences the availability of data on the wastage of bread waste, as discussed previously. Frequently, bakery drivers are exclusively responsible for deciding the amount delivered to each store (Ismatov, 2015), although store managers can influence the decision. Nevertheless, store managers might lack sufficient knowledge on the demand for bread and as the costs of waste are borne by the bakeries, managers might not take actions to reduce waste and might be more prone to negotiate a larger delivery to ensure a high service level (Eriksson et al., 2017). According to Cicatiello et al. (2020), risk factors such as demand forecasting and a high level of service could be influenced by individual choices at stores.

Some companies in Sweden have ended the TBAs in an effort to reduce bread waste in stores. One example was the collaboration between the bakery Fazer and the supermarket chain Lidl, in which the bakery became the only external supplier of bread and the retailer took full responsibility for ordering and disposal activities (Fazer, 2018, Lidl Sverige, 2018). The retailer stated that by owning the product, they were able to exercise more control over the ordering process and price reduction for products close to the best-before (Lidl Sverige, 2018). Lidl reported that bread waste decreased by half in the first month of operation (Lidl Sverige, 2021). Nevertheless, to the best of the author's knowledge, there is no publicly available data on the overall efficiency of waste reduction by the measure during the period in which the TBA was not in place. This agreement ended at the beginning of 2021, and currently Lidl works with other bread suppliers, and with the TBA model (Lidl Sverige, 2021).

The waste reduction observed at the stores could be attributed to other reasons besides the end of the TBA and it was likely achieved by a combination of measures. For example, since the bakery became the sole external supplier, there was a reduction in the assortment of bread products available at the store. Moreover, it is expected that the communication and information sharing between the companies has increased due to the collaboration. A large assortment of products and lack of information sharing have been identified as risk factors for high waste generation and measures dealing with such risk factors, in combination with the end of the TBA, might have contributed to some degree to the waste reduction observed. The conclusion is that a single action might not be sufficient to achieve waste reduction, and many actions acting in different risk factors are necessary.

TBAs in the context of retailer market power

El-Ansary and Stern (1972) define power in the supply chain as '*the ability of a channel member to control the decision variables in the marketing strategy of another member in a given channel at a different level of distribution*'. The Swedish market for retail and bakery is characterised by the dominance of a few large retail and bakery chains (Table 2). The three largest retail chains in Sweden control approximately 90% of the market, while the largest bakeries control 80% of the market. The brochure on the directive on unfair trading practices in the agricultural and food supply chain (EUagri, 2019) states that '*asymmetry in bargaining power may lead to the imposition of unfair trading practices on suppliers. Due to their weaker position, suppliers are often de facto forced to accept unfair practices to continue to sell their products and maintain commercial relations with buyers in the supply chain*'.

The EU Directive on unfair trading practices (European Commission, 2019) prohibits 16 specific unfair trading practices and distinguish between 'black' and 'grey' practices. Black unfair trading practices are prohibited, regardless of the circumstances, while grey practices are allowed if the supplier and buyer agree beforehand in a clear and unambiguous manner.

There are two possible ways to interpret the TBAs of unsold bread under the proposed legislation. TBAs could be categorised in the blacklist prohibition, due to the '*risk of loss or deterioration transferred to the supplier*' (EUagri, 2019). It is prohibited under the legislation for a buyer to ask the seller to pay the price of deteriorations or losses that have occurred after the transfer of the ownership of the goods unless the problem has occurred due to the seller's wrongdoings.

Alternatively, TBAs could be categorised in the grey list, under the 'return of unsold products'. It can be argued that since the supplier controls the volume delivered to the store, the retailer has no control over the bread being sold or any deterioration or loss that occurs in the store. Therefore, the supplier also takes the risk that not all bread will be sold.

The essential question on defining TBAs as black or grey unfair trading practices is whether the buyer requires the shelves to be always full or not. If the bakeries have the power in that regard, TBAs would probably be categorised as a grey practice, since it is the bakery that determines the volumes delivered to the stores. In that case, it would be the seller and not the buyer in control of potential losses. In the scenario where the retailer requires the shelves to be replenished, the practice would be considered a black practice, and therefore, would require the retailers to pay for the unsold products. In addition, if the seller proposes to buy back food waste to produce biogas or similar, the Swedish government has stated that it would probably not fall under the prohibition (Näringsdepartementet, 2021).

The influence of TBAs on waste levels and its fairness as a trade practice are disputed by actors in the supply chain. In a recent report by Brödinstitutet (Lundell and Sitell, 2021), a subsidiary of the Swedish Bakers & Confectioners trade association, the authors argued that the return system in Sweden is preferable for the low generation of bread waste. At the same time, as discussed previously, other stakeholders believe that such agreements must be replaced and argue that TBAs are a risk factor for high waste generation and the end of such agreements resulted in lower waste generation (Lidl Sverige, 2021, SVT, 2019).

Lastly, Ghosh and Eriksson (2019) proposed that TBAs might also lead to higher market concentration in the bakery sector because larger suppliers can more easily absorb the costs related to returns due to large sales volumes, high bargaining power and relatively good forecasting. It is argued that this system might be a barrier for new and smaller actors to enter the market.

4.3. Limitations and uncertainties

The results reported in this phase of the study were based on the best available data. Nevertheless, the data gathered had limitations and uncertainties. The main source of uncertainties that might influence the results is the relatively low coverage of **Paper I** since it quantified waste generation at a single supermarket in Sweden, which reduces the representativeness of the data. Nevertheless, the results were confirmed from the literature. Moreover, the LCA covered only a part of the products wasted at retailers, which could hinder the identification of some products as hotspots.

In **Paper II** the main sources of uncertainty are:

- Uncertainties related to the lack of primary data on small- and medium-sized bakeries. This research provided limited insights into waste quantities at such enterprises. Furthermore, the data were derived for one industrial bakery. However, because the bakery market in Sweden is highly concentrated, the extrapolation error is likely to be small.

- The use of market share to scale company data to the national level assumes that the percentage of total sales generated by a particular company is a good proxy for waste quantities across companies. This relationship is uncertain, and companies might have differences in logistics for the distribution, range of assortment, and other characteristics that might affect waste generation.
- The need to reconcile databases from bakeries and retailers. Product identification was not always identical in the different databases, and manual work was required to produce compatible datasets.

CHAPTER 5

PRESCRIPTIVE STUDY

In the previous chapter, the aspects linking TBA to waste generation were explored. Nevertheless, contrary to the risk of waste generation due to the ordering activities, reverse logistics supported by the collection of unsold products enable the valorisation of bread, which might have a superior environmental performance in comparison with the current waste management practices in Sweden.

Overall, TBAs can be divided into two parts. The first relates to the activities of forecasting, ordering, and placement of products on the shelf, referred to as ordering activities. As shown in **Paper II** and discussed previously, ordering activities are related to the risk of high waste levels. The second group of activities, which includes the collection and treatment of unsold products, is related to the concept of a reverse supply chain. The segregated collection of surplus bread creates opportunities to implement cradle-to-cradle rather than cradle-to-grave measures by allowing valorisation pathways for the segregated bread, which are not possible for mixed food waste fractions.

These two opposite effects indicate that reverse supply chains do not always provide sustainability to the entire supply chain, and they must be analysed individually to understand if the net impacts are positive or negative. This is explored in **Paper III and Paper IV**, where pathways for bread valorisation, which are enabled by reverse logistics, are compared to waste prevention and the current waste management of organic waste in Sweden.

This chapter proposes circular economy strategies for the management and valorisation of bread surplus based on the results from the waste quantification, the analysis of causes, and the return system. The solutions proposed are based on the characteristics of the material, the assessment of potential measures, and their alignment with the food waste hierarchy. Life cycle assessment was carried out to investigate the extent of benefits arising from prevention and valorisation measures in relation to the current waste management pathways (**Paper III**). Furthermore, this chapter describes and discusses a new process for the valorisation of surplus bread into a high value-added food product based on solid-state fermentation (**Paper IV**).

5.1. Prevention, waste management, and valorisation of surplus bread

The environmental assessment of prevention and valorisation pathways must consider the composition and properties of the surplus bread flow, and how this affects the performance of the valorisation or management pathways. Furthermore, the assessment must account for co-products generated from each pathway, which has a significant influence on the results of the LCA.

Figure 10 shows the mapped prevention, valorisation, and disposal pathways for surplus and waste bread. This highlights the different bread flows and suitable measures for each. For instance, the lack of segregation of bread waste in public canteens, restaurants, and households makes it more difficult to be valorised, although source reduction measures would still apply for them. Bakery and retail surplus flows are easier to segregate and collect using the return logistics that are in place in Sweden. Therefore, it is possible to send such flows to any of the proposed valorisation measures. Figure 10 also identifies the displaced products in the market, named avoided products, which are relevant for the calculation of environmental impacts through LCA.

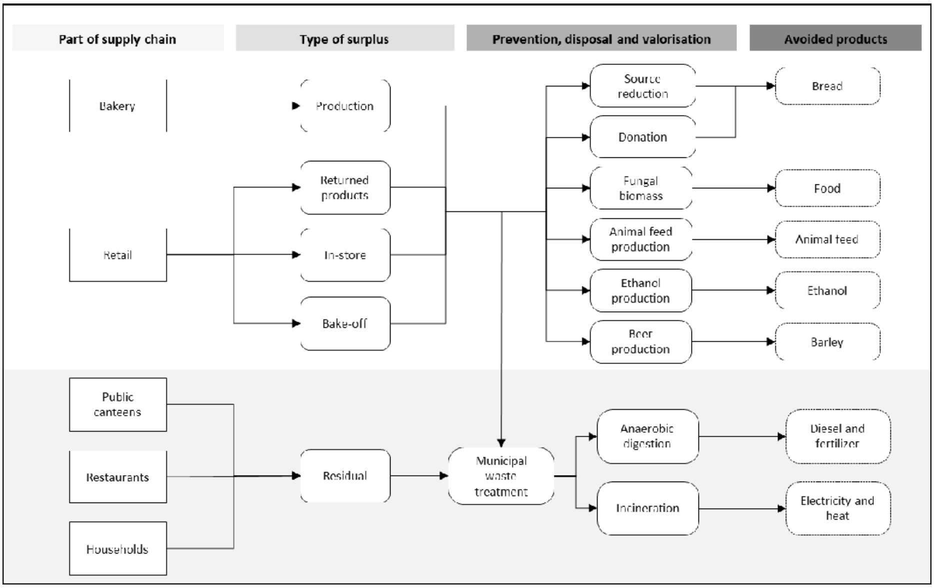

Figure 10 – Map of prevention, valorisation, and waste management pathways for surplus bread (Brancoli et al., 2020).

5.1.1. Food waste prevention

Preventing waste is the first step in any food waste hierarchy, and this is the priority of both Swedish and European waste legislation. From a resource conservation perspective, waste should be prevented rather than managed in any type of valorisation pathway. Prevention activities include improvements in waste reduction practices and donations.

There are various reasons for the wastage of bread, and the measures for reducing waste are dependent on the types of flow and the parts of the food chain where these occur. This section focuses on different strategies for the prevention of bread waste at the retail level.

Reducing the large assortment of prepacked bread has been identified as a potential waste reduction practice. As indicated in **Paper II,** the large range of products available to consumers, and the subsequent demand for being stocked, is a risk factor for high waste generation. This is because a large inventory is necessary to provide a high service level for a large range of products, and as demand is uncertain, this increases the risk of waste. The statistical analysis of the waste data in **Paper II** identified this risk by the negative correlation between sales and loss rates, in agreement with previous findings (Lewis et al., 2017, Mena et al., 2011, Schneider and

Scherhaufer, 2009, van Donselaar et al., 2006). Moreover, the largest loss rates were found for niche products, which are products often supplied in small quantities.

A recent initiative in a store in Gotland, Sweden, deals with the perceived consumer demand for full shelves. It is believed that a well-stocked shelf sells better than an empty shelf, but similar to having a large assortment range, this is a risk for waste generation. The solution was to produce pictures of the bread products and add them to the bottom of the shelf, creating a feeling for consumers that the shelf is well stocked (Friköpenskap, 2021). According to the store, sales were not affected, and bread loss was reduced by half within two months.

Another measure identified for bread waste prevention is the extension of the shelf life of products. Freshness is reported to be a particularly important characteristic for the purchase of bread, and consumers have reported the preference for bread to be as fresh as possible at the time of purchase (Lyndhurst, 2011, Østergaard and Hanssen, 2018). Companies have developed different strategies to extend the shelf life and increase the freshness of their products. Polfärskt Bröd AB, a company that distributes bread for different bakeries, transports bread frozen, allowing retailers to use the freezers of stores as buffers for managing demand. Another strategy is the use of a modified atmosphere in the packaging of bread, which is reported to being able to extend the shelf life by up to 200% (Rodríguez et al., 2000, Fernandez et al., 2006).

More specifically, in relation to the returned fraction, the take-back agreements (TBAs) are a risk factor for waste generation, as discussed in the previous chapter. Therefore, the prevention of bread waste can be achieved by shifting the process of ordering from bakeries to retailers, as indicated in **Paper II** and discussed in Section 4.1.2. Prohibiting the free return of bread has been highlighted as one of the three policy instruments with the highest effectiveness in reducing food waste in a publication from the Nordic Council of Ministers (Hanssen et al., 2021).

Bake-off bread is also a relevant category that generates a substantial amount of waste at retailers, and it is estimated that 9 000 tonnes of bake-off bread is wasted per year in Sweden (**Paper II**). These products have a short shelf life, due to the process of staling, which causes the textural and sensory attributes of bread to change during storage (Gmoser et al., 2020). The large range of bake-off items and consumers' expectations of their availability during store opening times are risk factors for waste. Retailers also perceived the need for a full shelf as a requirement to guarantee a sufficient number of sales. Interviews conducted by Stenmarck et al. (2011) indicated that the production of fresh bread is often increased by 7% on average to meet such demands. Besides the adjustment of production to meet demand, another possible solution to address this issue is the sale of bake-off products at a discounted price at the end of the day. This could be done at the store, or by the use of apps that connect surplus food from sellers or donors to buyers or charity organisations (de Almeida Oroski, 2020).

Lowering the price of sub-optimal products is a common strategy for reducing waste at the store (Aschemann-Witzel et al., 2017b). Schneider and Scherhaufer (2009) report that a common practice in Austrian bakeries is to progressively reduce the price of bake-off products during the day. However, there is an ongoing debate as to whether promotions can reduce waste or if they simply shift the burden of waste from the supermarket to consumers. Some researchers (e.g., WRAP, 2012, Mena et al., 2011) have reported that promotions could increase household waste due to over-purchasing. However, Aschemann-Witzel et al. (2017b) stated that deal-prone consumers are not, by definition, more likely to waste food. This was also supported by Giordano et al. (2019), who found no evidence of either a positive or a negative correlation between the purchase of discounted products and the quantity of household waste. Nevertheless, promotions have been attributed to waste at the retail level due to increasing forecast uncertainties (Eriksson, 2015). This is because promotions shift consumers' buying practices by purchasing the promoted product to the detriment of another product, contributing to more unpredictable demand patterns. This increased unpredictability might add complexity to the forecasting of not only to the promoted product but also to products that were substituted.

The redistribution of surplus bread is another strategy for prevention. This can be done for profit through digital tools or other commercial organisations, by selling the products on secondary markets, or the surplus can be donated to food banks and social organisations. Bergström et al. (2020) identified 18 different redistribution units in Sweden, including social supermarkets, soup kitchens, retail food bags that redistribute food to low-income people, and virtual markets, where companies can advertise surplus food to be sold or donated. Parfitt et al. (2016b) reported that redistribution might be limited due to the relatively short shelf life and concerns about the impact on brand integrity. Food safety is an important aspect of redistribution, and Scherhaufer and Schneider (2011) reported that bakeries often do not donate due to legal requirements and the impossibility of guaranteeing hygiene standards after the product has left their premises. As part of the Circular Economy Action Plan, the EU Commission has adopted food donation guidelines to facilitate the compliance of providers and recipients with relevant requirements, including safety, hygiene, and liability (European Commission, 2017).

The large scale of bread donation has been reported to exceed the needs of organisations. This phenomenon was observed in Austrian supermarkets and bakeries, where donated bread was left at stores due to the saturation of demand (Schneider and Wassermann, 2004a, Schneider and Wassermann, 2004b, Schneider, 2013). Similarly, it has been speculated that the market for bread donation is close to saturation in Sweden, and therefore, the efficiency of food donation for waste prevention is uncertain (Ungerth, 2021).

5.1.2. Food waste management

The municipalities in Sweden are responsible for their waste management plans, in accordance with national environmental objectives. The management of food waste in Sweden consists mainly of biological treatments and incineration. Landfilling of organic waste has been banned since 2005 (Miljödepartementet, 2001).

The most common method for managing organic waste fractions is anaerobic digestion, and only a small amount is composted in central composting plants or at home (Avfall Sverige, 2020). Anaerobic digestion involves the biological decomposition of organic matter into methane and carbon dioxide in an oxygen-deficient environment (Kabir et al., 2016). The process yields biogas, which is commonly upgraded to biomethane in Sweden and used as vehicle fuel (Larsson et al., 2016), and digestate, which can be applied on land as a biofertilizer, recovering nutrients such as nitrogen and phosphorus. The Swedish government has set a milestone to increase the sorting and biological treatment of food waste. The goal is that by 2023, at least 75% of food waste from households, commercial kitchens, shops, and restaurants must be sorted and treated biologically (Swedish Environmental Protection Agency, 2021).

Incineration is an alternative for the management of food waste. Incineration is a mature technology that involves the combustion of waste materials and the associated generation of electricity and heat, which is commonly used for district heating. The incineration of food waste alone is unusual due to the high moisture content and the presence of non-combustible materials (Pham et al., 2015). The incineration of food waste mainly occurs when there is no source segregation in place and the organic waste is mixed with other types of materials or when facilities for biological treatment are not available.

5.1.3. Valorisation pathways for bread

The mapping of valorisation pathways mostly focuses on the returned bread flow and its utilisation for alternative treatment methods at higher levels of the waste hierarchy. The advantage of returned bread is its relative homogeneity in comparison with mixed food waste from wholesalers, retailers, or households, which is heterogeneous in composition and therefore limits the scope for possible management solutions. Other surplus flows, including bakery waste, and in-store surplus, such as bake-off and non-TBA products (see definition in Section 2.2.2 and Figure 10) can also be valorised by the pathways proposed here. Nevertheless, there are some challenges, mainly related to the possibility of segregated collection. The cooperation between retailers and bakeries is necessary to enable such fractions to be transported together with surplus TBA products. Often, in-store bread surpluses, excluding TBA products, are managed in the same way that the stores deal with its mixed organic fraction, commonly via anaerobic digestion or incineration.

As the primary goal is the prevention of waste, priority was given to existing industries that could use surplus bread as a substitute for their regular raw materials, but would not be dependent on it. In other words, if the supply of surplus bread decreases, due to prevention measures, it would be possible for these industries to return to the use of the original raw material (e.g., wheat for ethanol or barley for beer). The technologies and justifications for their selection are discussed in detail below.

Ethanol

Bioethanol is a liquid biofuel produced by the fermentation of different biomass feedstock. Globally, bioethanol is the dominant biofuel for transportation, and it is suitable to be used mixed in gasoline engines due to some characteristics such as high octane number and high heat of vaporisation (Nair, 2017). Bioethanol is mainly produced from wheat in Sweden. The largest bioethanol plant is located in the south of the country. Currently, the plant has a capacity of 230 000 m^3 per year, consuming approximately 550 000 tonnes of mainly wheat as feedstock (Börjesson et al., 2012).

Bread is mainly constituted of starch, which is a generic fermentation feed that can be used for multiple applications (Ebrahimi et al., 2008). Ethanol can be produced from bakery residues, including dough, flour, breadcrumbs, and packaged (after packaging has been removed) and bake-off bread. The conversion of starch to ethanol is a two-step process. In the initial step, starch is converted into fermentable sugars, which are subsequently fermented into ethanol using *Saccharomyces cerevisiae.*

Ethanol production based on waste feedstock can help mitigate environmental impacts and the competition between energy and food crops, according to a policy brief from the European Union (Hirschnitz-Garbers and Gosens, 2015). The document also identified the organisational effort and logistics for obtaining waste as obstacles to its implementation. These obstacles can be overcome by the return logistics for bread that are already in place in Sweden.

Considering the total amount of bread waste generated in Sweden, as described in **Paper II,** 15% of the current Swedish ethanol production could be met by using bread as a feedstock, considering a yield of 350 g of ethanol per kg of bread (Ebrahimi et al., 2008). Realistically, considering only the returned bread flow and bakery waste, 5% of the national production capacity could be met. Nevertheless, it is important to note that this numbers refers to the domestic production and not the total consumption of ethanol, and a significant share of the domestic demand for ethanol is covered by imports, mainly from the EU, Brazil, and the USA.

There are few production facilities for ethanol in Sweden, and production is primarily undertaken by Läntmännen Agroetanol. The main implication of this is an increase in transportation distances, and consequently, costs and environmental impacts.

The life cycle assessment in **Paper III** considered that the production of ethanol from bread replaces ethanol produced from wheat as a feedstock. The main co-product of ethanol production is dried distiller's grains with solubles (DDGS), a high-protein animal feed, often used as cattle feed.

Beer

Beer is the largest segment in the alcoholic drink market. The market for beer has been growing worldwide, with sales values of US$587 billion in 2019, which is expected to increase to US$867 billion by 2025 (Statista, 2020). In Sweden, the annual production value of the beer industry was 673 million euros (EUROSTAT, 2021), with 465 million litres being consumed in 2018 (Brewers of Europe, 2021).

Beer is produced primarily from cereal grains, hops, and water. Barley is the most common cereal used as a source of starch sugars. The emissions of nitrous oxide from barley cultivation and the energy consumption during the processing of barley malt have been identified as key contributors to the environmental impacts of beer production (Amienyo and Azapagic, 2016).

Recently, small breweries have started to use surplus bread as a substitute for some of the malted barley. The emergence of such entrepreneurs dealing with bread surplus is often linked to a sustainable agenda that aims to address economic, social, and ecological goals. These companies are often driven by the goal of acting as catalysts for societal change (Matzembacher et al., 2019). Examples are found in the UK, by the Toast Ale brewery (Toast Ale, 2021), which states its alignment with UN Sustainable Development Goal 12.3 and works towards a transition to circular brewing. In Sweden, Crumbs beer (Crumbs, 2021) is also produced using surplus bread and uses a model that establishes local collaboration between breweries, bakeries, and retailers.

According to the most recent data, Sweden has 375 breweries and 369 microbreweries (Brewers of Europe, 2021, Bryggeriföreningen, 2021), producing 448 million litres of beer (Brewers of Europe, 2021). Considering the amount of returned bread in Sweden and a yield of 2.5 litres of beer per kg of bread, it is estimated that approximately 10% of the national production of beer could be covered by the use of surplus bread. If the total amount of surplus bread generated by retailers is considered, 15% of the domestic production capacity can be met.

The main advantage of brewing beer using surplus bread is the high number of facilities spread across the country, compared to other valorisation pathways, such as ethanol and animal feed production, which have a limited number of plants in Sweden. This could decrease the transportation distances, and consequently, the costs and environmental impacts. As noted in **Paper III,** transportation distances have a significant influence on the environmental performance of valorisation pathways.

Animal feed

Animal feed constitutes a major cost for livestock producers, and it is the largest contributor to environmental impacts in many livestock production systems, such as swine (McAuliffe et al., 2016). The production of feed from food waste is part of the EU sustainability package (European Commission, 2020), with the aim of promoting resource efficiency, sustainable development, and reducing vulnerability in the food supply chain. These goals can be met using food waste or surplus for feed production, entailing lower environmental impacts and economic costs, as well as a more resilient food chain that utilises resources more effectively.

The re-use of food, which would otherwise have been wasted, as animal feed is the third priority in the food waste hierarchy, after prevention and reuse for humans. Although food waste may be used to feed multiple livestock species, it is often used as feed for pigs (Kumar et al., 2014). The use of food waste as feed is limited by strict legislation aimed at ensuring animal and human health, preventing diseases such as bovine spongiform encephalopathy, and foot and mouth disease (Parfitt et al., 2016a). The goal is to decrease the risk of animal infection and the presence of harmful substances in the feed, which can have major economic consequences for society and cause suffering in animals. For example, the use of food waste from households, restaurants, and commercial kitchens as feed has been restricted (Lindow et al., 2020). In addition to legislation, there are some practical obstacles to the use of food waste and residues as feed, such as the difficulties of a stable delivery of a homogeneous flow for the feed producer.

Surplus bread has the potential to overcome both the legislative and practical requirements. Bread can be used as a feed ingredient for livestock, even if it is baked with milk or other dairy products, as long as it complies with legislation (Lindow et al., 2020). Moreover, the return system for bread allows the supply of a stable and homogenous material, which is desirable for feed producers. Nevertheless, bread is seldom used as the sole feed because it is considered to worsen carcass and meat quality (Sirtori et al., 2007). Feed producers rely on recipes to ensure that the feed product has sufficient nutrients and appropriate palatability. In such feed recipes, bread is supplemented with protein sources, such as barley, maize, and soybean. It was estimated that 10 000 tonnes of bread and pasta were used as animal feed for pigs in Sweden in 2017, indicating a reduction in comparison with 2005 (Landquist et al., 2020).

Paper III considered recipes that replaced different ingredients, including maize, barley, and wheat bran, by surplus bread. The recipes chosen ensured appropriate feed characteristics that did not affect animal growth performance (Sirtori et al., 2007, Kumar et al., 2014).

Animal feed is also produced as a co-product after bread processing in the ethanol and beer pathways. DDGS and brewers' grains, are the solid residues left after the processing of ethanol and beer, respectively. Both products have a long history of being used as animal feed because of their relatively high protein content, which ranges from 20% to 40% (Heuzé et al., 2017).

5.1.4. Fungal biomass production from surplus bread

The valorisation technologies assessed in **Paper III** were located at relatively low stages in the food waste hierarchy and were technologies with high levels of technology readiness.

Considering the nutritional aspects of bread, it is important to investigate pathways that maintain its function as food, as prioritised in the food waste hierarchy. Furthermore, considering that limited infrastructure available for the valorisation of surplus bread might hinder the environmental benefits of such measures due to the need for long transportation distances, it is necessary to develop technologies that could be implemented in a variety of configurations. This part of the research focused on the development of a novel solid-state fermentation technology that uses surplus bread as a substrate for the production of a high-protein food product using the filamentous fungus *Neurospora intermedia* (**Paper IV**). The technology has the potential to be applied in bakeries using the existent equipment, or in a dedicated plant.

Filamentous fungi have traditionally been used by some societies as food, for example, tempeh and oncom, traditional staple foods from Indonesia (Gmoser et al., 2020, Shurtleff and Aoyagi, 1979). The use of filamentous fungi as food has gained attention due to its nutritional profile, high protein content, presence of essential amino acids, and easy digestibility (Souza Filho et al., 2018, Moore and Chiu, 2001). Moreover, the market for meat substitutes has increased due to health motivations, ethical beliefs, environmental concerns, and religious reasons. The production of livestock, particularly cattle, is associated with multiple environmental impacts, including high greenhouse gas emissions, pollution of freshwater, and biodiversity loss due to land use changes (Steinfeld et al., 2006). There has been a shift in the consumption patterns and an increase in the number of vegetarians and non-vegetarians that want to decrease their meat consumption. Substitutes for meat protein products are also relevant for people with religious diets, such as Jews, Hindus, and Muslims (Hashempour-Baltork et al., 2020).

Figure 11 - The fungal product and possible application as a patty for burger.

The food company Quorn® commercialises a food product made from the filamentous fungus *Fusarium venenatum*. The fungi are grown in a synthetic medium of glucose and mixed with egg albumen, flavour, and colour compounds (Wiebe, 2002). The cultivation medium has been identified as a hotspot in terms of both economic costs and environmental impacts (Smetana et al., 2015, Souza Filho et al., 2018).

The technology assessed in **Paper IV** addresses this problem by substituting the synthetic medium with surplus bread. Bread is a suitable substrate for solid-state fermentation (SSF) because it contains nutrients, such as starch, that are easy for the fungi to consume. Furthermore, its porous structure is beneficial for fungal growth. The process is based on SSF, a process carried out in a solid medium in the absence of a free-flowing liquid. The main advantages of this technology are that it is a relatively simple and cheap technique, with lower energy requirements and less pre-treatment. Moreover, it also generates less wastewater and requires a lower reactor volume compared to submerged fermentation processes (Couto and Sanromán, 2006, Ferreira et al., 2016, Pandey et al., 2001).

The SSF is based on the cultivation of *N. intermedia*, a filamentous ascomycete. The fungus has been historically used in fermented food and it is listed as 'Generally Recognized as Safe' (GRAS) by several organisations, including the U.S. Food and Drug Administration (USFDA), and the World Health Organization (WHO), the agency of the United Nations specialised in public health (Nair, 2017). *N. intermedia* grown exclusively on dried bread produces a nutrient-rich product after six days of fermentation, increasing the content of protein, dietary fibres, and vitamins D and E (Gmoser et al., 2020) (**Paper IV**).

The improvement of this technology mainly focuses on the production of the inoculum, which was identified as a hotspot in the established benchmark technology. Different methods of inoculation were developed and assessed in relation to their environmental performance and technical aspects, including morphology, lag phase, and protein content.

5.2. Evaluation - Waste hierarchy for bread

The results from **Paper II** indicate that returned bread is a flow with desirable characteristics, such as quantity, composition, and logistics, for use as a raw material in multiple valorisation pathways. This highlights the possibility of diverting surplus bread from current waste management practices to alternatives higher in the food waste hierarchy. According to Parfitt et al. (2016b), in the UK, 83% of bakery surplus could be prevented or valorised since most of the waste bread does not present food safety risks at the point in time when it is generated. Yet, there was a lack of evidence on whether such pathways were preferable from an environmental perspective.

The results from the environmental assessment in **Paper III** show a trend where prevention was ranked the highest and the current waste management pathways, i.e. anaerobic digestion and incineration, were ranked the lowest. Donation and the remaining valorisation pathways differed in ranking depending on the impact category analysed (Figure 12).

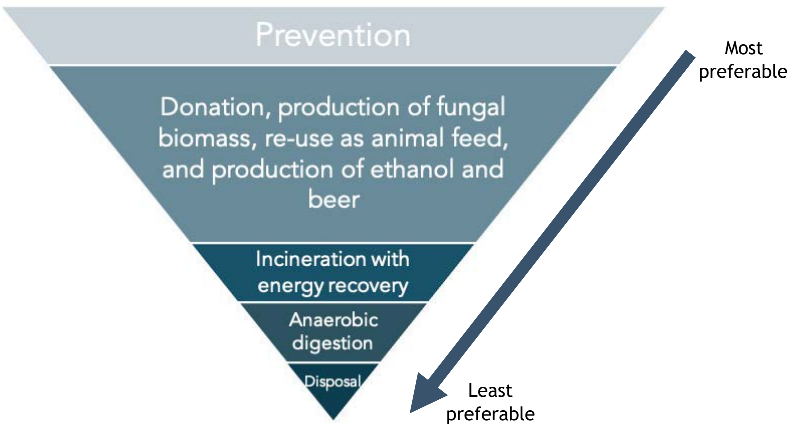

Figure 12 - Hierarchy for prioritisation of the management of bread surplus. The same colour used for the re-use for human consumption, re-use as animal feed, and production of ethanol and beer indicates that there is no single trend for prioritization between these measures.

When the LCA results were analysed by the weighted single score (**Paper III**), some divergences in the food waste hierarchy were found. For example, anaerobic digestion was ranked lower than incineration. This is because bread is a relatively dry product compared to other food products. Bread has a water content of approximately 30%, which is relatively low compared to that of other waste fractions, such as mixed fruits and vegetables, which have an average water content of 86% (Tonini et al., 2018). This makes bread more suitable for incineration compared with other organic wastes, as it produces relatively more electricity and heat per kilogram of waste incinerated. Another difference to the waste hierarchy was that donation was ranked lower compared with other valorisation pathways in many impact categories, and lower than animal feed in the single score results.

It is important to note that waste hierarchies are often not solely based on environmental performance, and often consider economic and social aspects. Therefore, the results from **Paper III** must be interpreted with caution, as the use of surplus bread by humans is generally ethically preferred. Nevertheless, the results highlight that the level of waste in donated products, both at the consumer level and at the donation point, is an important environmental hotspot, and must be reduced to achieve the potential environmental benefits from donation. The benefits of donation might be diminished if too much surplus is sent to this activity, particularly considering that the market for the donation of bread might be saturated, as has been highlighted previously (Schneider, 2013). It is hypothesised that the market for donated bread in Sweden is smaller than the amount of surplus bread currently being generated, emphasising the importance of considering other valorisation pathways. Thus, although preferable, the donation of surplus bread must consider such constraints, and companies should not transfer the burden of the waste to the recipient organisations or recipients of the donated product.

There are two main reasons why it is not possible to draw a single hierarchy for the proposed waste prevention and valorisation pathways. As presented in **Paper III**, the results from the environmental assessment indicate trade-offs between different impact categories (Figure 13). Moreover, the results are sensitive to the distances for which surplus bread needs to be transported to valorisation plants. Valorisation pathways located close to the generation site are preferred, highlighting the importance of bakeries to use of local infrastructure to valorise their surplus flow. The environmental benefits from valorisation can be lost if bread is transported over long distances. Overall, the results of the study indicated that shifting from waste management to valorisation pathways generally leads to environmental savings as long as transportation distances are short.

Figure 13 presents the results for two impact categories from the environmental assessment of the different pathways for surplus bread. A more comprehensive list of results is presented in **Paper III**. The results indicated that the prevention of surplus generation has the largest savings in the climate change impact category, mainly because of the avoided production of wheat. This is similar to ethanol production, which was ranked second, where the majority of the savings originate from the avoided wheat production, and less significantly, from the DDGS, which is a co-product of ethanol production. The results for animal feed were ranked third in the climate change impact category, and the savings originate from the substitution of ingredients in traditional feed production, particularly barley. The contributions in the marine eutrophication impact category were similar to those in the climate change category, but the ranking of the valorisation pathways was different. As expected, nutrient leaching from mineral fertilisers in agricultural production has a significant influence on the results of marine eutrophication, favouring animal feed production over the production of ethanol (Figure 13).

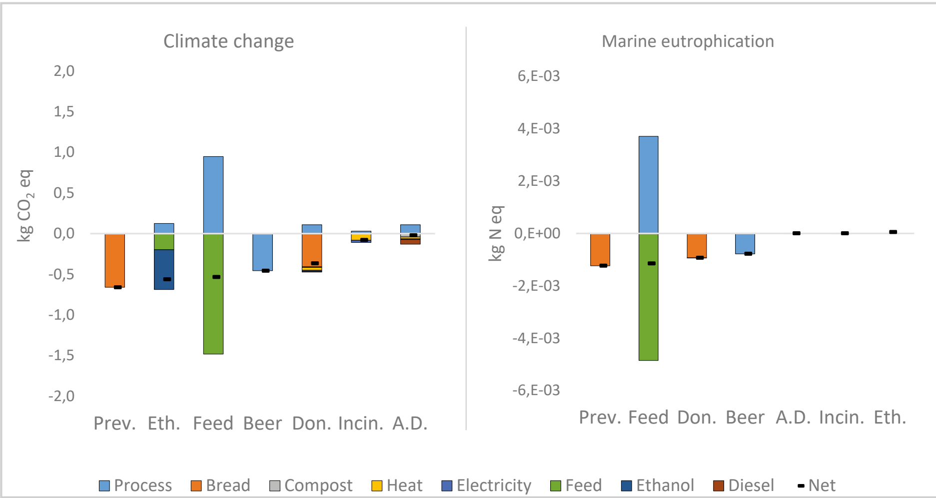

Figure 13 – Results at midpoint level for the climate change and marine eutrophication impact categories for prevention, valorisation, and waste management pathways for 1 kg of surplus bread. The results are ranked from the highest to lowest benefits for each impact category. The scenarios described are prevention (Prev.), donation (Don.), ethanol production (Eth.), animal feed (Feed), beer production (Beer), incineration (Inc.) and anaerobic digestion (A.D.) (Brancoli et al., 2020).

5.2.1. Evaluation of the developed fungal biomass product

Paper IV has a different goal from that of **Paper III,** in which established technologies with high technology readiness were assessed and compared. In **Paper IV**, the technology is being developed and is at a laboratory-scale. The goal of this study was to support the developers of the technology by incorporating environmental considerations during the design and development processes. The aim was to decrease the potential environmental impacts during the production stage, without compromising important characteristics of the product, including texture and protein content.

This study identified inoculum production and fermentation as hotspots in the system. The focus was on improving the environmental performance of the inoculum, particularly the carbon source. Different scenarios were assessed by substituting carbon sources by other substrates. The results indicated that in comparison with the baseline scenario, a maximum reduction of

60% of the environmental impacts was possible at a single score level, while maintaining desirable characteristics of the product, such as texture and protein content (**Paper IV**).

Nutritional analysis of the fermented fungal product showed an increase in protein content, total dietary fibre, and vitamin D and E in the fermented product in comparison with those in stale sourdough bread (Gmoser et al., 2020). However, in addition to the positive nutritional aspects, it is necessary to consider the acceptance of the product from the consumer's perspective. The results from a sensorial trial for the fungal product indicated that most participants accepted the appearance and taste of the product, but the responses also indicated potential improvements in relation to bitterness and texture (Hellwig et al., 2020).

The technology used to generate the fermented fungal product can be implemented in various scenarios, from a stand-alone business model that receives surplus bread from retailers and bakeries, to its implementation in small and medium bakeries that can use their own surplus to produce and sell the product, together with their existing range of products. One advantage of the process proposed here is that it can be produced using equipment similar to that used for baking bread, including proving ovens and dough mixers. This has the potential to increase the number of facilities available for the management of bread surplus, decreasing the necessary transportation, which has been identified previously as a hotspot in the valorisation of surplus bread. The economic feasibility of implementing a similar process in small-scale bakeries in Sweden has been previously investigated (Gmoser et al., 2021). The authors concluded that the process is economically feasible with an annual profit of € 62 000 and a payback period of four years at a discount rate of 7%. The authors also identified labour costs and the fermenter as the highest operation and capital costs, respectively.

5.3. Limitations and uncertainties

LCAs are traditionally divided into two categories, namely, attributional and consequential. Attributional LCAs quantify the impacts associated with a product within a chosen temporal window, whereas consequential LCAs are used to assess the marginal consequences of a change in the product system. Nevertheless, there are still debates in the community regarding how each approach should be used and its applications. The International Reference Life Cycle Data System (ILCD) Handbook (Wolf et al., 2010) proposes that when the study aims to support micro-level decisions, average data should be used to model small changes in production volume, while the use of marginal data is recommended to model changes with a large-scale effect on production capacity. Other authors state that marginal data should be used when the study aims to model consequences, regardless of its size (e.g., Ekvall et al., 2016, Finnveden et al., 2009, Weidema et al., 2009).

Paper III follows a similar approach to situation A for micro-level decision support described in the ILCD Handbook (Wolf et al., 2010). It is assumed that the additional input of bread as a raw material for the valorisation pathways is not expected to result in additional installed equipment, given the limited share of the raw material displacement by bread. Therefore, it was expected that there would be no or limited changes in the background system. Nevertheless, attributional LCAs have been argued to result in a higher errors due to a low accuracy (Weidema et al., 2009). The substituted products in the scenarios in **Paper III** have a large influence on the results and the use of an average market mix or marginal data could influence the results. The uncertainties in **Paper III** were reduced via a sensitivity analysis, which included scenarios of the best and worst cases.

The main uncertainties in **Paper IV** were attributed to the low technology level of the modelled process, which was translated into uncertainties in the environmental impacts. Depending on the setup through which the technology will be applied, for example, a dedicated plant or small-scale production in a bakery, there might be differences in the results due to differences in resource consumption at different scales of production.

CHAPTER 6

CONCLUSIONS

This chapter presents the main conclusions of this thesis, and the implications and recommendations for the stakeholders in the bread supply chain. Finally, directions for future research are suggested.

Although food waste in retail has been studied throughout the past decade, bread waste has remained relatively unexplored. The general objective of the present research was to propose and evaluate measures to reduce and valorise surplus bread at the supplier–retailer interface. Through a variety of approaches, this doctoral thesis investigated the quantities, causes, and measures for the prevention, valorisation, and management of surplus bread.

The thesis initially sought to quantify, characterise, and identify hotspots of the food waste generated by a supermarket in Sweden. The quantification measured the waste at 22 tons per year, accounting for a carbon footprint of 3500 kg CO2-eq t^{-1}. The results indicate that the discarded bread and beef products are the main contributors to the environmental footprint of the supermarket, of which the bread waste contributed with 30% of the total discarded mass. Bread was also the second most important product category in relation to the environmental impacts at a single score level for the analysed product categories. Furthermore, bread was identified as a product category with a high potential for waste prevention.

As the bread product category was identified as a major contributor in terms of generated waste and due to its high potential for waste prevention, the understanding of the waste level at the Swedish supply chain was of interest. The quantification for bread waste and surplus revealed that approximately 80 500 tons, the equivalent of 8 kg per person is wasted annually in Sweden.

The waste and surplus were concentrated at the supplier–retailer interface and at households. Considering the Swedish national target to reduce food waste by 20% from 2020 to 2025 and the commitment to fulfil target 12.3 of the UN's Sustainable Development Goals, surplus bread is a key waste fraction to be addressed to reach the government’s targets for food waste reduction.

The analysis of risk factors for waste generation in the supplier–retailer interface indicated that bread waste originates from a wide variety of reasons, such as the large assortment of bread products, demand for freshness and the lack of data on waste quantities. Moreover, the take-back agreements were also identified as a relevant risk factor and it was the sub-category with the highest waste generation in mass, estimated at 15 000 tonnes per year. The identification of risk factors has significant managerial implications for the bread supply chain, as they direct the attention of retailers and suppliers to the main causes for the large amounts of generated waste. Therefore, the stakeholders are able to adjust their strategies, as well as evaluate the opportunities arising from collaborative measures, potentially sharing the benefits and costs of such actions. Overall, it is crucial for retailers and suppliers to improve their ordering system and adjust the delivered quantities to reduce waste.

The take-back agreements were identified as a risk factor for waste generation, as well as an opportunity for valorisation of the surplus bread, since such agreements enable the segregated collection of bread. The results indicate that surplus bread management in the supplier–retailer interface should prioritize prevention and valorisation pathways over anaerobic digestion and incineration. Shifting from current waste management to valorisation pathways has the potential of a carbon footprint saving of 0.56 kg CO_2-eq per kg of surplus bread. Furthermore, it is concluded that while the waste generation can be reduced by shifting responsibilities in the ordering process, the segregated collection enabled by take-back agreements is still beneficial.

Finally, bake-off products is a relevant category to be addressed by retailers due to the high amounts of generated surplus. This thesis proposed a solid-state fermentation process suitable for dealing with such surplus flows, as it has the potential to be implemented in supermarkets, as well as small, medium-sized and industrial bakeries. Surplus bread can be used as a substrate to the cultivation of the fungus *N. intermedia* as a high-protein food product, with the potential to be a sustainable alternative to other protein-rich food products. The findings indicate priorities for the technical optimisation of the process, particularly the production of inoculum and the fermentation process. This study also exemplifies the application of LCA in the early design phase of technology development as an important tool to identify environmental hotspots and trade-offs between technical and environmental aspects of the technology.

The study concludes that the bread supply chain has a high potential of transitioning to circularity. A higher degree of collaboration between businesses across the supply chain is required to realise the benefits of bread prevention and valorisation strategies. The results from the environmental assessment can be further used to support investments by both business and policymakers to decrease the environmental impacts associated with the bread supply chain. Although the results obtained in this study are representative of the Swedish bread supply chain, it provides a basis to implement and evaluate similar initiatives in other countries, particularly those where bread is supplied under TBAs.

6.1. Future work

Considering the growing interest in food waste prevention and valorisation, the following areas are proposed for future investigations and improvements:

- Implementation and assessment of the efficacy of waste prevention and valorisation measures, including the amounts of waste prevented and the economic, environmental, and social benefits or burdens related to them. For example, the investigation of the effects of ending TBAs in selected stores, where the supermarket assumes responsibility for the ordering process.
- Further investigations through material flow analysis on the amount of surplus bread that is sent to different redistribution, valorisation, or disposal pathways. In particular, an investigation of the market for donated bread in Sweden, in relation to the measurement of the loss rates and estimates of the amounts of surplus bread suitable for redistribution.
- Sustainability assessment of prevention, valorisation, and disposal pathways with accompanying economic and social implications of different prevention, management, and prevention pathways for surplus bread.
- Further research on the development of fermentation processes for the production of fungal biomass using bread as a substrate. In particular, further improvements in the fermentation step to reduce energy consumption and the properties of the biomass, including texture and taste.

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Life cycle assessment of supermarket food waste

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ABSTRACT

Retail is an important actor regarding waste throughout the entire food supply chain. Although it produces lower amounts of waste compared to other steps in the food value chain, such as households and agriculture, it has a significant influence on the supply chain, including both suppliers in the upstream processes and consumers in the downstream. The research presented in this contribution analyses the impacts of food waste at a supermarket in Sweden. In addition to shedding light on which waste fractions have the largest environmental impacts and what part of the waste life cycle is responsible for the majority of the impacts, the results provide information to support development of strategies and actions to reduce of the supermarket's environmental footprint. Therefore, the food waste was categorised and quantified over the period of one year, the environmental impacts of waste that were generated regularly and in large amounts were assessed, and alternative waste management practices were suggested. The research revealed the importance of not only measuring the food waste in terms of mass, but also in terms of environmental impacts and economic costs. The results show that meat and bread waste contributes the most to the environmental footprint of the supermarket. Since bread is a large fraction of the food waste for many Swedish supermarkets, this is a key item for actions aimed at reducing the environmental footprint of supermarkets. Separation of waste packaging from its food content at the source and the use of bread as animal feed were investigated as alternative waste treatment routes and the results show that both have the potential to lead to a reduction in the carbon footprint of the supermarket.

1. Introduction

Food, during its life cycle, contributes to current global environmental challenges such as climate change, eutrophication, terrestrial and aquatic acidification, depletion of the stratospheric ozone layer, depletion of natural resources and loss of biodiversity (Garnett, 2013; Hertwich and Peters, 2009; Pretty et al., 2005). Furthermore, the global food demand is projected to increase by 70% by 2050 (FAO, 2009), which may well increase the environmental impact on the planet.

These issues are aggravated by the wastage of one third of all food produced for human consumption (Gustavsson et al., 2011), and raises ethical concerns since 795 million people suffer from undernourishment (FAO et al., 2015). According to the Food and Agricultural Organisation of the United Nations (FAO, 2013), the global food waste in 2007 was estimated to be 1.6 Gtonnes of primary product equivalents and the corresponding edible part was estimated as 1.3 Gtonnes.

Reduction of food waste is, therefore, important for food security and for reducing unnecessary economic costs. It will also reduce environmental impacts, and is less controversial than alternatives such as the reduction in consumption of some products like meat and dairy (Beretta et al., 2013; Gruber et al., 2015; Scholz et al., 2015).

Even though retail has lower amounts of food waste compared to other steps in the food value chain, (FAO, 2013; Naturvårdsverket, 2013), it has a significant influence on food waste generated throughout the supply chain. The European Commission (2013) reported that retailers have increased their bargaining power over other actors in the supply chain. Retailers also influence – and are influenced by – consumers that are downstream in the supply chain. For example, there is a large amount of waste in primary production due to the strict quality standards of consumers and retailers that define product classes. Food products that are classified as lower quality can yield lower profits, and it is therefore often not financially viable to harvest the product, even though it could be sold in supermarkets. For instance, each year 15% of lettuce grown in Sweden is not harvested (Livsmedelsverket, 2015).

Retailers are located towards the end of the supply chain, and hence a large environmental impact has already been generated

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from transportation, packaging and other processes before the food reaches the supermarkets. It also concentrates large quantities of waste at a few physical locations, which facilitates collection of the waste for further treatment as well as data of the waste fractions and quantities. Thus, supermarkets are potentially good targets to investigate food waste and subsequently implement measures for waste prevention. This would reduce the environmental impacts of the waste as well as economic losses suffered by the supermarkets (Scholz et al., 2015).

Statistics on the amount of food waste generated in the Swedish retail sector varies substantially depending on the source of the data. This is also true for the fraction of this waste that is avoidable, defined as "food that could have been eaten provided that it has been handled correctly and eaten by its use-by date" (Naturvårdsverket, 2013). For example, Naturvårdsverket (2013) reported that 70,000 t of food waste was generated by food retailers during 2012, of which 91% was avoidable. In a more recent report (Naturvårdsverket, 2016) they recalculate this figure to be 45,000 t. The reason for this large discrepancy is not clear, but may be due to differences in methods to obtain these figures. Of interest is that, using the newer method to calculate the food waste, there was a reduction from 45000 t in 2012–30000 t in 2014. Surprisingly, only 36% of this 30000 t was avoidable (compared to the value of 91% determined for 2012). Irrespective of this, there is substantial food waste in the Swedish retail sector.

This study analyses the food waste at a supermarket in Sweden in terms of product type, mass, environmental impacts and economic costs. The supermarket is located in the city of Borås, and has a sales area of approximately 410 m^2. It is therefore a typical mid-size urban supermarket (Gómez-Suárez and Martínez-Ruiz, 2016). The results presented here are therefore expected to be relevant for many Swedish supermarkets, and perhaps even for retailers in other developed countries. In addition, the information obtained in this study can be used to support the development of strategies and actions aimed to reduce supermarket food waste and to identify alternative waste treatment routes that reduce the supermarket's environmental footprint.

This study is divided into three parts:

(i) The categorisation and quantification of food waste in the supermarket over a one year period.
(ii) The results from the waste categorisation and quantification revealed which waste fractions were large and generated often. The environmental impacts of these waste fractions were assessed.
(iii) The results from the waste categorisation and quantification together with the life cycle assessment (LCA) supported the design of alternative waste treatment routes. The change in the environmental footprint for each alternative was quantified.

2. Material and methods

2.1. Waste categorisation and quantification

The type and amount of food waste at the supermarket was gathered from October 2014 to September 2015. Products that were considered unsellable due to defects or that have expired shelf-life were scanned by a bar code reader and the data was saved in the supermarket's database. Products that were sold without a bar code, such as fruits and vegetables, were weighed, and the mass of the waste was entered manually into the database. After the data collection, the food products were categorized. This study does not differentiate between avoidable and unavoidable food waste. It is assumed that the waste was edible at or before the time that it was disposed of, or that an excessive amount of products was ordered and that this led to waste.

The economic analysis is limited to the costs incurred by the supermarket and it includes the purchase costs of the wasted products and the disposal costs (including fetching the waste from the supermarket). Other costs borne by the supermarket, e.g., salaries, were not included. This analysis was incorporated as a secondary goal in this study since economic losses are important for motivating change in the retail sector.

Pre-supermarket waste, which is waste that is rejected upon delivery, is not included in this study since it was not recorded in the supermarket's database. Pre-supermarket waste consists of items that do not satisfy the quality requirements of the supermarket (Eriksson et al., 2012). According to Eriksson (2015) pre-supermarket waste can represent significant quantities of waste, particularly for fruits and vegetables, where it can correspond to three times the amount of waste in the supermarket.

2.2. LCA of the supermarket food waste

Food waste fractions that were generated often (high frequency) and in large amounts were included in the LCA. The frequency of waste generation was assessed as the number of days in the year that the product was disposed of. This parameter was used to avoid waste that was generated due to a single or infrequent occurrence, such as a breakdown of equipment. Waste fractions that occur in frequent and large amounts are of more interest than those that are generated in small amounts and less often since they are statistically more meaningful (they are more likely to represent annual waste fractions and quantities over several years).

2.2.1. Goal and scope definition

Based on the criteria discussed in the previous paragraph, the waste fractions that were selected for the LCA were beef, pork, chicken, bread, strawberries, bananas, tomatoes, lettuce, potatoes, carrots, cabbage and apples. Dairy waste was not included in the main LCA due to the large variety of wasted products and the lack of available data to accurately model all impact categories.

The functional unit for this study is the waste generated by the supermarket from October 2014 to September 2015 for the selected products, which corresponds 11.4 t from the total of 22.5 t of food waste generated during that period. This is similar to that used in comparable LCAs (Scholz et al., 2015), and was used since the aim was to compare the environmental impacts of different food waste fractions generated during an entire year. Impacts per kg of food waste are presented in Section 2.2 of the Supplementary material.

2.2.2. System boundaries

Fig. 1 shows the simplified flow chart of the waste fractions that were analysed and includes some of the relevant processes used to model each product. The life cycle was modelled from the cradle to the grave. It includes processes from the agricultural production, packaging production, transportation, retail and end of life treatment. The bakery stage is also included for bread waste. The city of Borås, Sweden was the geographical reference for the retail and for the waste treatment. A global dataset that is based on the average global data was used for the agricultural production. The current waste management practice in the city is to use the food waste for anaerobic digestion. The waste treatment is explained in more detail in Section 2.3. More details of the data and assumptions made for the life cycle modelling are available in Section 2, Supplementary material.

The fruits and vegetables sold in the supermarket come from different countries and some of products even have several countries of origin. Representative countries for each waste fraction were chosen to model the transportation routes. For each waste frac-

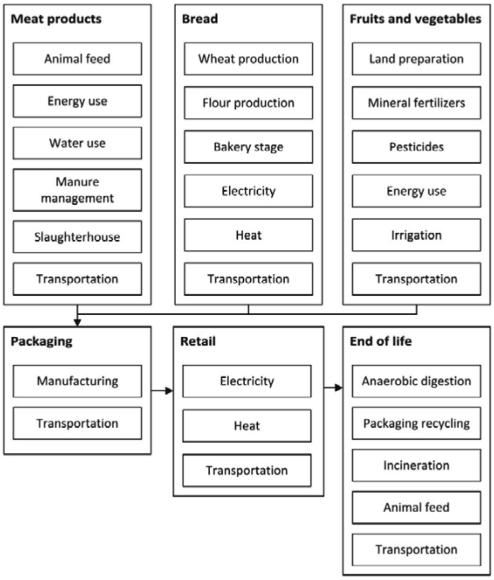

Fig. 1. Simplified flow chart of the LCA for the different waste fractions. Only a few processes are shown for the sake of clarity.

tion, the representative country is the country of origin for most products in this fraction at the supermarket (see Table 2, Supplementary material). No distinction was made between varieties of the same fruit or vegetable. Meat waste was assumed, for simplification of the model, as waste arising from fresh meat. For sausages and minced meat, the meat content was allocated to the relevant meat fraction (beef, pork or chicken). The transportation distances are given in Table 2 of the Supplementary material.

2.2.3. Inventory analysis

The inventory analysis was based on datasets that are relevant to the different waste fractions. Details are given in Section 2 of the Supplementary material.

2.2.4. Impact assessment

The life cycle impact assessment method that was used is the international reference life cycle data system (ILCD) for midpoint indicators (European Commission, 2011). This method is recommended by the European Commission to ensure quality and consistency of life cycle data (Wolf et al., 2012). The categories included in this study are climate change, ozone depletion, particulate matter, photochemical ozone formation, acidification, eutrophication, freshwater ecotoxicity, and resource depletion.

2.3. Life cycle assessment of alternative waste treatment scenarios

2.3.1. Goal and scope

In order to identify opportunities to reduce the carbon footprint at the supermarket, two alternative waste management scenarios (Sc. II and Sc. IV) were compared with the current practice at the supermarket (Sc. I and Sc. III). This gate-to-grave study includes the waste treatment scenarios for the food waste. The waste is assumed to enter the system burden-free and the co-products generated were considered to substitute other products on the global market.

2.3.1.1. System description of scenarios I and II. The functional unit for Scenarios I and II is the waste fractions described in Section 2.2.1, as well as those from dairy and ready meal products, generated from October 2014 to September 2015. The inclusion of the aforementioned products was possible on this assessment since the waste is assumed to enter the system burden-free. Thus the lack of data regarding primary production for dairy and ready meal products (that prevented the modelling of these products in the LCA done in Section 2.2) does not restrict the assessment on this section. The description of the packaging data is available in Table 4 of the Supplementary material.

(Sc. I) This scenario is the current waste treatment method in the city, where food waste is treated using anaerobic digestion. Anaerobic digestion is the most common method of treating food waste in Sweden (Avfall Sverige, 2014), and hence the results presented here are expected to be relevant to many Swedish supermarkets. The food waste is sent to an anaerobic digestion plant for co-digestion with manure. The food waste is not separated from its packaging before arriving at the plant, and pretreatment at the plant is required. The pre-treatment consists of optical sorting, a drum sieve, magnetic sorting of metal and a mechanical filter where most of the food waste is washed from non-food waste using water (Rousta et al., 2015). The slurry is sent to a biological reactor for biogas production. The remaining part, consisting of the rejected material from the pre-treatment, contains food and non-food waste (mainly the food packaging). Measurements made at the plant show that 44% by mass of the food waste is rejected with the non-food waste. This rejected material is too contaminated to be sent for material recycling of the packaging, and is instead used for incineration with energy recovery.

(Sc. II) This scenario models source separation of the food waste from its packaging at the supermarket. This would enable the packaging to be recycled at a material recovery facility (MRF), and for the food waste to enter the anaerobic digestion plant with less pre-treatment. This pre-treatment requires less energy compared to the mixed waste line in Sc. I, since there are fewer pre-treatments steps compared to Scenario I. Also, there is less food waste in the rejected material (modelled as no waste in this study), meaning that, in comparison with Sc. I, more food waste is used for anaerobic digestion, instead of incineration. The packaging material and sizes, shown in Table 4 of the Supplementary Material, were assessed by measurements made at the supermarket and represent packaging of different products.

2.3.1.2. System description of scenarios III and IV. The functional unit for Scenarios III and IV is the bread waste produced from October 2014 to September 2015, which corresponds to 6.7 t. As discussed below, the results from the waste quantification and impact assessment indicate that bread is an important product both in terms of the total waste produced and in terms of the environmental impacts. Therefore, these results support the decision to model a scenario specifically for bread waste.

(Sc. III) This scenario assumes that the bread waste is treated using the current waste management practice, i.e., it is not sorted from the packaging and that the mixed food-packaging fraction is sent to the anaerobic digestion plant, where a fraction of the bread goes through anaerobic digestion and the rest is lost in the pre-treatment mixed with non-food materials and is incinerated, as in Sc. I.

(Sc. IV) This scenario assumes that the bread waste is separated from other food fractions and its packaging at the supermarket, and that the bread waste is used as animal feed and the packaging is recycled.

2.3.2. System boundaries

The system boundaries considered for all scenarios are illustrated in Fig. 2. It contains the main mass flows and the substituted products (dashed lines).

The environmental impacts of the products from the waste treatment scenarios (i.e., energy feedstock, secondary materials and animal feed) were assessed using system expansion. That is, these products are considered as alternatives to other products on the global market that are used to generate energy, the secondary products and animal feed. The heat and electricity produced at the incineration plant and at the anaerobic digestion plant were assumed to substitute the production of these products for the Swedish national energy matrix. The digestate produced at the anaerobic digestion plant was assumed to be used as organic fertilizer and substitute the use of mineral fertilizer. The nutrient content for nitrogen, phosphorus and potassium was retrieved from typical values for co-digestion anaerobic plants (WRAP, 2012). The mineral fertilizer that was substituted was assumed to be urea, diammonium phosphate and potassium chloride (Martinez-Sanchez et al., 2016; Tonini et al., 2015). The greenhouse gas emissions for the secondary production of materials were retrieved from a previous study (Hillman et al., 2015).

The bread waste used for animal feed (Sc. IV) is assumed to substitute wheat, a conventional feed for the energy fraction in pig feed (Federation of Swedish Farmers, 2015). Bread has a low protein content of ~9% (Danish Food Composition Databank, 2016), lower than the protein content of wheat (Feedipedia, 2016), and much lower than that of soya meal, which is typically used as the protein fraction in animal feed (Tonini et al., 2015). In this study, 1.0 kg of bread waste replaced 0.4 kg of wheat owing to its nutrient composition and the suggested maximum inclusion rate of bread in a pig diet (Boggess et al., 2008; Dalgaard et al., 2008; Danish Food Composition Databank, 2016; Kumar et al., 2014). The bread waste is assumed to be transported 200 km for use as animal feed. This distance is the average distance from the city of Borås to the south region of Sweden, where the majority of pig farms are located (Marquer, 2010). According to Scherhaufer and Schneider (2011), EU Regulation No. EC/1069/2009 allows bread waste to be used as pig feed.

2.3.3. Inventory analysis

The inventory analysis was based on datasets that are relevant to the different waste fractions and its packaging. Details are given in Section 3 of the Supplementary material.

2.3.4. Impact assessment

Due to lack of data for the recycling processes for other impact categories, the only impact category assessed in this part of the study was climate change, using the ILCD assessment recommended method (European Commission, 2011).

3. Results

3.1. Waste categorisation and quantification

The supermarket wasted 22.5 t of food from October 2014 to September 2015. The waste was categorized into the following fractions: bread, meat, fruits, vegetables, pastry, ready meals, dairy and other. The bread fraction consists of different types of bread produced at the supermarket bakery or from external suppliers. The meat fraction includes waste such as beef, pork, chicken, lamb and processed meat (e.g., sausages). The fruits and vegetables fractions

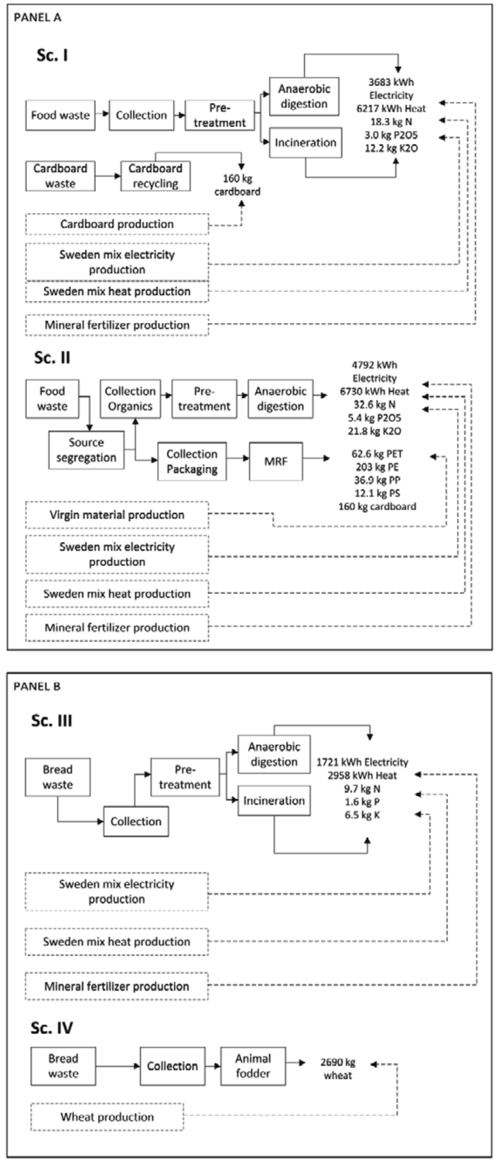

Fig. 2. Overview of the waste treatment scenarios assessed in this study. Scenarios I and II (Panel A) are for all waste fractions included in this LCA and Scenarios III and IV (Panel B) include only the bread waste fraction. Scenarios I and III model the current treatment in Borås city, and the alternative scenarios II and IV require separation of food waste from its packaging at the supermarket. Two comparisons are made in this part of the study. The first is between Scenarios I and II (Sc. I and Sc. II) and the second between Scenarios III and IV (Sc. III and Sc. IV).

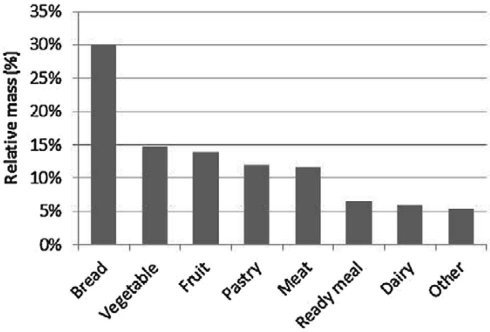

Fig. 3. Contribution of each fraction to the total food waste at the supermarket.

are comprised of waste from fresh products sold in the supermarket. Pastries are mainly produced at the supermarket chain's bakery and include products such as pies and rolls. Ready meals are frozen or chilled meals that usually come as a single portion and require little preparation. The dairy fraction includes several wastes, such as milk, creams, cheese, skim milk and yoghurt. Food waste products that were not included in the previous categories, such as snacks, egg and beverage wastes were grouped in the 'other' category. Non-food products, such as medicines and cleaning products, were not considered in this study.

Fig. 3 shows that bread waste is the largest waste fraction by mass, with an annual quantity of 6.7 t. Most of the bread waste is from fresh bread baked at the supermarket chainís bakery. Fruits and vegetables together have a share of 29% by mass, which corresponds to 6.4 t. Since bread waste is less dense than other waste fractions (e.g., meat and dairy), the volume of bread waste far exceeds the volume of other waste fractions. The total waste in the meat fraction was 2.6 t, which is 12% of the total waste by mass. The quantities of the different waste fractions are presented in table 1of the Supplementary material.

3.2. LCA of the supermarket food waste

The mass of all of the waste fractions included in the LCA (beef, pork, chicken, bread, strawberries, bananas, tomatoes, lettuce, potatoes, carrots, cabbage and apples) was 11 t, which is 49% by mass of the entire food waste at the supermarket during the year of study. The relative contribution of each waste fraction to the total mass of all products included in the LCA is given in the first column in Fig. 4. Similar data is given for the economic costs and selected impacts categories in the remaining columns in the figure, and in Section 2 of the Supplementary Material. In Fig. 4 all of the fruits and vegetable waste is grouped into one fraction for the sake of clarity. The environmental impacts per kilogram of each waste fraction and the impact from different parts of the supply chain are also discussed in Section 2.2 of the Supplementary material.

Fig. 4 reveals that bread, beef and pork waste typically have the highest environmental impacts among the food waste fractions that were included in the LCA. Beef waste has the largest environmental impact in six of the nine environmental impact categories, while bread waste has the largest contribution in the remaining three environmental impact categories. Bread waste also has the largest contribution to the total mass of the waste and the economic costs suffered by the supermarket.

In spite of its fairly low mass fraction, beef waste has the majority share of the emissions in the climate change impact category (third column in Fig. 4). This fraction is responsible for 22.4 t of CO_2eq. emissions per year, mainly due to the enteric fermentation of cattle and the use of fertilizer when growing animal feed. Beef waste has the largest contribution to particulate matter, photochemical ozone formation, acidification as well as terrestrial and freshwater eutrophication. Pork waste has an intermediate impact in most of the categories, and has the second largest contribution to economic costs. Chicken has lower impacts in all categories compared to beef and pork.

Bread waste has the largest contribution in mass, economic costs incurred by the supermarket, ozone depletion, freshwater ecotoxicity and resource depletion. In the climate change category, bread waste was responsible for the emission of 5.5 t of CO_2eq. year^{-1}.

The eight fractions of fruit and vegetable waste selected for this study have relatively low contributions to all categories. The total waste in these eight fractions is responsible for 0.94 t of CO_2eq. year^{-1}. The types of waste, within the fruit and vegetable waste

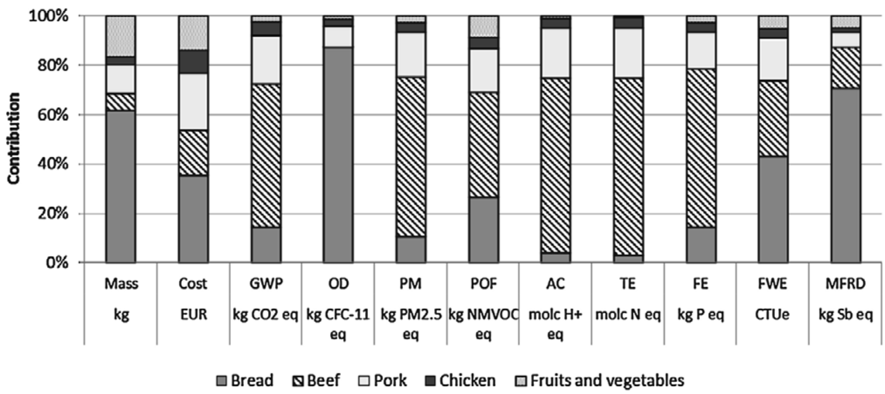

Fig. 4. Relative contribution of each waste fraction to the total mass, economic cost, climate change (GWP), ozone depletion (OD), particulate matter (PM), photochemical ozone formation (POF), acidification (AC), terrestrial eutrophication (TE), freshwater eutrophication (FE), freshwater ecotoxicity (FWE), and resource depletion (MFRD).The data is for waste fractions that were selected for the LCA and that, together, contribute to 49% of the total annual food waste at the supermarket.

fraction, which contribute the most to climate change are banana, strawberry and tomato. Together they are responsible for 70% of the emissions from the fruit and vegetable waste fraction.

3.3. LCA of alternative waste management scenarios

One of the objectives of this study is to provide information for supermarket top management to support development of strategies and actions that can reduce the environmental footprint of the supermarket. Therefore, the two alternative waste treatment scenarios discussed above with reference to Fig. 2 (Sc. II and Sc. IV) were compared with the current one practiced by the supermarket (Sc. I and Sc. III).

Both of the alternative waste treatment scenarios resulted in lower emissions of greenhouse gases compared to the waste treatment that is currently used.

The comparison of Scenarios I and II shows that the emissions from Sc. II are 1027 kg CO_2eq·year^{-1} lower than from Sc. I. Fig. 5 shows that most of the savings in Sc. II came from the higher efficiency of the system, with more food waste going to anaerobic digestion instead of incineration. Therefore, it led to an increase in the energy and fertilizer production, which substituted electricity, heat and mineral fertilizer production. Sc. II also avoided the production of virgin material due to increased recycling rates of the packaging materials. The positive part of the graph represents the emissions from the anaerobic digestion, transportation and the recycling processes.

The waste treatment in Sc. IV reduces emissions of CO_2eq. by 1549 kg year^{-1} compared to

Sc. III. The reduction on the emissions in Sc. IV is primarily due to the avoided production of wheat used as animal feed (Fig. 5). The process emissions for Sc. IV represent the emission from the bread transportation to be use as feed. For the reference scenario (Sc. III), the emissions are from the anaerobic digestion process and the savings are from energy and fertilizer production, similar to Sc. I.

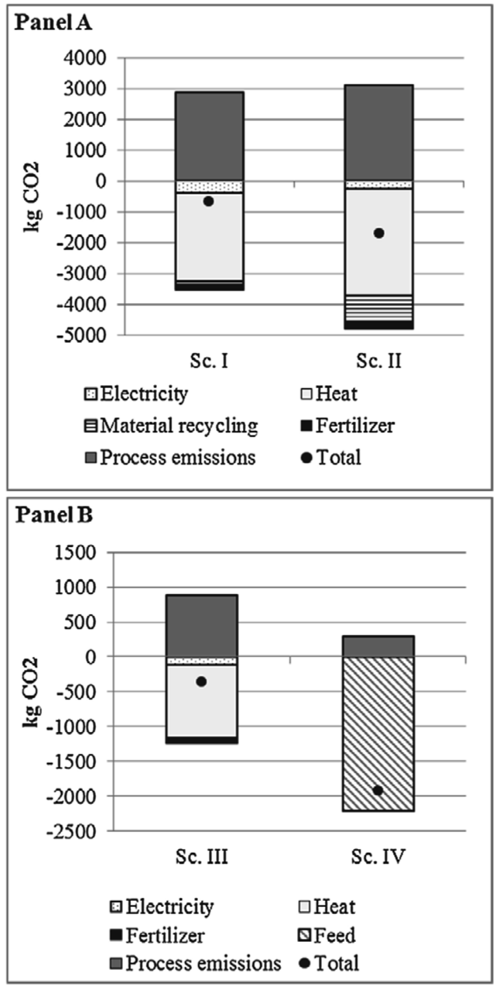

Fig. 5. Climate impact for the waste treatment Scenarios I and II (Panel A) and III and IV (Panel B). The scenarios are discussed in the text with reference to Fig. 2.

4. Discussion

Many studies (Scholz et al., 2015; Stenmarck et al., 2011) of supermarket food and food waste focus on a single measurement, such as mass of the food or food waste, or assess only a few environmental impact categories. The present study focuses on various fractions of supermarket food waste and their contribution to the total mass of food waste, the economic costs for the supermarket and the environmental impact for all waste included in the LCA. The different relative contribution of each waste fraction to the total mass, economic costs and the environmental impacts highlights the importance of analysing the effects of food waste in terms of several indicators. Fig. 4 shows that beef and bread waste contribute the most to supermarketís environmental impact and economic losses, but have different relative contributions in different categories.

Comparing the present results with previous studies is hindered by the fact that different methods and measurement techniques have been used in the studies. Many studies of food waste present the results as a percentage of the total sales, and the present study did not have access to this data. Also, some studies include pre-supermarket waste in the analysis.

It should also be noted that the present study is for a single supermarket. Although this has the strength that recent and relevant data for this supermarket is used, it has the weakness that the results may not be directly generalizable to other supermarkets. Also, the LCA reported here excluded dairy waste. Although this waste has a low mass compared to other fractions (see Fig. 3), dairy products have high environmental impacts per kg. The dairy waste may therefore have a significant environmental impact. However, a study by Scholz et al. (2015) estimates that supermarket dairy products contribute to approximately 5% of the supermarket's carbon footprint.

A study by the Swedish Environmental Research Institute indicates that fresh bakery products are the most wasted product in retail (Stenmarck et al., 2011). This is in agreement with the results of the present study. Interviews with the retail management revealed that use of bread waste as animal feed is not practiced by the supermarket chain studied in that work and, according to the Swedish Waste Management Association, anaerobic digestion is the most common method to treat food waste in Sweden (Avfall Sverige, 2014). This indicates that the results obtained here (both amounts and effects of bread waste as well as the proposed alternative waste treatment scenarios) are relevant for many supermarkets in Sweden.

Previous studies on the environmental impacts of supermarket waste did not include bread in the scope of products studied (Scholz et al., 2015). The results obtained in the present study show that bread waste is key when developing strategies to reduce the environmental impact of supermarkets.

As discussed by Scherhaufer and Schneider (2011), the high amount of bread waste may be connected to the demands and behaviour of the consumers. The demands for freshness and for a wide range of types of bread necessitate frequent production in sufficiently large quantities so that fresh bread is available throughout the day (Scherhaufer and Schneider, 2011). Interviews performed by Stenmarck et al. (2011) indicate that supermarkets often produce 7% more than the expected sales in order to meet the consumer demands. Reducing overproduction may be challenging, since predicting consumer demands is difficult. Factors such as weather, which are in themselves hard to predict, can cause significant variations in the demand (Stenmarck et al., 2011). Thus, the use of bread waste as animal feed has an advantage, since it decreases the environmental impact at the same time as being a revenue stream for the supermarket.

A study by the Waste and Resource Action Program (WRAP, 2013) showed that bakery products are the fourth largest fraction in household food waste, accounting for 800,000 million tonnes per year in the United Kingdom. A questionnaire used in the study revealed that consumers believe that they produce only small amounts of bread waste and that its environmental impact is low. Therefore, communication of the results from the present life cycle assessment to consumers has the potential to increase awareness of the impacts of bread waste, perhaps reducing the waste not only at supermarkets, but also by households.

The alternative treatment routes (Sc. II and IV) led to a decrease in greenhouse gas emission. However, these routes may demand more investment in staff training than the current waste treatment scenarios (Sc. I and III), since source separation at the supermarket is required. The increase in energy demand and in cost was not assessed in this study. Moreover, the environmental assessment of alternative waste treatment routes is limited by the impact assessment of the carbon footprint, since some data that is needed for modelling other impact categories is not available. The energy used for separating the waste at the supermarket was not assessed, but it is expected that it will be done by the staff of the supermarket, and is therefore expected to be small.

5. Conclusion

Food waste leads to loss of valuable resources, such as energy, water, land and labour and to unnecessary emissions of pollutants. The research described here investigated supermarket food waste by categorising and quantifying the waste at a supermarket, assessing the environmental impacts of the waste and suggesting alternative ways to treat the waste in order to reduce its environmental footprint.

The life cycle assessment results reported here reveal that the annual wastage of bread and beef products have the largest contribution to the environmental footprint of the supermarket. Compared to the other waste fractions included in the LCA, the annual bread waste has the largest contribution to the total mass of the food waste, the economic costs incurred by the supermarket and the environmental impacts in ozone depletion, freshwater ecotoxicity and resource depletion categories. Beef waste has the largest contribution to particulate matter, photochemical ozone formation, acidification as well as terrestrial and freshwater eutrophication categories.

Alternative waste treatment scenarios, that require separation of the food waste from its packaging at the supermarket and that allows for material recycling of the packaging and the use of bread waste as animal feed, have the potential to reduce the emissions of CO_2eq. by as much as 1027 and 1549 kg per year respectively, in comparison with the current waste treatment practiced by the supermarket.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.resconrec.2016.11.024.

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Paper II

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Full length article

Bread loss rates at the supplier-retailer interface – Analysis of risk factors to support waste prevention measures

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ABSTRACT

This paper quantifies bread waste throughout the Swedish supply chain and investigates the loss rate of pre-packaged bread products at the supplier-retailer interface. The goal is to understand the extent of bread waste in Sweden and to identify risk factors for high quantities of waste at the supplier-retailer interface, in order to provide information supporting waste prevention measures. The study uses primary data, in combination with national statistics and data from sustainability reports and the literature. Primary data were collected from 380 stores of a Swedish retail company and a bakery. Bread waste was calculated to be 80 410 tons/year in Sweden, the equivalent of 8.1 kg per person/year, and was found to be concentrated at households and in retail, specifically at the supplier-retailer interface. The results provide evidence that take-back agreements between suppliers and retailers, where the retailer only pays for sold products and the supplier bears the cost of the unsold products and their collection and treatment, are risk factors for high waste generation. Current business models may need to be changed to achieve a more sustainable bread supply chain with less waste.

1. Introduction

Food waste is gaining increasing attention from the media, researchers, companies and policymakers, due to the large amounts generated globally and the associated environmental, social and economic impacts (Garnett, 2011; Steinfeld et al., 2006). It is estimated that one-third of the food produced in the world is not consumed (FAO, 2013). The European Union generates approximately 88 million tonnes of food waste (Stenmarck et al., 2016), with Sweden generating an estimated 1.3 million tonnes (Westöö et al., 2018). Food waste generated throughout the supply chain represents a significant loss of the resources invested in production, transport and storage of food products. These processes involve significant use of energy and resources and generate emissions with environmental impacts. All of these impacts are to no avail when the food is not consumed.

Bread waste represents a significant share of food waste. It is estimated that bakery waste corresponds to 10% of all food waste in the United Kingdom and that 32% of all bread purchased is wasted in households (WRAP et al., 2011). Brancoli et al. (2017) quantified and assessed the environmental and economic impacts of food waste in a Swedish supermarket and concluded that bread was one of the main contributors in relation to mass, environmental impacts and economic costs. Similarly, Stensgård and Hanssen (2016) identified bread as the product group with the highest waste percentage in relation to sales value at Norwegian retailers.

Compared with other parts of the supply chain, retailers are reported to generate lower amounts of food waste (Stenmarck et al., 2016; Westöö et al., 2018). However, retailers are in a unique position to influence the generation or prevention of waste in other parts of the supply chain, such as primary production, distribution and end consumption (Brancoli et al., 2017; Kulikovskaja and Aschemann-Witzel, 2017). An example of retailer influence is the current model adopted by the largest bakeries for bread distribution in Sweden, which involves a full take-back agreement (TBA) between retailer and supplier. This means that the bakeries are responsible for forecasting, ordering, placement and removal of products from supermarket shelves. Moreover, the bakeries are financially responsible for unsold products (including their collection and waste management), operating in a reverse or circular supply chain, unlike the majority of products sold in retail.

Although TBAs are not unique to Sweden, their presence in Sweden is often attributed to the high market power of retailers (Ghosh and Eriksson, 2019). Sweden has one of the highest concentrations of retailers in the European market (Rohm et al., 2017), with three companies having a combined market share of 86% (DLF et al., 2018).

Some studies suggest that TBAs are a risk factor for waste generation. For example, Lebersorger and Schneider (2014) found higher loss

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rates for TBA products than for in-store waste, with no significant correlation between loss rates and other retail characteristics such as purchases per year, total sales and sales per product range. Eriksson et al. (2017) compared return levels at retailers and non-retailers and found higher levels at retailers than at non-retailers such as hotels, schools and restaurants that do not have a TBA with their suppliers.

Only a few studies have been carried out on bread waste at the interface between suppliers and retailers. As reported in Eriksson et al. (2017), most previous studies on retail food waste do not consider the occurrence of bread waste due to TBAs (e.g. Beretta et al., 2013; Cicatiello et al., 2017; Mena et al., 2011). This may be because this type of agreement simply does not exist in the sector investigated, or because such data are not accessible since unsold TBA products are not recorded in retailer databases (as they are disposed of by the supplying bakeries). For instance, Cicatiello et al. (2017) reported that less than 1% of the waste was recorded in the bakery department of the retailer. However, some studies mention the occurrence of TBA in Norway (Stensgård and Hanssen, 2016), Sweden (Eriksson et al., 2017; Ghosh and Eriksson, 2019) and Austria (Lebersorger and Schneider, 2014). There are also references in the waste treatment literature to using returned bread as a potential substrate for specific treatment pathways, suggesting the occurrence of TBA in other countries such as the Netherlands (Weegels, 2010) and Germany (Brosowski et al., 2016). However, these sources do not reveal the extent to which TBAs are implemented.

Most potential solutions for reducing waste require cooperation between at least two actors in the supply chain (Liljestrand, 2016). A recent report published by three Swedish government agencies discusses the risks that TBAs create in relation to generation of food waste (Livsmedelsverket et al., 2018). Nevertheless, there is a lack of evidence to support either side of the argument, and it is essential to find evidence on whether this business practice actually increases the risk of food waste generation.

The production of food, irrespective of whether it is consumed or wasted, is associated with environmental impacts. Since waste management options involve downcycling and consume resources, prevention is a priority action over waste management or valorisation activities (Dutch Ministry of Economic Affairs, 2014; USEPA, 2018).

Primary data on bread waste, particularly at the supplier-retailer interface, are scarce in the literature. Measurement of food waste is necessary to determine its extent and identify hotspots. Information on waste quantities would allow definition and prioritisation of prevention measures and monitoring of progress and effectiveness. Nevertheless, waste data is still considered a highly sensitive matter for both suppliers and retailers. Confidentiality agreements often hinder a comprehensive investigation on the topic of food waste (e.g. Garrone et al., 2016; Lebersorger and Schneider, 2014; Mena et al., 2011; Mourad, 2016; Redlingshöfer et al., 2017; Stenmarck et al., 2011). Moreover, in a recent study of food waste, Xue et al. (2017) reported that only 20% of the 202 reviewed publications were based on primary data, highlighting the importance of studies which use direct measurements “to improve the accuracy and reliability of the data”. One can speculate that one of the reasons for the few number of studies using primary data is confidentiality requirements by the data holders. This study is a first step to analyse waste data from the Swedish bread industry and represents the best available data after privacy and confidentiality requirements were met.

The objectives of this study were to investigate the loss rate of prepackaged bread products at the supplier-retailer interface. Moreover, since one of the reasons for food waste quantification is to understand the magnitude and location of waste, and hence to support waste prevention efforts, primary and secondary data were combined to quantify bread waste generation throughout the Swedish supply chain. The overall aims were to determine the extent of bread waste in Sweden, to estimate the contribution of waste at the supplier-retailer interface to the total waste, and to identify the risk factors for high waste generation at the supplier-retailer interface. This information can be used to provide information supporting waste prevention measures.

2. Methods

2.1. Database and definitions

2.1.1. Bread waste

Bread waste was defined as bread removed from the food supply chain and was considered avoidable waste, since in general it has no inedible fraction. The Fusions Project defines waste as the fraction removed from the food supply chain and sent to disposal or recovery, i.e. anaerobic digestion, composting, crops ploughed in or not harvested, bioenergy, incineration, co-generation, sewer, landfill and discards (Ostergren et al., 2014). Bread removed from the food supply chain to be reused, recycled, e.g. as animal feed, reworked or incorporated into other food products is not considered waste under this definition. The primary data collected in this study fitted the Fusions Project definition, since they showed that in-store and bake-off bread waste is disposed of by anaerobic digestion or incineration and returned bread is used in ethanol production. However, for bread waste generation that is extrapolated to the national level, it was not possible to determine the destination of bread from remaining bakeries and retailers that did not provide data regarding the amounts and destination of their bread waste. It is therefore possible that some returned bread is recirculated into the food supply chain as food donations, yeast production or animal feed, and is thus not defined as food waste according to Östergren et al. (2014). Since the amount of recirculation is unknown, in this study all returned bread was considered to be food waste.

2.1.2. Bread categories

The scope of this study was bread, defined as a staple food prepared from dough of flour and water, typically by baking. The quantification of bread waste at the national level considered pre-packed bread and bake-off products. Bake-off products, including pastry products, are baked from pre-made dough in supermarket stores or supplied by a bakery in a nearby store. Pre-package bread and bake-off products have a relatively short shelf-life, typically ranging from one to ten days for the former and one day for the latter. Products with longer shelf-lives, such as crispbread (shelf-life up to one year), are outside the scope of this study.

There are several categories of pre-packed bread in supermarkets, such as white, wholegrain and dark bread. Products within these categories can also differ in several aspects, such as type of flour used (wheat, maize, barley, wholegrain, gluten-free, etc.), type of leavening and presence of other ingredients. In the present analysis, pre-packed bread waste generated in retail was further divided into two different categories as regards its distribution system, namely products with and without a TBA.

2.1.2.1. Products under the take-back agreement (TBA). Take-back agreements between bakeries and retailers state that the supplier is financially responsible for the unsold bread, also referred as ‘returned bread’. As a result, retailers only pay for the products they sell, while suppliers are responsible for forecasting, ordering, placement and withdrawal of products from retailer shelves, and perform reverse logistics operations by collecting and disposing of the unsold products (Eriksson et al., 2017).

2.1.2.2. Products without take-back agreements (No-TBA). No-TBA products are often products branded by the retailer, i.e. private label products. For such products, the retailer is financially responsible for the unsold products and for their disposal as waste. Typically, such products are discarded together with the other fractions of in-store food waste, but since such waste is owned by the supermarket it is also subject to in-store actions with the potential to prevent waste, such as discounting or donating to charity.

2.1.3. Data collection

There is a lack of studies using primary data for the retail sector (Møller et al., 2014). A common problem is that food waste data are sensitive information for retail companies. Since many companies and supermarkets will not disclose their data on food waste, confidentiality was granted to the companies participating in this study.

Bread waste data were collected from 380 stores of a Swedish retail company and a bakery. The data were available at the product level and aggregated for all stores during a period of one year. The databases of the retail company and bakery included sales, delivery amounts, percentage returns, primary package size, shelf-life and type of distribution system, i.e. TBA or No-TBA.

The supermarket stores included in this study belong to the same parent company, but differ in their characteristics. The parent company is one of the three largest supermarket chains in Sweden and has different types of stores, including a discount chain of stores with a smaller range of products and "premium brand" stores that have a comparatively wide range of products. The sales area varies from 300 to 4000 m^2. The bakery company is one of the three largest bakeries in the Swedish bread market, which together have more than 80% market share for pre-packed bread.

When quantifying bread waste at the national level, the primary data described above were used, in combination with national statistics and data from sustainability reports and from the literature.

2.2. Data processing and quality

2.2.1. Primary data collection

2.2.1.1. Losses at bakery level. The calculation of bread waste at the bakery level considered three of the major bakeries, as well as the production of private-label products, representing 87% of the Swedish market for bakeries (Company confidential, 2018) (Table 1). Besides bread waste, the waste at bakery level included waste dough, flour and other ingredients. Only one bakery provided quantitative data about its losses and production, but it was possible to retrieve data for another bakery from its sustainability report (Polarbröd, 2016). In Eq. 1, these companies are represented by the index $i = 1$ and $i = 2$. The remaining bakery and the private label production are represented by $i = 3$ *and* $i = 4$.

The total waste for the three companies and private label was calculated using their market share as a scaling factor:

$$Waste\ [ton] = \left(\sum_{i=1}^{2} Waste_i \Big/ \sum_{i=1}^{2} Market\ share_i \right) * \sum_{i=1}^{4} Market\ share_i \tag{1}$$

2.2.1.2. Losses for retail TBA products. The calculation of TBA bread waste considered the three major bakeries, excluding the private label production since those products do not have a TBA. TBA bread production was calculated using data from one bakery ($i = 1$), by subtracting the amount of private label products produced by this bakery from the total bread production and extrapolating it for each of the remaining companies using the market share as the scaling factor (Eq. 2). The loss rates were based on primary data collected during the study for this bakery and data obtained in interviews with representatives of Swedish bread suppliers by Ismatov (2015). The TBA production level for each individual company i was calculated using Eq. 2 and total waste was then calculated as the sum of total TBA production multiplied by the loss rate of each company (Eq. 3).

$$Production\ TBA_i\ [ton] = \left(Production\ TBA_1 \Big/ Market\ share_1 \right) * Market\ share_i \tag{2}$$

$$Waste\ [ton] = \sum_{i=1}^{3} Production_i * Loss\ rate_i \tag{3}$$

2.2.1.3. Losses in retail No-TBA products and bake-off products. The waste calculation for products with no TBA and for bake-off bread considered six of the largest retailers in Sweden, representing a combined 89% of the total retail trade in Sweden (DLF et al., 2018). Primary data were retrieved for one retail chain, denoted Retailer A in Eq. 4, and the market share for Swedish retailers (Table 1) was used as a scaling factor:

$$Total\ waste\ [tons] = Waste_{Retailer\ A} \Big/ Market\ share_{Retailer\ A} \tag{4}$$

2.2.1.4. Losses in convenience stores. Since there was no available data on bread waste in convenience stores, data for this sector were obtained through in-store interviews. Based on the interviews, an average daily waste of ten pieces of bakery products with an average weight of 90 g was assumed. The number of stores for the five largest convenience store chains (Table 2) was used as the scaling factor. The total waste in convenience stores was hence calculated as:

$$Total\ waste\ [tons] = Waste \left[tons \Big/ day \right] * N_{stores} * 365\ [days] \tag{5}$$

2.2.2. Secondary data collection

2.2.2.1. Losses in households. The calculation of bread waste in households used data retrieved from the Swedish national statistics. Total food waste considered three different waste fractions, namely residual waste, separately collected food waste and home composting (Westöö et al., 2018). It was assumed that bread is fully avoidable and therefore only the percentage of avoidable waste was considered in the calculation, using composition analysis data from Jensen and Viklund (2015). Finally, the percentage of bread waste in the avoidable fraction was calculated using composition analysis data from Andersson (2012). For the home composting waste fraction, there were no data available regarding the avoidable fraction and the bread percentage in the avoidable stream. Therefore, similar values to those reported for the separately collected food waste stream were assumed. Table 3 shows the data used for calculation of household bread waste in Eq. 6, where the index i represents each of the household food waste fractions:

Table 1
Swedish market share held by different bakeries and retailers (Company confidential, 2018; DLF et al., 2018).

Company	Market share
Bakery	
Pågen	39%
Fazer	25%
Polarbröd	16%
Private label	7%
Retailers	
ICA	44.9%
COOP	16.2%
Axfood	15.3%
Bergendahls	6.7%
Lidl	3.9%
Netto	2.0%

Table 2
Number of convenience stores in Sweden (Dagab Inköp and Logistik AB, 2018).

Convenience chain	Number of stores
Reitan	510
Emab	350
Cirkle K	300
OKQ8	310
Preem	100

Table 3
Household waste data.

	In residual waste	In separately collected food waste	In home composting
Food waste	400 000	271 000	43 000
Avoidable food waste (%)	40%	20%	20%
Bread in avoidable food waste (%)	12%	17%	17%

$$Waste\ [tons] = \sum_{i=1}^{3} Food\ waste_i * Avoidable\ food\ waste_i * Bread\ in\ avoidable\ food\ waste_i \quad (6)$$

2.2.2.2. Losses in schools and restaurants. As for households, data for restaurants and schools were retrieved from references cited in the Swedish national statistics. Total food waste data were taken from Jensen et al. (2017) and Westöö et al. (2018) for schools and restaurants, respectively. The percentage of avoidable waste and the percentage of bread in avoidable waste were retrieved from an auxiliary document for national waste quantification (SMED, 2015). Total food waste for restaurants and schools in Sweden is reported to be 71 000 and 49 000 tons/year, respectively (Jensen et al., 2017; Westöö et al., 2018). The percentage of avoidable waste is 63% for restaurants and 54% for schools (SMED, 2015). The percentage of bread in the avoidable fraction is 16% and 10% for restaurants and schools, respectively (SMED, 2015). Eq. 6 was used to determine the total waste from schools and restaurants.

2.2.3. Bread loss rates at the supplier-retailer interface

The database was compiled with information from the bakery and retailer. Bread loss rates, at the product level, were calculated as the ratio of the mass of bread waste to the total mass of bread delivered to the supermarket. All masses stated in tons refer to metric tons and represent products wasted in-store or returned to the suppliers. The data were analysed using the software MINITAB 17 (Minitab Inc., State College, PA, USA).

Regression models were used to investigate the effects of three independent variables, namely sales, shelf-life and pack size, on the bread loss rates for individual products. Box-Cox transformation was applied to reduce the data skewness prior to the analysis, and an optimal power transformation (λ) was defined for each model. In total, seven models were tested, one for each independent variable, three pairwise combinations of the independent variables and one combining the three independent variables. Moreover, non-parametric correlation coefficient Kendall’s tau-b was used to investigate the correlations between the loss rate and the three independent variables. A two-sample *t*-test was used to determine whether the loss rate means for products under different distribution systems, i.e. with and without TBA, differed significantly. In all statistical analyses, a significance level of $p < 0.05$ was used.

3. Results

3.1. Bread loss rates at the supplier-retail interface

In total 40 240 tons of bread were generated in the interface between the bakeries and retailers in the Swedish value chain and the majority of the waste occurred after production (Fig. 1). TBA products was the category with the highest share, contributing 39% of the waste. The second highest category was production waste (30%), followed by bake-off products, corresponding to 24% of the total share. Products that were not subjected to TBAs accounted for 7% of the waste generated in the supplier-retailer interface.

Since the majority of products sold in the stores are subjected to TBAs, the types of bread returned represent the average assortment supplied by the bakeries (whole wheat, sourdough, rye bread, etc.). Products without TBA (No-TBA) on the other hand have a smaller assortment, and consist mainly of private label products, i.e. products that are manufactured by the bakery for sale under retailers' brand.

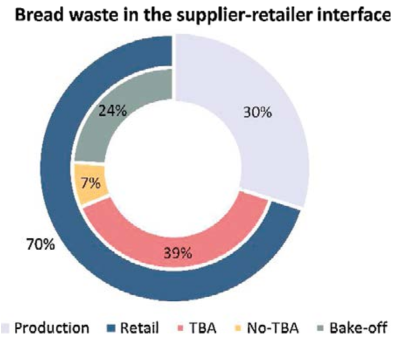

Fig. 1. Bread waste generation in the supplier-retailer interface (results from the primary data collection).

For the stores included in the study, 21% of the product range was responsible for 80% of the total waste (Fig. 2). Products supplied in large quantities to the stores had the largest impact on the total waste in terms of mass, but had relatively small loss rates. Large loss rates were associated with niche products, i.e. products that appeal to a particular market subgroup and that often have relatively small sales volumes, such as gluten-free products and breads containing unusual ingredients.

The 10 products with the highest waste in terms of mass were TBA products, with an average loss rate of 9.4% and contributing 60% of total waste. The average loss rate for the 10 products with the highest loss rates was 35%, and their contribution to the total waste was 2.4%.

3.2. Bread waste in Sweden

Primary data for the supplier-retailer interface was combined with secondary data to provide an overview on the total bread waste in the Swedish supply chain. Bread waste in Sweden was calculated to be 80 410 tons/year, the equivalent of 8.1 kg per person/year. Data for the Swedish population used the reference period of November 2018, which was 10 million people (SCB, 2018). Fig. 3 shows the distribution

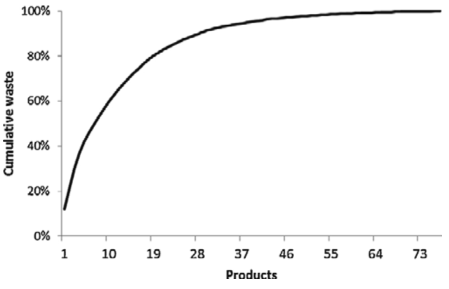

Fig. 2. Cumulative bread waste generation by product, sorted by largest to smallest waste in mass.

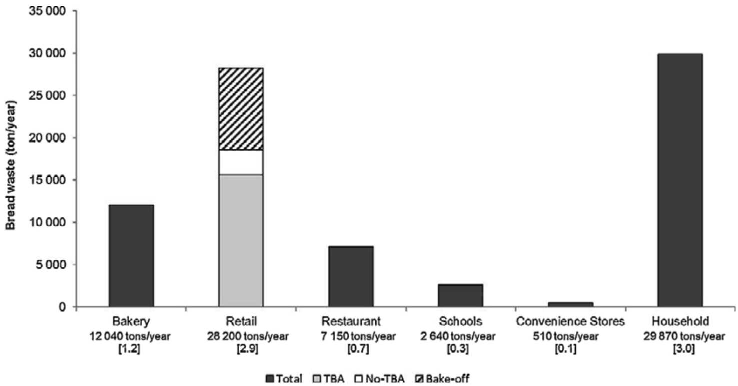

Fig. 3. Bread waste generated yearly in different parts of the value chain in Sweden (primary and secondary data combined). Numbers in brackets represent bread waste generation in kg per person per year.

of bread waste generated in different parts of the value chain. The majority of waste was observed at households and in retail, especially at the interface between supplier and retailers.

Losses during production were calculated to be 12 040 tons/year, corresponding to 15% of the waste in the value chain. Bread waste in retailers was calculated to be 28 220 tons/year and 55% of that amount, corresponding to 15 510 tons/year, was delivered to retail with a TBA. The second largest category of bread waste in retail was bake-off bread, representing 34% of the waste generation.

Restaurants and schools had relatively low amounts of waste, corresponding to 9% and 3% of the total waste, respectively. The total bread waste in households was 29 870 tons/year, which is 13% of the avoidable waste fraction in households.

3.3. Relationship between sales, shelf-life, pack size and loss rate

Multiple linear regression analysis was used to develop a model for predicting bread loss rates based on the following product characteristics: sales (tons/year), shelf-life (days) and primary pack size (g). For all models, the Box-Cox transformation estimated $\lambda = 0$, which represents the natural logarithmic transformation. The results from the statistical analysis described in Section 2 showed that loss rate had a significant, but low correlation with sales, shelf-life and pack size, but no significant relationship was found for the combined effect of the independent variables (Table 4). The highest adjusted coefficient of determination (R^2) was found for the sales variable, which explained 13.94% of the variance in the loss rate (Kendall's tau b = -0.343).

The results from the regression models and correlation coefficients showed significant, but low, coefficients of determination for sales and pack size. Shelf-life showed significance, but none of the variance was explained by this variable (Table 4), and Kendall's tau-b correlation for shelf-life was not significant (Table 5). This indicates that a low percentage of the variance in the loss rates was explained by these three variables. Therefore, bread loss rates can be assumed to be influenced by other factors, such as TBAs and staff behavior.

3.4. Relationship between TBA and loss rate

The analysis of waste quantities indicated larger loss rates for products with a TBA than for No-TBA products. In order to test whether the difference between the means of the two categories was significant, a two-sample *t*-test was conducted. The results showed a significant difference ($p = 0.021$) in the loss rates for products with a TBA ($M = 0.189$, $SD = 0.112$) and for No-TBA products ($M = 0.1353$, $SD = 0.0519$). These results indicate that the type of distribution system has a statistically significant effect on loss rate levels for bread products. Specifically, the results indicate that when a product is subject to a TBA, it produces more waste. However, it is important to note that there is a difference in the sample size for TBA products ($N = 71$) and No-TBA products ($N = 7$).

4. Discussion

The most recent food waste quantification in Sweden (Westöö et al., 2018) revealed that food waste in retailers is 13-fold lower than that in households. However, our quantification of bread waste in the Swedish supply chain showed that bread waste is not concentrated in households, with waste rates of similar magnitude in retail and households. These results demonstrate the importance of taking actions in the retail sector to prevent bread waste. The monetary loss due to bread waste in the Swedish supply chain was estimated to be 240 million EUR per annum, based on an average retail price of 3 EUR/kg (Statista, 2018).

Bakeries generate a large amount of waste, which is similar to the waste generated by TBA products. It is noteworthy to mention, however, that part of the bread production is exported. Nevertheless, the rationale behind food waste quantification is a step towards the development of prevention efforts (Tostivint et al., 2016). The waste generated at the bakeries during production, regardless of the destination market, is under the influence of these companies, which are the ones with the ability to reduce the waste. In retail, two main waste categories were identified, namely TBA and bake-off products. The risks factors for high waste generation for TBA are discussed later. Regarding bake-off bread, in-store bakeries are important departments in retail, since they attract customers to the store. High waste of bake-off products can be related to the typical one-day shelf-life and also to the challenge of keeping shelves full with newly baked bread during the entire opening time of the supermarket.

The present quantification used a variety of sources and data collection methods, which added uncertainty to the results. For instance, local bakeries were excluded from the quantification. Although they have a relatively low market share, some of these bakeries are important in specific regions. The study covered 87% and 89% of the market for suppliers and retailers, respectively. The results were extrapolated assuming that the remaining companies have the same waste generation pattern as those included in the study. This resulted in estimated bread waste of 86 040 tons/year, in comparison with 80 410 tons/year in the original analysis (Section 3.2).

Literature data of bread waste at different stages of the supply chain

Table 4
Regression models using Box-Cox optimal transformation. Loss rate as dependent variable and standard error in brackets. Equation in the form: $\boldsymbol{ln}(\textbf{Loss rate}[\%]) = \textbf{C} + \textbf{C}_1 * \textbf{sales}\left[\frac{\textbf{ton}}{\textbf{year}}\right] + \textbf{C}_2 * \textbf{shelf life}\ [\textbf{days}] + \textbf{C}_3 * \textbf{package size}\ [\textbf{g}]$.

Independent variables tested	Sales	Shelf-life	Pack size	Sales and shelf-life	Sales and pack size	Shelf-life and pack size	Sales, shelf- life and pack size
(C_1) Sales (ton/year)	−0.000741 [0.000202]**			ns	ns	ns	ns
(C_2) Shelf-life (days)		−0.0393 [0.0610]*		ns	ns	ns	ns
(C_3) Package size (g)			−0.000683 [0.000311]*	ns	ns	ns	ns
(C) Intercept	−1.7496 [0.0649]**	−1.521 [0.475]*	−1.500 [0.168]**	ns	ns	ns	ns
R^2 (adjusted)	13.94%	0%	4.73%				

ns: not significant.
* $p < 0.05$.
** $p < 0.001$.

Table 5
Kendall's tau-b correlation coefficients between bread loss rates and sales, shelf-life and pack size.

Bread loss rate	
Sales (tons/year)	−0.343**
Shelf life	ns
Pack size	−0.158*

* $p < 0.05$.
** $p < 0.001$.

and the data obtained in the present study are compared in Table 6. However, as pointed out by Lebersorger and Schneider (2014), such data cannot be accurately compared since the methodology and the reference unit used for quantification of the waste may differ. Another aspect that might hinder comparison between studies is the risk of intentional or unintentional exclusion of returned bread waste from the waste quantification analysis, since this information is often not included in the supermarket database. Hence, TBAs not only increase the risk of waste, but also increase the risk of under-reporting the mass of the waste.

The statistical analysis revealed that the adjusted coefficient of determination was highest for the sales variable, as also found by Eriksson et al. (2014) and Lebersorger and Schneider (2014). Products supplied in large amounts and with higher sales in terms of mass were found to have relatively lower loss rates. Larger loss rates were found for niche products supplied in small quantities, which is similar to the pattern found by Eriksson et al. (2012) for fresh fruit and vegetables.

The negative correlation between sales and loss rates implies that the large range of products available to consumers, and its subsequent demand for being stocked, is a risk factor for high waste generation. The reason is that a large inventory is necessary to provide a high service level for a large range of products and, since demand is uncertain, this increases the risk of waste. This is in line with previous findings in the literature (Lewis et al., 2017; Mena et al., 2011; Schneider and Scherhaufer, 2009; van Donselaar et al., 2006). This finding indicates that reducing the product assortment has the potential to decrease bread waste. This requires customer willingness to buy another product if the preferred item is out of stock. Van Woensel et al. (2007) report that, on average, 84% of Dutch customers will substitute their bread purchase. The effectiveness of reducing the assortment as a measure to reduce bread waste has also been reported by a Danish retailer, which achieved a 67% decrease in bread waste by decreasing its range of pre-packed white and dark bread by 25% (Schrøder, 2017). Although a 2% decrease in sales was reported, the measure resulted in an increase in profits due to the reduction in waste generation (Schrøder, 2017).

Overall, a few products were responsible for the majority of the waste in terms of mass in the present study, even though these products had lower loss rates. This suggests that, in addition to reducing the range of the products, measures aiming at waste prevention of such products should be prioritised, due to the potential to achieve a larger general waste reduction.

The low adjusted coefficient of determination for the relationships between loss rates and the predictor variables (sales, shelf-life, pack size) indicate that other factors, such as TBAs and staff behaviour, influence the loss rates. This finding was corroborated by statistical analysis of the waste data for different distribution systems, which indicated that products with a TBA have higher waste levels than No-TBA products, TBAs were identified therefore as a risk factor for increased volumes of waste. Similarly, Eriksson et al. (2017) identified TBAs as a risk factor for high waste generation when comparing waste levels in retailers and non-retailer actors. As the data in Table 5 show, studies that characterised in-store and return waste individually (EHI, 2011; Lebersorger and Schneider, 2014) reported higher loss rates for returned products. The differences in the average loss rates for products with and without a TBA in this study were small, but still large enough to provide reduction potential.

TBAs influence and shape the relationship between suppliers and retailers (Eriksson et al., 2017; Ismatov, 2015). In general, due to short lead times in the bread industry, it is common for suppliers to deliver the products. TBAs extend the suppliers responsibilities to forecasting, assortment negotiations and quantities delivered. These agreements have similar terms for the different suppliers and retailers in Sweden, i.e. the majority of the stores have a full service agreement, meaning that the suppliers deliver and collect the unsold bread. Moreover, the negotiated time for take back is similar, and is approximately 2–3 days from expiry date (Ismatov, 2015). Although the TBAs themselves are very similar, different criteria may be required by the bakeries and supermarkets before they enter into a TBA. These criteria may be, for example, the minimum amount of bread delivered to the supermarket or the distance travelled to deliver the bread. For instance, stores with low turnovers or that are located far from urban centres might have different terms in the contracts, such as minimum-order requirements or that they are responsible for the waste treatment of the unsold products (Ismatov, 2015).

Typically, for stores under TBA the tasks related to the supply of bread are the responsibility of the drives, which also collects the returned products. These drivers act as sales representatives for the bakeries for specific stores, for which they negotiate and supply bread on a regular basis. In order to adjust their forecast, drivers use historical data and information on events that can influence sales, such as campaigns and strategies from competing companies (Eriksson et al., 2017; Ismatov, 2015). Since the suppliers (drivers) are the responsible for the replenishment of the shelves, stores seldom have a bread department manager. This means that the drivers have to negotiate with a store manager that is not primarily responsible for the bread department. This highlights the importance of sharing information, which is essential to good planning and forecasting. Moreover, as identified by

Table 6
Comparison of bread waste data reported in the literature and in the present study.

	Bread loss rate	- and basis	Country	Source
Bakery	5.2%	of total production	Sweden	Present study
	6.9%	of total production	Sweden	(Polarbröd, 2016)
	5.1%	of total production	Switzerland	(Beretta et al., 2013)
	1.2% [2]	of total production	Norway	(Stensgård and Hanssen, 2016)
Retail				
Bake-off	8.5%[1]	of total mass delivered	Sweden	Present study
	27%	of total waste mass	Sweden	(Brancoli et al., 2017)
	6.5%[1]	not specified	Germany	(EHI, 2011) apud. (Lebersorger and Schneider, 2014)
In-store	3%	of total waste mass	Sweden	(Brancoli et al., 2017)
	2.8%[1]	of sales in cost price	Austria	(Lebersorger and Schneider, 2014)
	1%	not specified	Germany	(EHI, 2011) apud. (Lebersorger and Schneider, 2014)
TBA	8.8%	of total mass delivered	Sweden	Present study
	5 – 14%	of mass supplied	Sweden	(Eriksson et al., 2017)
	30%	of supplied bread loaves	Sweden	(Ghosh and Eriksson, 2019)
	10.4%	not specified	Germany	(EHI, 2011) apud. (Lebersorger and Schneider, 2014)
	12.5%[1]	of sales in cost price	Austria	(Lebersorger and Schneider, 2014)
Not specified	0.5 – 8%	of sales value	Unknown	(Holweg et al., 2016)
	30.6%	of total waste mass	Italy	(Cicatiello et al., 2017)
	9.4%[2]	of sales value	Norway	(Stensgård and Hanssen, 2016)
Restaurants	16%	of avoidable waste mass	Sweden	Present study
Schools	10%	of avoidable waste mass	Sweden	Present study
	12 %	of total waste mass	Italy	(Falasconi et al., 2015)
	3%	plate leftovers	Finland	(Silvennoinen et al., 2015)
Households	13%	of avoidable waste mass	Sweden	Present study
	32%	of purchase of bread	UK	(WRAP et al., 2011)
	13%	of total waste mass	Norway	(Stensgård and Hanssen, 2016)
	27%	of edible food waste mass	Norway	(Hanssen et al., 2016)
	13%[3]	of avoidable waste	Finland	(Katajajuuri et al., 2014)

[1] Bread and pastries.
[2] Fresh bakery products [3] Bakery and grain products.

Ismatov (2015), TBAs create a constant struggle between suppliers and retailers regarding the service level. Retailers, which have no costs for unsold bread, require shelves that are full most of the time, while the suppliers have an interest in maintaining their return levels as low as possible.

Many factors may be associated with the positive correlation between higher loss rates and TBAs. Under such agreements, retailers have no financial stake in unsold bread and thus have no incentive to minimise retail losses by optimising demand planning and ordering (Lebersorger and Schneider, 2014). Moreover, bread is one of the staple products that brings customers into the stores and therefore retailers endeavour to keep their bread shelves full, even though this generates waste from unsold products. On the supplier side, the incentive to keep the bread shelves full derives from the risk of losing free shelf space to competitors (Ghosh and Eriksson, 2019). Bakeries claim that they produce more to meet this demand than they would if they had control over the amounts delivered (Rohm et al., 2017).

Accurate demand forecasting is a complex task owing to the influence of numerous factors, e.g. weather, marketing campaigns, holidays and promotions. Although some waste is inevitable in the supply chain, bakeries believe that retailers could forecast sales more accurately since they have better control over events such as campaigns, stock and sales numbers. Rohm et al. (2017) claim that TBAs may increase waste due to the distribution routes and to the ordering being done by the bakeries, and not the stores themselves. For example, stores at the end of the distribution line often receive the remaining products in the distribution trucks, regardless of their needs, since the drivers, who also act as the bakery's sales representative, do not want to return fresh products (ibid.).

Freshness is reported to be particularly important for consumers who prefer bread to be as fresh as possible at the time of purchase (Lyndhurst, 2011; Østergaard and Hanssen, 2018). Since freshness is a major factor for consumers when buying bread, the bread stock is usually purchased in a last-in, first-out pattern, meaning that consumers are most likely to choose fresher products and leave old products unsold. Moreover, stores routinely remove products from their shelves up to three days before the best-before date and the majority of these products are wasted due to their proximity to this date. A TBA prevents retailers from selling TBA products close to their best-before date at a lower price, and when a consumer has the opportunity to buy a fresher product for the same price, they probably do so. In order to discount products close to their best-before date, they must be labelled in-store, which takes time for the seller and shelf space from the retailer. Moreover, promotions must be negotiated between suppliers and retailers, adding complexity to the process. The bakery company involved in the present study reported that initially there were concerns that promotions would decrease the sales of fresh products, but internal research has shown that this is not the case. According to Kulikovskaja and Aschemann-Witzel (2017), lowering the price of suboptimal products is a common and successful practice for shifting such products. However, promotions could transfer the waste from retailers to consumers, increasing the waste in households due to over purchasing (Brook Lyndhurst, ESA, 2012; Mena et al., 2011).

TBAs create a problem concerning ownership of the unsold products. Hence, in addition to the barrier to selling the products at reduced prices, TBAs also prevent any other action by retailers (e.g. donations to charity, reworking into new products or serving in staff canteens), since they do not own the products. On the positive side, the collection of waste bread by suppliers has the advantage of operating reverse logistics, enabling a clean flow of bread, i.e. not mixed with other food fractions. This provides opportunities for different waste treatment pathways such as ethanol production, production of animal feed, reworking and production of yeast, which are not always viable for a mixed waste fraction. These pathways are an alternative to incineration or anaerobic digestion, which are the most common disposal methods for mixed food waste in Sweden (Avfall Sverige, 2017). However, environmental assessment of alternative pathways is necessary to determine the potential benefits and impacts of each alternative.

The present study indicates that TBAs play an important role in bread waste generation. One can speculate that changing the existing business model for suppliers may be difficult for individual companies, so policy changes may be necessary to encourage several actors in the food chain to implement food waste prevention measures or alter their commercial practices. If the TBAs in the Swedish supply chain would be categorized as an unfair trade practice, legislation could be used to ban such systems, as has been proposed by the European Union (European Commission, 2018). The European Union defines “forcing the supplier to pay for wasted products” and the returning of unsold food products for the supplier, as unfair retailer practices. It is argued that retailers' concentrated market power enable such practices. The proposal, however, is aimed to “to ensure fairer treatment for small and medium sized food and farming businesses”, and it is possible to argue that the major bread suppliers in Sweden are large corporations with relatively high bargaining power to negotiate such agreements. Legislation can also be used to reduce the high levels of return waste by increasing taxes on such residual flow. Consumers are also fundamental in the efforts to reduce the amount of returned bread. As public indignation over food waste grows, suppliers and retailers will have to respond to such consumer pressure by means of collaboration. For instance, if retailers' point of sale (POS) data is shared with suppliers, more accurate forecasting of the demands for bread products can be achieved.

5. Conclusions

There was a negative correlation between sales and loss rates, suggesting that reducing the range of products has the potential to decrease waste. However, the coefficient of determination was low and the variable explained only 14% of the variation in bread loss rate. The other variables tested showed lower coefficients of correlation, indicating that other factors, such as take-back agreements, pose a risk for generating large amounts of waste. Shifting responsibility for unsold bread from bakeries to retailers could decrease waste levels by changing product ownership and increasing the incentive for retailers to optimise ordering and forecasting, and to implement waste-reducing actions, such as price reductions.

The bread loss rates found in this study fell within the range reported in the literature. The results highlighted the importance of individual, but especially joint, actions from different actors in the supply chain. Comparison of bread waste amounts at retailers and households revealed similar waste levels, which indicates that reducing unnecessary waste at retail has the potential to achieve significant decreases in the environmental, economic and social burdens associated with bread waste in the value chain. TBA products was the category with the highest waste generation in retail (15 510 tons/year). This can be compared with the bake-off products (9 588 tons/year) with much shorter shelf life or all waste generated in the bakery industry (12 040 tons/year). This should make the TBA a priority focus for waste reduction actions.

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Environmental impacts of waste management and valorisation pathways for surplus bread in Sweden

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ABSTRACT

Bread waste represents a significant part of food waste in Sweden. At the same time, the return system established between bakeries and retailers enables a flow of bread waste that is not contaminated with other food waste products. This provides an opportunity for alternative valorisation and waste management options, in addition to the most common municipal waste treatment, namely anaerobic digestion and incineration. An attributional life cycle assessment of the management of 1 kg of surplus bread was conducted to assess the relative environmental impacts of alternative and existing waste management options. Eighteen impact categories were assessed using the ReCiPe methodology. The different management options that were investigated for the surplus bread are donation, use as animal feed, beer production, ethanol production, anaerobic digestion, and incineration. These results are also compared to reducing the production of bread by the amount of surplus bread (reduction at the source). The results support a waste hierarchy where reduction at the source has the highest environmental savings, followed by use of surplus bread as animal feed, donation, for beer production and for ethanol production. Anaerobic digestion and incineration offer the lowest environmental savings, particularly in a low-impact energy system. The results suggests that Sweden can make use of the established return system to implement environmentally preferred options for the management of surplus bread.

1. Introduction

Food waste leads to a loss of the resources that are invested in the supply chain, such as water, energy, fertilizers, and land, which are used in the production, transport and storage of food products. These materials and processes have different environmental impacts, such as global warming, acidification, and eutrophication. Therefore, the loss of food results not only in the loss of the product itself, but also of all of the resources used in the supply chain. This problem has been recognized by the United Nations in the Sustainable Development Goals (United Nations, 2015). Specifically, the goal "sustainable consumption and production patterns" has a sub-target 12.3 that aims to halve food waste in the consumer and retail levels and to reduce food loss in the production and supply chains.

Bread waste is a large part of the global food waste. The Waste and Resources Action Programme (WRAP) has estimated that bread waste is 10% of all food waste generated in the United Kingdom (WRAP et al., 2011). Brancoli et al. (2019) have estimated that 80,410 tons of bread is wasted in Sweden each year, which is the equivalent of 8.1 kg per capita/year.

Besides the large quantities of bread waste, the distribution scheme for bread in Sweden and other countries such as Norway (Stensgård and Hanssen, 2016), Austria (Lebersorger and Schneider, 2014), the Netherlands (Weegels, 2010) and Germany (Brosowski et al., 2016), makes it an interesting product since it is not mixed with other food waste and can therefore be managed separately to other waste streams. The bread distribution is conducted within a full service scheme that involves a take-back agreement (TBA) between retailer and supplier. This means that the bakeries are responsible for ordering the bread that is supplied to the supermarkets, which includes the forecasting, placement, and removal of products from the supermarket shelves. More importantly, the bakeries are also financially responsible for the unsold products and its collection and treatment, thereby operating in a reverse supply chain (Eriksson et al., 2017; Ghosh and Eriksson, 2019). This reverse logistics enables a clean flow of bread, i.e. not mixed with other food fractions, and provides opportunities for different waste treatment pathways such as ethanol or animal feed production, which are not always viable for a mixed waste fraction. These pathways are an alternative to incineration or

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anaerobic digestion, which are the most common treatment methods for mixed food waste in Sweden (Avfall Sverige, 2017).

In order to comply with legal frameworks and environmental quality objectives set by governments, waste management systems have progressively become more complex (Brancoli and Bolton, 2019; Manfredi et al., 2011). The European Union have established the EU Waste Framework Directive (European Commission, 2008), which sets basic concepts and defines a priority for different waste management alternatives. According to this hierarchy, prevention of waste is the preferred option, followed by reuse, material recycling, energy recovery and disposal as the least preferred alternative. Nevertheless, the hierarchy does not always provide the best waste management alternative due to differences in local conditions such as energy supply mix and the waste treatment technology. The life cycle assessment methodology (LCA) is often used to support deviations from the waste hierarchy in a robust scientific way. LCA uses a holistic approach, which includes the whole life cycle of the material or process under study, thereby reducing the risk of shifting environmental impacts from one part of the waste management system to another (Hauschild and Barlaz, 2010; Manfredi et al., 2011).

Although previous studies have assessed the environmental impacts of different food waste management options (e.g. Albizzati et al., 2019; Khoo et al., 2010; Saer et al., 2013; Xu et al., 2015), few have studied systems dedicated to bread waste. Vandermeersch et al. (2014) studied the production of animal feed from bread waste and concluded that the valorisation of bread waste into animal feed was environmentally preferable in comparison with anaerobic digestion. Eriksson et al. (2015) assessed the environmental impacts of six different waste management options for different food fractions, and concluded that among the products studied, bread had the highest potential to reduce greenhouse gas emissions.

This study builds on the existing literature and contributes further by systematically assessing, using LCA, the environmental impacts associated with different options for managing bread surplus in Sweden. The goal of the LCA is to compare the following options: source reduction, donation, animal feed production, ethanol production, beer production, anaerobic digestion and incineration. Although the exact amounts sent to each treatment is unknown, the alternatives included in this study are the ones which are already implemented in Sweden or that would be possible to implement (Fazer, 2016; Polarbröd, 2016; Pågen, 2018). Currently, bread surplus is also used for yeast production in Sweden (Pågen, 2018), however, it was not included in this study due to lack of data. The environmental savings offered by these waste management schemes are also compared to reducing the production of bread by the amount of surplus bread. The relative environmental savings offered by the different waste management options and their comparison with waste prevention is compared to the waste hierarchy (European Commission, 2008).

2. Materials and methods

2.1. Definitions

The FUSION Project (Östergren et al., 2014) defines food waste according to its disposal or valorisation route. Food waste is defined as "any food, and inedible parts of food, removed from the food supply chain to be recovered or disposed (including composted, crops ploughed in/not harvested, anaerobic digestion, bioenergy production, co-generation, incineration, disposal to sewer, landfill or discarded to sea). The FUSIONS framework does not classify products that are converted to animal feed or bio-based materials as food waste. Products used in biochemical processing are not defined as food waste either. While acknowledging this definition, in this study all bread which is produced for human consumption and that is not consumed by humans is referred to as waste, regardless of its final disposal or valorisation route. The only discrepancy with the FUSIONS definition is when the bread is used to produce animal feed. We justify this classification by the fact that no bread is baked for animal feed, and that animal feed can be produced directly from wheat, without the need of the processing required for bread production. The use of wheat as animal feed is common, and although most wheat is grown for human consumption (with low quality and surplus wheat being redirected to produce feed), it is also grown for feed (Blair, 2018; Inra et al., 2016; Lalman and Highfill, 2011). Bread that is donated is, of course, not defined as waste. This study uses "surplus" as a general term to define both bread that is wasted or donated.

2.2. Goal and scope definition

The functional unit of the study is the management – or prevention – of 1 kg of surplus bread in Sweden. The production and disposal of packaging is not included because it is assumed similar in all treatment and valorisation scenarios assessed. As mentioned above, the surplus bread can be reduced at the source, donated, or sent to different waste management pathways, namely animal feed production, ethanol production, or beer production (Fig. 1). The return bread flow, described in Fig. 1, is the flow that is currently the most suitable for all the scenarios assessed in this study, since it is collected separately. Nevertheless, the remaining flows in the supplier-retailer interface could also be treated via the scenarios described in this study if the proper agreements between the actors are reached. These flows are in-store, which are products normally branded by the retailer, and are not submitted to take-back agreements, and bake-off, which are products baked from pre-made dough in supermarket stores or supplied by a bakery in a nearby store. The residual flow comprises the flows outside the supplier-retailer interface, which are not under take-back agreements and are usually disposed together with the other organic waste fractions and unfit for the scenarios in which segregated bread is necessary.

The system boundaries includes receiving the surplus bread and the further valorisation, waste management, or donation. The geographic scope is Sweden. The temporal scope is the waste management and valorisation pathways that are currently used, or possible to be used, in Sweden. The ecoinvent database version 3.5 was used for the background processes. The characterization methods are based on ReCiPe 2016 v1.1 midpoint and endpoint method, Hierarchist version (Huijbregts et al., 2016). The impact categories are presented in the Supplementary material in Table SM.1.

This study has the goal to compare the prevention of baking surplus bread and the different technologies for treatment and valorisation of surplus bread, and can be classified as an account study in the International Reference Life Cycle Data System (JRC, 2010). Therefore, this study uses an attributional approach and, consequently, average data for the upstream and downstream processes are used, including the product substitution.

2.3. Inventory of waste management and valorisation pathways

As shown in Fig. 1, seven scenarios were investigated in this study: source reduction, donation, animal feed production, ethanol production, beer production, anaerobic digestion, and incineration. The multi-functionality of the scenarios, i.e. the co-products and services, were treated using system expansion, so that the management options are credited by accounting for the environmental impacts of the substituted co-products (using the market average).

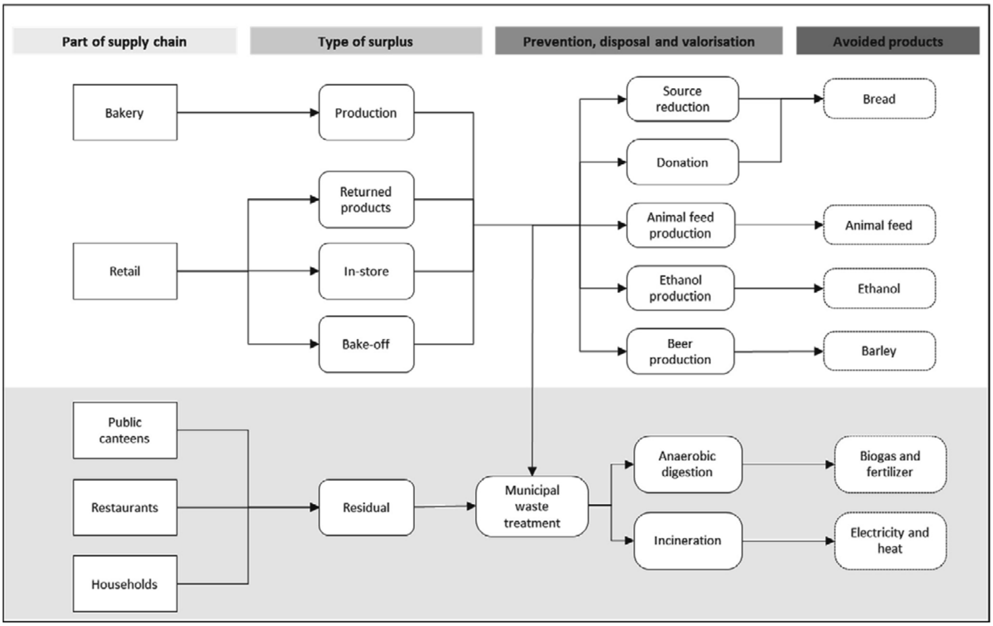

Fig. 1. Bread surplus flows, waste treatment and valorisation scenarios, and avoided products. The area highlighted in grey shows common flows of mixed food waste, which are not feasible for the management pathways highlighted in the white area.

Fig. 2 shows the system boundaries and the main activities for the waste management and valorisation scenarios considered in this study, including the avoided products in the system expansion.

2.3.1. Donation and source reduction

Donation systems are often complex and dynamic. This study uses a scenario to represent a hypothetical donation system, which is a simplification of the large variety of activities that take place in Sweden. The modelling includes transportation between the retail and the donation centre (40 km) and from the later to the households (5 km), based on Bergström et al. (2020). The bread losses before and at the donation centre was assumed to be 20%, similar to Bergström et al. (2020). The bread losses in households was estimated to be 32%, since this is this figure reported by WRAP for UK households (WRAP et al., 2011). This loss rate is representative for average consumers, and it might be different when considering donation recipients. The loss rate can be affect for example by the relative lower shelf life of donated products and by differences in the behaviour of people who purchase the bread or received it via donation.

The wasted bread was assumed to be treated by the average waste technologies used in Sweden for the organic waste fraction, described in Fig. 1 and Fig. 2b as ''Municipal waste treatment''. The ''Municipal waste treatment'' is composed by anaerobic digestion (40%) and incineration (60%), which are the most common methods for household waste treatment in Sweden (Avfall Sverige, 2017).

The modelling of source reduction is used in this study as a benchmark, since surplus bread should, in principle, not be manufactured. The maximum level of prevention is modelled, i.e. avoiding the production of one kg of bread.

2.3.2. Animal feed production

Feed production was modelled using a pig feed recipe from Sirtori et al. (2007), where some ingredients from the original recipe were replaced by bread waste. The recipes are described in Table SM.4 in the Supplementary Material. The recipe described in Sirtori et al. (2007), ensured that the feed with bread waste (bread feed) was equivalent to the original recipe (original feed) in terms of the conversion rate and chemical composition parameters such as gross energy, protein content, dry matter and crude fibre. Thus, it was ensured that the growth performance of the pigs was not affected by the substituted ingredients. Consequently, the total intake for each type of feed is different and was modelled accordingly. Sirtori et al. (2007) have reported higher average daily gain and back fat thickness for the animals fed with the bread feed. System expansion was used accounting for the fact that the bread feed avoids the production of the original feed.

The LCI includes the ingredients that are necessary for the production of the bread feed and the downstream processes. Ingredients with a contribution lower than 5% in mass were not considered. The valorisation of bread waste for feed production starts with the receiving of the bread, unpacking, crushing, mixing with the other ingredients and packing (Fig. 2c).

2.3.3. Ethanol production

The ethanol production was based on a typical Swedish ethanol plant modelled by Brancoli et al. (2017a). Data for the inventory was obtained by lab scale experiments (Ferreira et al., 2015; Nair et al., 2015), a pilot scale unit and computer simulation (Rajendran et al., 2016) done in Aspen Plus® (v.8.4) (Aspentech: Burlington, MA, USA).

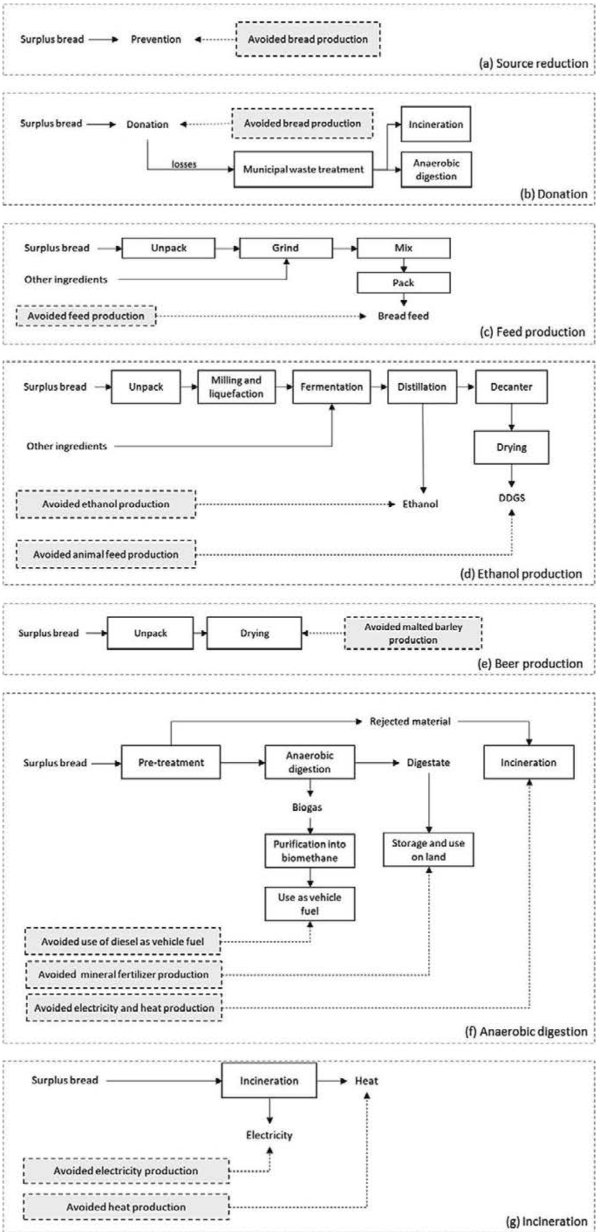

Fig. 2. System boundaries for prevention and the valorisation and waste management scenarios considered in this study. The dashed lines and dashed boxes represent the substitution of products in the market. (a) Shows the source reduction of bread. (b) The donation of the surplus bread. (c) The use of surplus bread as animal feed. (d) The valorisation of bread via ethanol production. (e) The production of beer. (f) Shows the anaerobic digestion process, and (g) the incineration process.

The ethanol production yield was based on data from Ebrahimi et al. (2008), where the separate hydrolysis and fermentation of waste wheat bread was studied. They obtained overall ethanol yield of 350 g per kg of bread. This value is similar to that obtained by Pietrzak and Kawa-Rygielska (2014), who reported a yield of 354.4 g of ethanol per kg of bread waste.

The production of ethanol has dried distillers grains with solubles (DDGS) as the main co-product, which is used as animal feed. The amount of replacement feed products was calculated using system expansion according to its energy and protein content. It was considered that the by-product (DDGS) was used as cattle feed and the digestible energy for ruminants was used (Inra et al., 2016). The digestible crude protein (dCP) was determined from the crude protein content by subtracting the unavailable protein content (Eq. (1)) as done in (Tonini et al., 2016). It was assumed that DDGS replaced average market for energy and protein feed, as described in ecoinvent database (Wernet et al., 2016). The biochemical parameters used for the calculations are presented in Table 1.

$$dCP\ \left(\mathrm{kg\ \ kg^{-1}\ \ DM}\right) = \frac{(0.93 * CP * 100 - 3)}{100} \quad (1)$$

where DM is dry matter and CP is crude protein.

2.3.4. Beer production

Beer is produced primarily from a cereal grain, hops and water. Barley is the most common cereal that is used. Recently, small breweries have started to use surplus bread in their recipes, substituting part of the malted barley, originally used as a source of sugar for fermentation (Almeida et al., 2018; Connolly, 2019). Two recipes were analysed in this study, one from Almeida et al. (2018) and another from a brewery in the United Kingdom (UK) which made their recipe public (Toast Ale, 2020). In both recipes, 25–28% of the original malt was substituted with dried bread. The remaining ingredients were assumed the same as those used in the production of the standard beer (Almeida et al., 2018).

It was assumed that the energy consumption for the drying of bread is similar to the drying of the malted barley. Therefore, for simplification and assuming that the beer produced with bread surplus will avoid the production with malted barley, the only process that was modelled was the substitution of barley by bread. The changes in weight of barley when malted is 0.807 kg malt/kg barley based on Briggs (1998).

2.3.5. Anaerobic digestion

The anaerobic digestion was based on the inventory from a generic plant treating organic waste and manure that was sorted at the source. The characteristics and the inventory analysis of the plant is shown in Table SM.2 in the Supplementary Material. The biogas potential of bread was based on Dubrovskis and Plume (2017) and calculated as 0.6 N m^3 kg^{-1} VS. The yield was set to 70%, which means that the biogas generated was 0.42 N m^3 kg^{-1}. Based on Dubrovskis and Plume (2017), the biogas composition is 48% methane and 52% carbon dioxide. It was assumed that 2% of methane is emitted as gas leakage from the digester (Andreasi Bassi et al., 2017; Møller et al., 2011). Pre-treatment is necessary due to the presence of the bread packaging. Based on studies done by Brancoli et al. (2017b), it was assumed that 44% of the bread waste is lost in the pre-treatment process. The losses in pre-treatment can vary between different plants depending on the technology employed. According to Bernstad et al. (2013), 2–45% of incoming wet waste is lost as refuse and Møller et al. (2011) indicate losses of 41% in wet weight or 55% in relation to volatile solids. The material that is rejected in the pre-treatment process is sent to incineration. The model for incineration is similar to the model described in Section 2.3.6, with the difference that it takes into account the increase in the water content of the bread after the pre-treatment.

The biogas that was produced was upgraded to methane in order to be used as vehicle fuel. This is the most common use of biogas in Sweden (Larsson et al., 2016). The details on the use of upgraded biogas and digestate are described in the Supplementary Material.

Table 1
Biochemical composition for relevant parameters of the by-products and replacement products (Inra et al., 2016).

Biochemical parameter	Unit	Dried distillers grains with solubles
Crude protein	% DM	37.3
Digestible crude protein	% DM	34.7
Gross energy	MJ/kg DM	20.5
Digestible energy[a]	MJ/kg DM	16.1

[a] Ruminant digestible energy.

2.3.6. Incineration

The waste-to-energy process was adapted from the ecoinvent process "*treatment of biowaste, municipal incineration with fly ash extraction*" which represents the incineration of a mixture of garden and food waste (Wernet et al., 2016). For simplification, the emissions to air, water and land used the model for mixed organic waste, based on the waste composition and transfer coefficients.

The energy production was modelled according to the calorific value of bread. The higher heating value (HHV) for bread was calculated as 14.84 MJ/kg, which is derived from the correlation shown in Eq. 2 (Eboh et al., 2016). The elemental compostion of bread was based on Tonini et al. (2018). The lower heating value (LHV) was calculated as 13.44 MJ/kg dry matter, and 10.35 MJ/kg as received.

The net thermal efficiencies for heat and electricity based on the LHV was modelled as 59% and 19% respectively, which is in accordance with the base-line plant in Eboh et al. (2019). The net energy production was thereby estimated as 1.94 MJ/kg of electricity and 6.10 MJ/kg of thermal energy. The produced heat and electricity was assumed to substitute the Swedish mix for electricity and heat, described in Section 2.3.7.

2.3.7. Energy modelling

The results of an LCA are usually strongly dependent on both the energy used and avoided in each process. Electricity was modelled as the average mix of technologies used to produce and distribute the electricity in Sweden according to the ecoinvent database (Wernet et al., 2016).

The modelling of the mixed technology for heat is based on the heat statistics published by the European Commission (2019), where heat production is divided into six categories, namely solid fossil fuels, oil and petroleum products, natural gas and manufactured gas, renewables and biofuels, wastes non-renewable, and other. Each fuel category was further divided into specific fuels, as shown in Table 2. When the specific fuel was not identified from the European Commission (2019) statistics, data from Statistics Sweden (2019) was used. This is similar to the approach described by Andreasi Bassi et al. (2017). The detailed description of the modelling of the energy systems are described in Section 2.4 of the Supplementary Material.

Fuels that contributed less than 5% to its fuel category were not considered. Table 2 shows the gross heat production by major fuel groups and the specific fuel composition of the major fuel groups in 2013. This data is based on the energy statistical country datasheets from Eurostat (European Commission, 2019) combined with the ecoinvent processes (Wernet et al., 2016).

Table 2
Correspondence between the fuel in the Eurostat data (European Commission, 2019) and the processes imported from ecoinvent (Wernet et al., 2016).

Fuel	Process from ecoinvent	Average mix heat
Solid fossil fuels		
Hard coal	Heat and power co-generation, hard coal (SE)	3.9%
Oil and petroleum products	Heat and power co-generation, oil (SE)	1.3%
Natural gas and manufactured gas		
of which natural gas	Heat and power co-generation, natural gas, conventional power plant, 100 MW electrical (SE)	1.8%
others	Treatment of blast furnace gas, in power plant (SE)	3.1%
Renewables and biofuels		
Solid biofuels and renewable wastes	Heat and power co-generation, wood chips, 6667 kW, state-of-the-art 2014 (SE)	71.2%
Ambient heat	Air-water heat pump 10 kW heat production (Europe without Switzerland	6.1%
Wastes non-RES	Heat, for reuse in municipal waste incineration only market for (SE)	12.7%

2.3.8. Transportation

The different valorisation or waste management schemes studied in this work require different infrastructures, e.g., plants for manufacturing ethanol or beer and donation centres. The availability of these infrastructures are different for the various valorisation or waste management options. For instance, the infrastructures for anaerobic digestion and incineration are present in all regions in Sweden, while the number of ethanol plants and animal feed producers are lower and located in specific regions. For this reason, the transportation distances for anaerobic digestion and incineration are expected to be lower than those for ethanol or feed production.

Transportation of the surplus bread to the valorisation or waste management site was not included in this study. Instead, a threshold for maximum distances that bread could be transported to a certain valorisation or waste management before the environmental benefits of the specific valorisation or waste management option would be lost was calculated.

3. Sensitivity analysis

A sensitivity analysis was performed based on a scenario analysis in order to investigate the influence of assumptions and sensitive input parameters to some of the results (Clavreul et al., 2012).

Different products may be substituted by the surplus bread when using it to produce animal feed. It is possible that the bread substitutes single products, such as barley or soymeal, or that it substitutes different ingredients in a feed recipe. Therefore, a sensitivity analysis was performed using scenarios where different feed products were substituted. The substitution was based on the digestible crude protein and digestible energy, similar to what was done for the DDGS in ethanol production. One of the scenarios used an ecoinvent average market for protein and energy feed (market for feed ecoinvent) (Wernet et al., 2016) and the second considers barley as the energy feed and soymeal as the protein feed. Moreover, two recipes with different replacement rates of bread from Kumar et al. (2014) was also assessed. Due to the functional unit of this study, namely 1 kg of surplus bread, and since the percentage of the replacement by bread in each recipe varied, the amount of feed produced per kg of bread also varied. For instance, in Kumar et al. (2014), between 25% and 50% of the ingredients were substitute by bread resulting in the production of 4.0 kg and 2.0 kg of feed per functional unit respectively. The detailed descriptions of the scenarios are available in Section 2.2 of the Supplementary Material.

The sensitivity analysis for anaerobic digestion used a scenario where the biogas was used in stationary engines for electricity and heat production. The emissions were based on Nielsen et al. (2010). In this scenario, it is not necessary to upgrading the biogas.

For energy production systems, namely anaerobic digestion and incineration, the electricity and heat consumed and produced have a significant impact on the results. Therefore, two scenarios were modelled for the source of electricity and heat for the foreground processes - one with a clean source and another with a dirty source. The energy sources were chosen from the Swedish energy mix. The dirty technology that was chosen was hard coal for heat and electricity. The clean energy was modelled with wind for electricity production and wood chips for heat. A sensitivity analysis for ethanol production included a scenario which assumed that the ethanol substituted petrol as vehicle fuel and the DDGS replaced soybean meal and barley.

4. Results

The results from the LCA are the potential environmental impacts, and not a precise prediction of absolute values for each impact category.

Fig. 3 shows the characterization results for each impact category. The environmental impact for each valorisation or waste management option is shown in relation to the best result in each impact category, which is arbitrarily given an impact of −100%.

The trend seen by the results in the eighteen impact categories supported the waste hierarchy: source reduction of bread waste is the preferred option followed by feed production, donation, beer production and ethanol production. There is no clear preference between these four latter valorisation pathways. The worst waste management options, with the exception of four impact categories (ionizing radiation, marine eutrophication, land use and water consumption), are anaerobic digestion and incineration, which are the most common waste management schemes in Sweden. Source reduction has the highest environmental savings in sixteen impact categories (all except ionizing radiation and fossil resource scarcity).

The waste management and valorisation pathways yield environmental savings in most of the impact categories. The environmental savings from source reduction come in its majority from the avoided production of wheat, similarly to donation and ethanol production. Moreover, source reduction has a relative higher environmental performance in comparison with the other scenarios because it does not have any environmental impact associated with downstream processes, such as transportation, or waste treatment and valorisation pathways. In terms of global warming (GW), preventing the production of surplus bread avoids the emissions of −0.66 kg CO_2 eq per kg of bread. Donation is similar to source reduction, although transportation and losses were included in this scenario. The majority of the environmental impacts during ethanol production come from crops used as feedstock, and the environmental savings determined in the present study are obtained when these crops, modelled as wheat, are substituted by bread waste. This resulted in a net savings of −0.56 kg CO_2 eq per kg of bread valorised as ethanol. The environmental savings from feed production come from the substitution of ingredients, such as maize and barley, by surplus bread and result in an avoided burden of −0.53 kg CO_2 eq per kg bread. The savings observed from beer production, e.g. −0.46 kg CO_2 eq in the GW category, are a result of the substitution of malted barley by surplus bread.

The relatively poor performance of anaerobic digestion is due partially to the amount of material that is rejected in the pretreatment. For the GW category, it resulted in a savings of −0.02 kg CO_2 eq per kg of bread. Anaerobic digestion has the high-

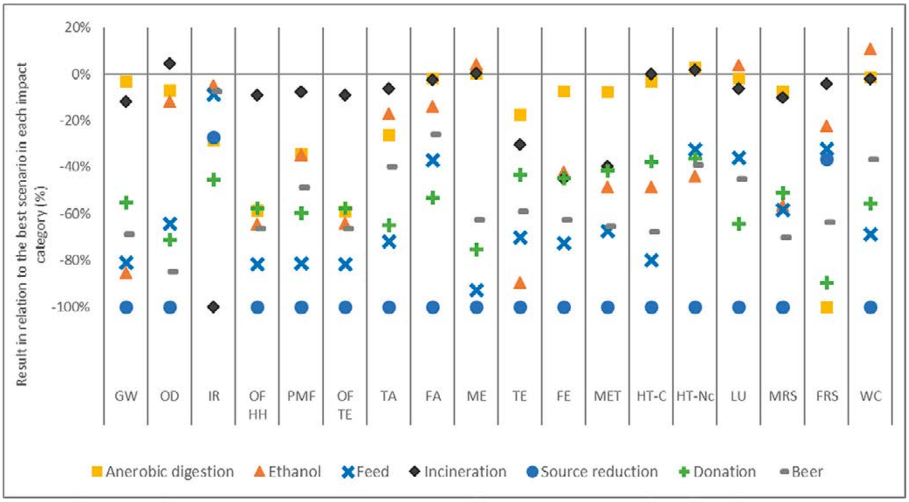

Fig. 3. Characterized results per kg of bread for the 18 impact categories analysed in relation to the best scenario in each category: Global warming (GW), Stratospheric ozone depletion (OD), Ionizing radiation (IR), Ozone formation, Human health (OF, HH), Fine particulate matter formation (PMF), Ozone formation, Terrestrial ecosystems (OF, TE), Terrestrial acidification (TA), Freshwater eutrophication (FA), Marine eutrophication (ME), Terrestrial ecotoxicity (TE), Freshwater ecotoxicity (FE), Marine ecotoxicity (MET), Human carcinogenic toxicity (HT-C), Human non-carcinogenic toxicity (HT-Nc), Land use (LU), Mineral resource scarcity (MRS), Fossil resource scarcity (FRS), Water consumption (WC).

est environmental savings in the fossil resource scarcity impact category due to the substitution of diesel by the upgraded biogas produced.

Electricity and heat production in Sweden have relatively low environmental impacts, which results in lower credits for the substituted electricity and heat studied here. Therefore, the low credits from substituted energy and the relatively low heating value of bread yields a small environmental savings for incineration (e.g. -0.08 kg CO_2 eq per kg of bread in the GW category). The detailed results for the characterization step in Fig. 3 is described in Section 3.1 of the Supplementary Material.

The valorisation and management pathways for surplus bread can be divided into three clusters with regards to their environmental savings. Source reduction of surplus bread has the highest performance in the majority of the impact categories. The next cluster includes donation as well as ethanol, beer and feed production. After source reduction, these pathways offer the highest environmental savings, although the relative value of the savings varies between the different impact categories. The cluster with the lowest environmental savings includes incineration and anaerobic digestion, which had the smallest savings in more than 80% of the impact categories.

The results from the different impact categories cannot be compared since they have different units. It is also not easy to ascertain the relative importance of the categories, and hence it is difficult to conclude which valorisation or management pathway offered the most environmental savings (across all impact categories). Although the use of weighting is controversial due to the underlying judgments, the method was used here to simplify and to facilitate analysis of the results. Fig. 4 shows the weighted results, where the results from the different impact categories are aggregated into a single score based on ReCiPe weighting sets (Goedkoop et al., 2009). The weighted results supports the hierarchy of source reduction as the best practice, followed by the cluster containing feed production, donation, beer production and ethanol

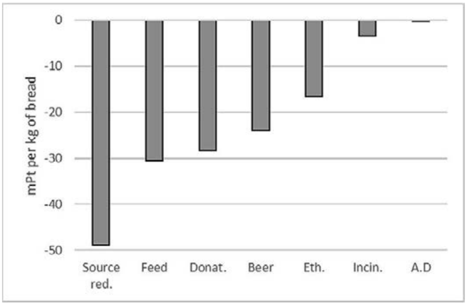

Fig. 4. Weighted results in milli points (mPt) per kg of bread for the scenarios analysed: source reduction (Source red.), feed production (Feed), donation (Donat.), beer production (Beer), ethanol production (Eth.), incineration (Incin.) and anaerobic digestion (A.D).

production and the final cluster of anaerobic digestion and incineration that offer the lowest environmental savings.

Besides the scenarios of prevention, namely source reduction and donation, it is possible to aggregate the results for the management of surplus bread in two groups based on the availability of infrastructure and the environmental savings offered by the scenario. The first group contains ethanol, beer and feed production, which have larger environmental savings than anaerobic digestion and incineration but at the same time, having limited available infrastructure in Sweden, and consequently might require long transportation distances. The second group includes anaerobic digestion and incineration, which have widely available local infrastructure but comparatively lower environmental savings.

Thus, assuming that the current infrastructure persists, it is important to define the threshold on the maximum transport distance that would still allow the management options in the first group to be superior. The calculation was done by comparing the two closest management scenarios in each group, i.e. incineration from the second group and ethanol production from the first group, using the weighted values (see Fig. 4). The calculations show that the surplus bread can be transported an additional 730 km, including return journeys, to the ethanol facilities before this valorisation pathway is no longer preferred to incineration. Of course, the threshold distances are longer for the other valorisation pathways and when compared to anaerobic digestion.

4.1. Sensitivity analysis

The results for the scenarios analysed in the sensitivity analyses are summarized in Fig. 5, and extend the results for the base scenarios, which are also show in Fig. 4. The characterization results are described in the Section 3.2 of the Supplementary Material. This is an attributional study and, as pointed out by Andreasi Bassi et al. (2017), the sensitivity analysis results should be used to better understand the current situation and not to assess potential impacts of future choices.

The highest percentage variation in the results was when changing the type of energy that was substituted during anaerobic digestion and incineration. When dirty energy is substituted by energy recovery from the system, there are larger environmental benefits, and *vice versa* for clean energy. The change in the energy mix from clean to dirty affects the results significantly. This indicates that the actual substitution that takes place in the market is critical for assessing the environmental benefits of using anaerobic digestion or incineration for bread waste treatment. Feed and ethanol production were less sensitive to the scenarios that were assessed. For feed production, the market average described in the ecoinvent database (Wernet et al., 2016) resulted in the lowest benefit comparing to the other feed scenarios assessed.

5. Discussion

Bread waste arises in different parts of the supply chain (Fig. 1). These flows are more or less suitable for the different waste management or valorisation practices. For example, returned bread, which is not contaminated with other food waste fractions, is a flow that is suitable for the majority of the waste management and valorisation options. In contrast, household waste, which is a mixed flow of different organic fractions, is suitable for less alternatives and is treated as a common municipal waste fraction. The most common methods for treating mixed food waste in Sweden are anaerobic digestion and incineration (Avfall Sverige, 2017), and these are the options studied in this work (Fig. 1).

The results from this study indicate that the return system, which is already implemented in Sweden, can be used to explore segregated waste management and valorisation pathways for surplus bread. These include ethanol and feed production, which have higher environmental savings compared to the municipal waste treatment methods. For example, the weighted results in Fig. 4 show that producing animal feed impacts the environment 8 and 160 times less than incineration and anaerobic digestion, respectively.

Surplus bread has been historically used as feed, and bread is characterized as a low-risk product provided it does not contain products from animal origin. Therefore, it is essential to ensure proper waste separation to avoid contamination with animal by-products. The main concerns with the use of surplus bread as animal feed are the moisture content and the nutrient variability. The latter risk can be minimized by introducing bread waste as part of a feed recipe (Inra et al., 2016). Besides the environmental benefits, selling the surplus bread as feed or as an ingredient to feed producers can provide an additional revenue stream to the bakeries and retailers. Moreover, it can reduce the costs associated with the waste treatment. Barriers to using surplus bread as animal feed

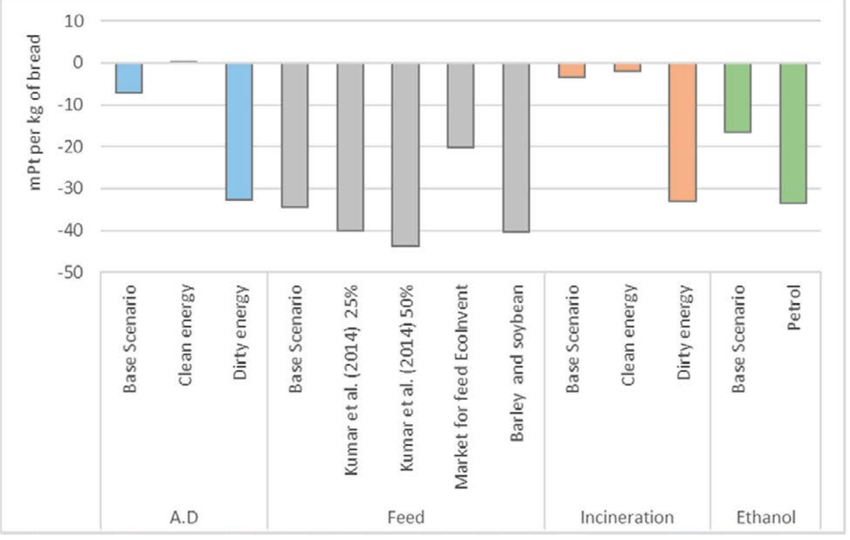

Fig. 5. Weighted results in milli points (mPt) per kg of bread for the scenarios assessed in the sensitivity analyses: anaerobic digestion (A.D), animal feed production (Feed), incineration, and ethanol production (Ethanol).

are identifying suitable selling channels for the products and complying with legislative requirements, particularly when the core business of bakeries and retailers are not the production of animal feed. The regulations are aimed to control the quality and safety of the feed produced, in order to protect animal health (Parfitt et al., 2016). It includes, among other things, registering the seller as a Feed Business Operator with their local authority, ensuring proper control of chemical or physical contamination with appropriate facilities and processes, and ensuring traceability processes that cover production and distribution.

A policy brief from the European Union (Hirschnitz-Garbers and Gosens, 2015) states that waste-based production of bioethanol can help mitigate environmental impacts and the competition between energy and food crops. The former claim is in agreement with the findings of this study. The document also states that one of the major challenges is the organisational effort and logistics for obtaining the waste. Bread waste is a good feedstock since the logistics of returns are already in place. Brancoli et al. (2019) estimated that 40,240 tons of bread is wasted each year by bakeries and retailers in Sweden. This is a waste flow that already has a logistics system in place and can thereby easily be used for bioethanol production. The potential of ethanol production from bread waste is estimated to be 12,000 tons per year, corresponding to approximately 8% of the Swedish annual production (SEA, 2019).

Beer production is an alternative to increase the diversity of infrastructure for bread valorisation pathways and consequently to decrease the transportation distances required. Moreover, microbrewery business models, such as the Toast Ale in the UK (Toast Ale, 2020), is a growing sector that requires relatively lower investments in infrastructure compared to the other scenarios. Donations are important from the environmental perspective and in the reduction of food waste, yet its relevance is demonstrated through its social implication on food security. However, due to the high amount of bread surplus in Sweden (Brancoli et al., 2019), it is unlikely that donations would be sufficient to solve the issue. Therefore, it is necessary to develop strategies on how to avoid surplus bread to be produced in the first place. Source reduction can be achieved, as pointed out by (Brancoli et al., 2019), through changes in the distribution system for bread, as products that are sold under take-back agreements have higher loss rate in comparison with products that are not governed by such agreements. However, such changes might require policy changes to encourage the actors involved to implement such actions (Brancoli et al., 2019). Anaerobic digestion and incineration have the advantage of being well-established technologies in Sweden. These processes can stabilize and convert different types of organic products to valuable products such as biogas and fertilizer from anaerobic digestion, and electricity and heat from incineration. However, as highlighted in this study, the uncontaminated flow of bread waste should be used at higher levels in the waste hierarchy due to the increased environmental benefits.

As shown above, differences in the availability of infrastructure for the waste management scenarios studied here might influence the environmental savings. Transportation distances must be taken into account when using the results from this study for a specific case. The results presented here estimate that surplus bread can be transported 730 km further to plants that produce animal feed, ethanol or beer that to plants for anaerobic digestion or incineration. If longer transportation distances are required then these options lose their benefits. It is important to assess specific cases individually. The transportation distances can be even more critical when selecting between the option to produce feed, beer or ethanol. A limitation of this study is the exclusion of the packaging material in the analysis. However, it has no influence in the treatment and valorisation scenarios assessed, since the production and disposal of the material is the same for all scenarios. The exception is source reduction, which would benefit from the avoided production of packaging. A previous study (Brancoli et al., 2017b) shows that packaging contributes around 10% in the results of the impact assessment in the life cycle of bread, depending on the impact category assessed. Nevertheless, this would not change the conclusions of this study regarding the ranking of the scenarios.

The results presented here can be compared with previous studies. For example, Eriksson et al. (2015) have done an LCA, limited to the global warming potential impact category, where different options for managing bread waste were compared. They found that incineration has larger environmental savings than donation, which in turn, has larger savings than anaerobic digestion. In contrast, the present study found greater environmental benefits for donation than for incineration and anaerobic digestion. This is also true when only considering the GW impact category, as in Eriksson et al. (2015). This difference can be explained by the choice of substituted products for anaerobic digestion and incineration. For instance, the incineration scenario in Eriksson et al. (2015) assumed that the energy produced by bread waste substituted fossil peat. As expected, these results are similar to those obtained here when the surplus bread substitutes dirty energy (see the sensitivity analysis discussed with respect to Fig. 5). The results of this study are also in agreement with Brancoli et al. (2017b), who concluded that using bread waste as animal feed has higher environmental benefits in comparison with anaerobic digestion.

6. Conclusion

The results from the life cycle assessment conducted here indicate a clear hierarchy for the valorisation and management of surplus bread. Source reduction, donation, or production of ethanol, beer or feed are favoured over anaerobic digestion and incineration. These two least preferred methods are currently used for treating municipal food waste in Sweden. Shifting from the current waste management pathways to the more environmentally friendly schemes can lead to environmental savings of 0.56 kg CO_2 eq. kg^{-1} surplus bread in the global warming potential category.

The current distribution system for bread in Sweden, where surplus bread is not mixed with other food waste fractions, facilitates implementation of the more environmentally friendly valorisation options studied here. Hence, two of the largest barriers for implementing these valorisation options – organisation effort and proper logistics – have already been overcome. Although the results presented in this study are valid for Sweden, they are also applicable to other countries, observing the discrepancies described in the sensitivity analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.wasman.2020.07.043.

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Paper IV

fermentation

Article

The Use of Life Cycle Assessment in the Support of the Development of Fungal Food Products from Surplus Bread

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Abstract: The use of food waste as feedstock in the manufacture of high-value products is a promising avenue to contribute to circular economy. Considering that the majority of environmental impacts of products are determined in the early phases of product development, it is crucial to integrate life cycle assessment during these phases. This study integrates environmental considerations in the development of solid-state fermentation based on the cultivation of *N. intermedia* for the production of a fungal food product using surplus bread as a substrate. The product can be sold as a ready-to-eat meal to reduce waste while generating additional income. Four inoculation scenarios were proposed, based on the use of bread, molasses, and glucose as substrate, and one scenario based on backslopping. The environmental performance was assessed, and the quality of the fungal product was evaluated in terms of morphology and protein content. The protein content of the fungal food product was similar in all scenarios, varying from 25% to 29%. The scenario based on backslopping showed the lowest environmental impacts while maintaining high protein content. The results show that the inoculum production and the solid-state fermentation are the two environmental hotspots and should be in focus when optimizing the process.

Keywords: life cycle assessment; *Neurospora intermedia*; bread; process development

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1. Introduction

In food production, extensive resources such as water, land, energy and nutrients are used with significant impact to the environment. It is forecasted that by 2050, the demand for agricultural products will increase by 35–50% due to a growing population and rising incomes, leading to higher pressure on the environment [1]. The problem is aggravated by the significant wastage of food throughout the supply chain. The wastage of food represents not only the loss of the product itself, but also all the resources used in food production, transportation and packaging.

The provision of animal-based protein generally has a higher environmental impact compared with vegetable or fungi-based proteins [2]. Concerns regarding the ethical and environmental implications of meat consumption have increased the demand for meat substitutes, such as those based on legumes and fungi [3]. Recently, the use of filamentous fungi as a commercial food product has gained considerable attention, due to its high protein content, the presence of essential amino acids and easy digestibility [4,5]. Filamentous fungi have traditionally been used by different societies as food, e.g., tempeh and oncom are indigenous staple foods in Indonesia [6,7]. One example of a product currently sold in the European market is Quorn™, which is made from the filamentous fungus *Fusarium venenatum* cultivated in synthetic medium with glucose, ammonium and mixed with egg albumen [8]. One problem with this product is the high cost of the cultivation medium [4]. Moreover, Smetana et al. [9] performed an environmental assessment of a similar product based on mycoprotein cultivated in molasses and concluded that the substrate is an environmental hotspot in the system. Therefore, it is relevant to consider other types of substrates to decrease production costs and environmental impacts.

One alternative substrate for fungal cultivation is surplus bread. The valorisation of food surplus into high-value products has gained interest as part of the strategies of transition to a bio-based economy grounded on a sustainable use of resources. Bread is rich in carbohydrates and its porosity makes it an ideal substrate for fungal cultivation [6,10]. Moreover, bread is a product with high waste generation, particularly at the level of the consumer and at the supplier-retailer interface. In the UK, it was estimated that 10% of all food waste are bakery products [11]. Brancoli et al. [12] studied food waste from retailers in Sweden and concluded that bread is a product category with large environmental impacts and economic costs [12]. In Sweden, the quantity of bread waste generated was estimated to be 80,410 tons/year, of which 28,200 tons/year are generated at the retail level [13]. The large waste volumes of uniform composition and quality, and the distribution model in which bread is sold makes it a good substrate for manufacturers who strive for stable delivery of raw materials.

There are mainly two different categories of bread, bake-off and pre-packaged. The technology proposed in this article can be used for both categories. Bake-off are products baked from pre-made dough in supermarkets or by a bakery. Pre-packaged bread is produced by the bakeries, and in some countries, it is often sold to retailers under takeback agreements (TBAs) [13–15]. In such agreements, the bakeries collect the bread that is not sold in the supermarkets, and here, bread waste is segregated from other food waste fractions from the retailers. This enables alternative pathways for the waste management and valorisation of the unsold bread. These pathways should preferably be located at higher levels in the food waste hierarchy, such as the production of fungal biomass for food, ethanol or animal feed [16,17]. Nevertheless, it is necessary to ensure that such technologies are sustainable and that they minimize the use of natural resources and the generation of waste as well as emissions of pollutants over their life cycles.

The life cycle assessment (LCA) methodology can be used to assess the environmental performance of a product to ensure its sustainability. The methodology can also be applied to support the early design stages in the development of the product by assessing the implication of design choices on the environmental performance of the technology [18]. Such studies are crucial since they can identify and prevent environmental impacts before the technology has been embedded by path dependency and lock-in [19]. Decisions made in the early stages of product development can have a significant influence on its subsequent environmental impacts. McAloone and Bey [20] estimated that 80% of the environmental performance of a product is determined by decisions made in the early stages of the technology development. For this reason, the European Union [21] recommends the use of LCA in product development to guarantee its sustainability.

Research on life cycle assessment of valorisation of bread waste is primarily focused on the lower stages of the food waste hierarchy. Few studies address bread waste valorisation into high-value products. Moreover, most of the research focuses on technologies with high technology readiness levels. Previous studies have investigated the environmental performance of bread waste management and valorisation alternatives, such as anaerobic digestion and incineration [22], conversion into biofuels [17], animal feed [17,23], and value-added chemicals [24]. There is a large body of research on the development of new technologies to valorise bread to high value products [6,25,26], but most of this research does not integrate lifecycle thinking into the early stages of the process development, increasing the risk for sub optimization of the technology.

The aim of this study was to integrate environmental considerations in the early stages of the development of a solid-state fermentation (SSF) process using bread as feedstock for the production of a protein-rich food product. The SSF process is proposed to be implemented as a stand-alone business, or on-site in small-scale bakeries to recover their otherwise discarded surplus bread. The technology can also be implemented in supermarkets that have a fresh bakery department in house, i.e., supermarkets that bake their own bread. The food product can be sold as a ready-to-eat meal directly to customers to reduce waste while generating additional income from the fungal product and provide a new sustainable food alternative that can replace other less sustainable options.

The goal of this study is to support the technology developers during the design and development phases by assessing the environmental sustainability of a range of possible scenarios in which the technology can operate. This will enable a better understanding of the relation between design choices and the environmental performance of the technology, as well as allow the steering of the technology towards solutions with a lower environmental impact. Moreover, research on fungal growth patterns in solid-state cultures are limited and this study investigates whether morphological differences of the inoculation culture influence the performance in the subsequent SSF step.

2. Materials and Methods

The process development was iterative and used the concept of proxy technology transfer-process [27], which considers that emerging technologies are not often based on completely new processes. Instead, it relies on a new application of existing technologies. The development of the process started with the assessment of an incumbent technology, namely the production of mycoprotein cultivated in molasses to identify hotspots in the system. The first results were then communicated to the researchers involved in the process development, starting an iterative approach wherein the hotspots in the system are identified and alternatives were proposed and assessed. In total, four different scenarios for the inoculation were proposed and compared in relation to the environmental performance and characteristics of the final product, such as protein content and morphology. A detailed description of the scenarios is available in Section 2.1.

2.1. Process Description

The basic steps in the process are the drying and grinding of the bread, mixing with water and the starter spores (inoculum), and fermentation. The later processes after fermentation, such as frying, and freezing for later consumption are excluded in the study (Figure 1).

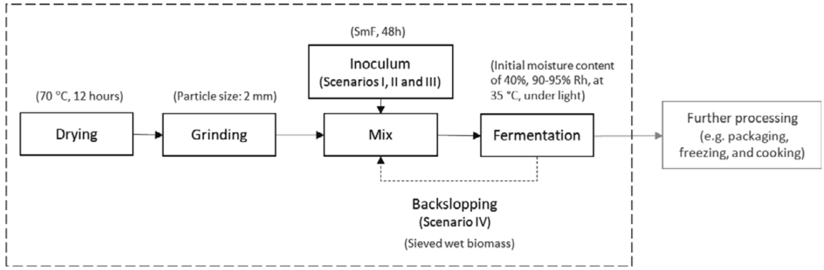

Figure 1. Basic process flow chart. The conditions of each process are described in brackets. The dotted box represents the system boundaries of the study. The further processing of the fungal biomass after fermentation is not included in the system boundaries.

2.1.1. Substrate and Fungus

Steinbrenner & Nyberg (Mölndal, Sweden) kindly provided unsold fresh sourdough bread that showed no signs of mould contamination. The bread was prepared according to the process described in Gmoser et al [25]. Samples were autoclaved and stored in airtight containers at room temperature prior to use. The edible food grade filamentous fungus *Neurospora intermedia* CBS 131.92 (Centraalbureau voor Schimmelcultures, Utrecht, The Netherlands) was used. The fungus was maintained on potato dextrose agar (PDA) plates containing 20 g/L glucose 20, 15 g/L agar and 4 g/L potato extract. The PDA plates were prepared by incubation for 5 days at 30 °C followed by storage at 4 °C. The spore solution was prepared by flooding each plate with 20 mL of distilled water, using a disposable plastic spreader to gently release the spores and transferring them to a sterile slant tube. The spore solution (2.5×10^6 spores/mL) was used to prepare the four different inoculations (one for each of the scenarios). A detailed description of the conditions of fermentation is given in Section 2.1.3.

2.1.2. Design of the Inoculation

The choice of the nutrient source was based on existing technologies. The inoculum for Scenario I is produced in a similar way to the inoculum production for tempeh as described in Wiloso et al. [28], but instead of using white rice or other grains, it is based on stale bread. Production of the inoculum for Scenario II, which is based on a similar process to cultivate yeast [29], used a synthetic medium based on molasses and ammonium chloride. Scenario III is based on glucose, and it was adapted from Jungbluth et al. [30], which describes the production of mycoprotein for Quorn™ (Table 1).

The three submerged fermentation (SmF) processes (Scenarios I, II and III) were inoculated with 10 mL/L of the spore solution, cultivated aerobically for 48 h in cotton plugged 250 mL Erlenmeyer flasks containing 100 mL of medium autoclaved at 121 °C for 20 min. The initial pH of the medium was adjusted to 5.5 with 2 M HCl or NaOH. The different medium compositions are summarized in Table 1. A water bath (Grant OLS-Aqua pro, Cambridge, UK) was used to maintain the temperature with orbital shaking at 125 rpm (radius of 9 mm) for 48 h. At the end of the fermentation, the wet biomass from each shake flask was harvested by pouring the cultivation medium through a 1 mm² pore area fine mesh and washed with distilled water. The liquid fraction was collected separately and frozen for further analysis. The moisture content of the fungal biomass was measured gravimetrically and sieved wet biomass was used directly as inoculum for the solid-state fermentation step.

Scenario IV relies on the recirculation (backslopping) of part of the fermented product from a previous batch as the inoculum. The solid-state fermentation (SSF) on stale bread used as inoculum was prepared according to the SSF process below, except that 0.5 mL of spore solution was used as inoculum instead. Based on an assumption that the fungal fermented final product contained 35% fungal biomass, with a moisture content of 60%, in the substrate-fungal matrix after 5 days of solid-state fermentation [25], the amount needed for backslopping was set to 0.01% dry weight of the substrate.

Table 1. Inventory analysis for the different scenarios for the production of the fermented fungal product per functional unit. The indented substances in italic refer to the inputs related to the production of the fungal biomass (inoculum) used for the inoculation.

	Unit	Scenario I	Scenario II	Scenario. III	Scenario IV
Inputs					
Bread	kg	2.0	2.0	2.0	2.0
Inoculum	g	10	10	10	29 *
Water	l	2.63	1.81	1.92	-
Glucose	g	-	-	38.50	-
NH_4Cl	g	-	16.28	14.50	-
NaOH	g	-	-	0.62	-
Molasses	g	-	0.36	-	-
Bread	g	52.60	-	-	-
Electricity	kWh	0.065	0.033	0.033	0.033

Water for mixing	l	1.34	1.34	1.34	1.34
Electricity for drying	kWh	0.53	0.53	0.53	0.53
Electricity for grinding	kWh	0.024	0.024	0.024	0.024
Electricity for SSF	kWh	1.55	1.55	1.55	1.55
Outputs					
Fermented fungal product *	kg	1	1	1	1

* Calculation based on the mass balance revealed by Wang et al. [31] and the starch, protein and moisture content in the samples.

2.1.3. Solid State Fermentation

Cultivation of *N. intermedia* was conducted in sterile petri dishes (100 mm × 20 mm). A total dry weight of 15 g breadcrumbs was inoculated with *N. intermedia* separately for each petri dish. The initial moisture content was adjusted to 40% (on a dry basis (*w*/*w*, db) with distilled water, after which each sample was mixed evenly, and a lid was placed on the petri dish. Solid-state fermentation was conducted batch wise for 2–10 days in a climatic test cabinet (NUVE test cabinet TK 120, Ankara, Turkey) at 90%–95% relative humidity (Rh) ± 1%, at 35 °C under light. The light source was applied to increase pigment production in the fungal biomass in order to form a visually appealing product with improved nutritional value [6,25]. The petri dishes were flipped every second day. At the end of fermentation, samples were dried at 45 °C. Effects of days of fermentation on the final weight and protein content in the fungal-bread product were investigated as well as a visual observation on morphology after the SmF step.

2.2. *Analyses*

The fungal spore concentration was determined using a Bürker counting chamber. The final moisture content, the amount of fungal biomass harvested after SmF, and fungal-bread product used as inoculum in Sc. IV were determined gravimetrically. Preparation of samples and measurement of the protein content by Kjeldahl digestion using a 2020 Kjeltec Digestor and a 2400 Kjeltec Analyser unit (FOSS Analytical A/S, Hilleröd, Denmark) was performed following the same procedure as in Gmoser et al. [25]. Total starch analysis was accomplished by α-amlylase/amylogluco-sidase/hexokinase method following the procedure provided in the Total Starch HK Assay Kit (Megazyme International Ltd., Ireland).

2.3. *Statistical Analysis*

Experiments on the SSF were performed in duplicate ($n = 2$) and two analytical replicates were performed on each batch (a = 2). Values present the mean value of the measurement ± SD. Raw data were statistically analysed using the software package MINITAB®® (version 17.1.0, Minitab Inc., State College, PA, USA). Analysis of variance (ANOVA) was determined using general linear model with a 95% confidence level followed by Tukey's multiple comparison test. A linear factorial regression model of the data was also conducted with the four inoculation methods as factor levels and days of SSF (2–8 days) as continuous control with protein concentration as output.

2.4. *Life Cycle Assessment*

The functional unit for the environmental assessment is 1 kg of fermented fungal product after the fermentation step. The selection of a mass unit was connected with the ability to compare results with those available in the literature. The system boundaries are cradle-to-gate as described in Section 2.1. The geographical scope is Sweden. The characterization method ReCiPe (ReCiPe 2016 v1.1 midpoint and endpoint methods, Hierarchist version) [32], was used due to its broad set of midpoint impact categories and the possibility of using both midpoint and endpoint methods [32].

The inventory was built using primary data flows from the laboratory experiments, which were initially based on data from the literature [28–30,33,34] and further developed based on the results from the trial experiments. The energy consumption of the equipment was estimated based on the methodology described in Piccinno et al. [35] for scaling up laboratory data. Capital goods are not included in the inventory, as all scenarios are expected to use similar equipment. The ecoinvent database version 3.6 was used for the background processes. Inventory data for the scenarios is available in Table 1.

3. Results

This section describes the experimental results of the solid-state fermentation, in relation to the increase of the protein content over time and the morphology of the inoculum. Moreover, the result for the life cycle assessment is presented.

3.1. Experimental Results

There is a significant difference in protein content of the final fungal product after SSF for the different inoculation methods (p value 0.00). The inoculation scenario using molasses (Sc. II) resulted in a higher protein content after SSF compared to glucose (Sc. III), with no significant difference between Scenarios I and IV. The inoculum scenario based on glucose medium (Sc. III) also resulted in a longer lag phase and slower initial exponential growth, possibly due to change in substrate during SSF, while *N. intermedia* grown on bread (Sc. I) and molasses (Sc. III) was in a more active phase, adapted to bread as substrate and hypothetically, already producing enzymes for complex carbohydrate assimilation, when applied to the SSF step. However, considering the natural variation in biological systems, the relatively small differences between the inoculation methods do not contribute with sufficient support for choosing one scenario over the other.

As expected, the protein content increased over time for all four groups, with a significant increase over every second day (p value = 0.00) (Figure 2). The highest increase in protein content happened between day four and six for all inoculation methods based on SmF, varying between 38%, 49% and 51%, for Scenarios I, II and III respectively. Inoculum based on backslopping resulted in a slightly higher increase in protein content between day two and four (39% increase) compared to the other days. The linear regression equations of the four inoculation methods revealed that SmF based on molasses gave the highest average protein increase per day of 4.0, compared to SmF based on bread, backslopping and SmF based on glucose of 3.6%, 3.4% and 3.2% increase per day respectively (consider a 6.6% lack of fit of the model). By comparing the linear regression equations, it can be seen that the daily increase based on glucose medium is only slightly offset in x-axis (timescale), whereas the protein value reaches approximately the same amount as the other inoculation methods after 8 days SSF.

The fungal morphology in the SmF based on molasses medium (Sc. II) resulted in fine, dense pellets, whereas the other SmF cultures resulted in loose filamentous mycelium. SSF based on inoculum from molasses medium gave the highest mean percentage of protein.

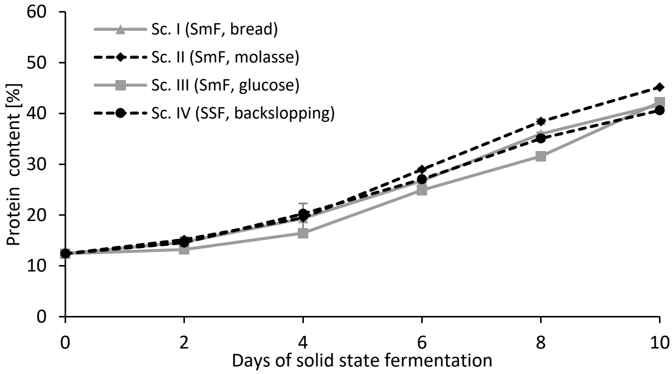

Figure 2. Solid-state cultivation on stale sourdough bread using *N. intermedia* after 0–10 days cultivation under light at 35 °C, 90% Rh, and 40% initial moisture content using different inoculation methods: backslopping (black circle, dotted line), submerged fermentation with synthetic medium based on glucose (grey square, solid line), submerged fermentation based on molasses (black diamonds, dotted line) and submerged fermentation based on breadcrumbs (grey triangles, solid line). Results are expressed as the mean value ± 95% confidence level.

3.2. Life Cycle Assessment

The relative results for five selected impact assessment categories for the scenarios and separately for the inoculum production process are presented in Figure 3 and Figure 4, respectively. The impact categories described in Figure 3 and Figure 4 are global warming, terrestrial acidification, freshwater eutrophication, marine ecotoxicity and fossil resource scarcity. The results for the remaining thirteen impact categories available in the ReCiPe characterization method [32] is available in Table 2.

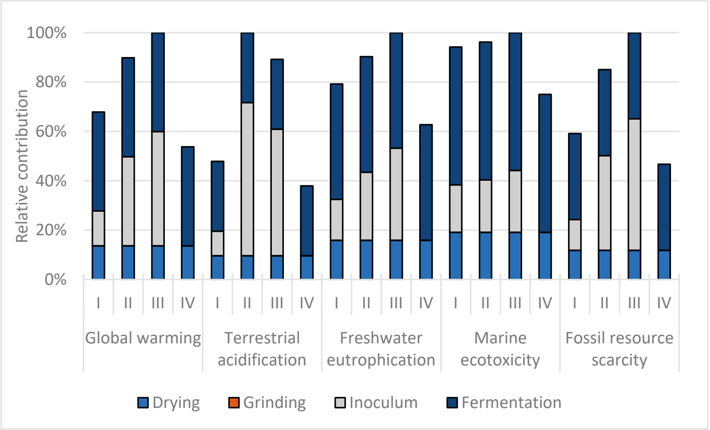

Figure 3. Selected LCIA results for the scenarios for the fermented fungal product production. Sc. I uses bread as the carbon source. Sc. II uses molasses. Sc. III is based on glucose and Sc. IV relies on backslopping, as described in Table 1.

Figure 3 indicates that the fermentation step and the choice of inoculum production has a large influence on the environmental burdens associated with the fermented fungal product.

The production of inoculum had a contribution higher than 40% in 9 out of 18 impact categories assessed in both Sc. II and Sc. III. More specifically, the carbon source used for the inoculum production for these scenarios is a hotspot, as shown in more detail in Figure 4. In Scenario I the inoculation has a relatively lower contribution, ranging from 20% to 29%. The reason is that it uses stale bread as the carbon source, which is modelled as waste and therefore burden free.

The results indicate that Scenario IV, which is based on backslopping, has the lowest environmental impacts in all of the eighteen categories assessed, followed by Scenario I. Both scenarios substituted glucose or molasses for stale bread, significantly reducing the environmental impacts on the inoculum production. Scenarios II and III had the worst performance in the categories assessed. Scenario II had the worst performance in five impact categories, while Scenario III had the highest impact in 12 environmental impact categories (Table 2).

The solid-state fermentation is also a hotspot, and it contributes significantly to the majority of the impact categories. These impacts are related to the electricity consumption during the process, as shown in Table 1. The impact is the same for all scenarios, but it has a different relative contribution as shown in Figure 3. Overall, the fermentation step, drying of bread and the inoculation were identified as hotspots in the production of the fungal product (Figure 3).

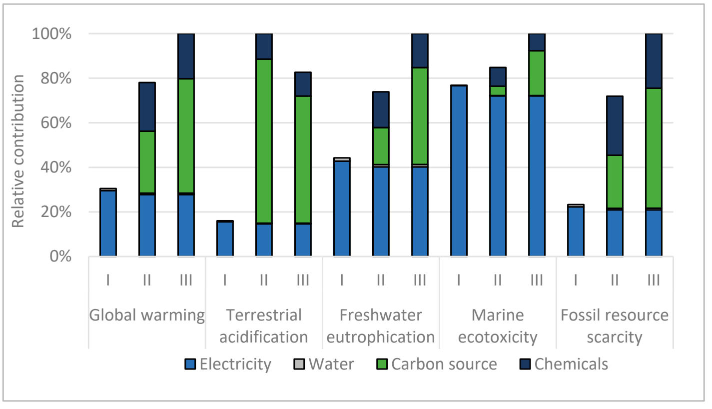

Figure 4. Selected LCIA results for the different inoculation production processes. Sc. I uses breadcrumbs as the carbon source. Sc. II uses molasses. Sc. III is based on glucose.

Figure 4 shows the detailed impact assessment for the different inoculation methods. Sc. IV has no contribution from the inoculum production since it is obtained from backslopping. This assumes that the contribution of the initial inoculum required to make the first batch of fungi is insignificant.

The results indicate that the carbon source used in the inoculum production is a hotspot in the majority of impacts categories in both Scenario II, which uses molasses and Scenario III, based on glucose (Figure 4). The carbon source is responsible for 36% and 51% of the impacts on the global warming impact category for Sc. II and Sc. III, respectively. In terrestrial acidification, the contribution is higher, 74% and 69% for Sc. II and Sc. III, respectively. Scenario I, conversely, uses stale bread as the carbon source, which is modelled as waste and is, therefore, burden-free.

Electricity consumption has a significant contribution for both the inoculum production and the overall process. The geographical scope is Sweden, which has a relatively low impact energy system. Therefore, the results of the study are sensitive to the energy mix used.

Table 2. Impact assessment results for Scenarios I, II, III and IV based on the ReCiPe midpoint impact categories.

Impact Category	Unit	Sc. I	Sc. II	Sc. III	Sc. IV
Global warming	kg CO_2 eq	1,4E-01	1,9E-01	2,1E-01	1,1E-01
Stratospheric ozone depletion	kg CFC-11 eq	3,0E-07	6,2E-07	5,2E-07	2,4E-07
Ionizing radiation	kBq Co-60 eq	8,0E-01	7,9E-01	7,9E-01	6,4E-01
Ozone formation, Human health	kg NOx eq	3,5E-04	4,8E-04	5,1E-04	2,8E-04
Fine particulate matter formation	kg $PM_{2.5}$ eq	2,3E-04	3,7E-04	3,8E-04	1,9E-04
Ozone formation, TE	kg NOx eq	3,6E-04	4,9E-04	5,2E-04	2,8E-04
Terrestrial acidification	kg SO_2 eq	5,8E-04	1,2E-03	1,1E-03	4,6E-04
Freshwater eutrophication	kg *p* eq	9,4E-05	1,1E-04	1,2E-04	7,5E-05
Marine eutrophication	kg *n* eq	1,8E-05	1,0E-04	8,5E-05	1,5E-05
Terrestrial ecotoxicity	kg 1,4-DCB	2,2E+00	2,4E+00	2,7E+00	1,8E+00
Freshwater ecotoxicity	kg 1,4-DCB	9,4E-02	9,6E-02	9,9E-02	7,5E-02
Marine ecotoxicity	kg 1,4-DCB	1,1E-01	1,2E-01	1,2E-01	9,1E-02
Human carcinogenic toxicity	kg 1,4-DCB	1,9E-02	2,0E-02	2,1E-02	1,5E-02
Human non-carcinogenic toxicity	kg 1,4-DCB	6,0E-01	6,4E-01	6,5E-01	4,8E-01
Land use	m^2 a crop eq	6,1E-02	9,2E-02	8,7E-02	4,8E-02
Mineral resource scarcity	kg Cu eq	2,0E-03	2,4E-03	2,5E-03	1,6E-03
Fossil resource scarcity	kg oil eq	2,6E-02	3,8E-02	4,5E-02	2,1E-02
Water consumption	m^3	2,1E-02	2,3E-02	2,2E-02	1,5E-02

4. Discussion

The environmental impacts of a product are often determined by the decisions made at the early stages of development. For this reason, the development of the product should integrate environmental considerations to minimize the environmental impacts throughout its life cycle without compromising essential characteristics such as function, cost and quality. The estimation of environmental impact at this early stage is challenging due to the limited amount of information available, for example, the energy consumption in the processes. In contrast, the later stages in the product development provide more quantitative data but fewer opportunities to change the system.

Table 3 summarizes and ranks the main technical and environmental differences between the four scenarios studied here, namely the environmental impacts, protein content, lag phase and morphology. These aspects are critical to support the decision of the more suitable method for fungal biomass production using bread as substrate. The environmental impact ranking in Table 3 is based on the single score from ReCiPe endpoint (H) as available in the SimaPro software version 9.1.

Table 3. Summary of technical and environmental indicators for the scenarios assessed. The protein content refers to SSF after 6 days.

Scenarios	Environmental Impact (mPt)	Protein Content	Lag Phase	Morphology
Scenario I	9.7	27%	Short	Dispersed mycelium
Scenario II	12.5	29%	Short	Pellets
Scenario III	13.0	25%	Long	Dispersed mycelium
Scenario IV	7.6	27%	Short	Dispersed mycelium

Filamentous fungi grow in different morphological forms in submerged cultures, including freely suspended filamentous mycelia and pellets depending on the genotype of the strain and culture conditions [36]. The growth mode affects the rheological properties of the cultivation medium, and consequently, the overall process performance and final product yields. Generally, the excessive growth of free filamentous mycelia is connected to practical and technical difficulties, such as lower oxygen diffusion, laborious harvesting, high medium viscosity and a relative lower yield of products [37]. These problems associated with the filamentous morphology can be overcome when the fungi grow in the form of pellets, which also have a higher potential for cell reuse and higher productivity due to the possibility of using high-density cultivations [38]. However, including in the pellet form, critical characteristics such as size and compactness versus fluffiness can influence oxygen and substrate transfer rates. The trend in research works has been towards the production of small fluffy pellets. Among the scenarios assessed, only Scenario II grew in the pellet morphology. The slightly shorter lag phase in Scenario II compared to Scenario III further promote this inoculation method since it promotes a faster initiation of the subsequent SSF step. Nevertheless, the environmental impact of Sc. II is significantly higher in comparison with Sc. I and IV (Table 3).

Scenario IV had the lowest environmental impact, and the protein content was sufficient for the intended use. However, one potential drawback with this method is the higher risk of contamination since it is more problematic to sustain a sterile condition at a larger scale when backslopping part of the final product as inoculum compared to inoculum using pure fungal biomass from SmF. Thus, microbiology analysis of the final product and industrial trials need to be assessed before this method can be applied.

Considering the weighted results at Table 3, Scenario I had an environmental impact that is 28% higher in comparison with Scenario IV, but the results indicate a lower impact in comparison with Scenarios II and III. Moreover, the main advantage of this scenario in comparison to Scenario IV is that it is less sensitive to contamination. Scenario I also yields filamentous mycelia morphology. However, further development of this scenario by changes in cultivation conditions such as pH and aeration can potentially alter the morphology.

The fermentation time is another important aspect influencing both the protein content and the environmental impacts of the final product. The results showed that the protein content for all scenarios reached similar values after 8 days of SSF. However, based on previous studies [6] on the visual appearance and smell of the final product, the SSF step should not be longer than a maximum of eight days and preferably six days if the protein content is sufficient for the intended purpose as food. A direct comparison of the environmental performance of this fungal product with established food products based on mycoprotein requires special attention due to scale-up issues and technology uncertainties inherent in the majority LCAs of products in an early stage of development [18]. Moreover, it is expected that the technology under assessment will need further development in relation to technical aspects and consumer's acceptance.

Jungbluth et al. [30] reported an electricity consumption of 1.17 kWh and a heat consumption of 13 MJ for the industrial production of 1 kg of mycoprotein at plant. Smetana et al. [9] reported 6 kWh/kg of mycoprotein ready to eat. The electricity consumption in this study ranged from 2.61 to 2.64 kWh/functional unit for the different scenarios. Nevertheless, it is important to note that a direct comparison is not possible, due to differences in the system boundaries of each study. The system boundaries in this study and Jungbluth et al. [30] end at the plant gate, while Smetana et al. [9] consider the system boundaries until the consumer's plate. There are also discrepancies in the technology readiness level between this study and Jungbluth et al. [30] in which inventory is based on an operational plant.

A huge advantage is that we now have an LCA model that can be used to get updated environmental impacts when the scenarios are adjusted to improve factors such as protein content, lag time and morphology. This allows for environmental impacts to be assessed throughout the development process.

5. Conclusions

The current analysis, which included different production pathways for the valorisation of stale bread into fungal biomass according to the environmental impacts, protein content, lag phase and morphology, indicated the potential use of the fungal biomass as a substitute for traditional food products. The technology proposed uses an abundant waste flow and has the potential to be implemented in supermarkets, small and medium-sized bakeries and industrial bakeries on its returned flows.

Technical and environmental analyses were performed to determine potential trade-offs of four proposed scenarios to valorise stale bread into a fungal based food product rich in protein. The results from the hotspot analysis indicate that the choice of the medium used in the inoculum production has a large influence on the environmental and technical performance. Inoculum production using molasses or glucose as medium, Sc. II and III respectively, are not preferred from an environmental perspective. Sc. I and Sc. IV, which uses surplus bread and backslopping, showed the lowest environmental impacts in all categories studied here, while maintaining important characteristics such as high protein content. Therefore, it is recommended that such alternatives should be further investigated for the technical feasibility of the process. Moreover, the hotspot analysis indicated that for all scenarios assessed, the fermentation step and drying of bread have a significant impact on the environmental burdens. Therefore, improvements aimed at environmental gains in the fungal biomass production should focus on alternatives for decreasing the fermentation time and consequently the electricity consumption, and on the development of alternative processes in which the drying of bread is not required.

None of the scenarios performed best in all of the parameters analysed. Therefore, it is not possible to draw a simple conclusion regarding the preferred scenario. The decision will ultimately be made according to the technical, environmental or economic agenda of the decision-makers. The contribution of this study is to highlight the trade-offs inherently involved in the decision process within the product development and to provide guidance for further development of the process.

Bread waste has the potential to be retained in the food chain by applying a SSF process using the edible fungus *N. intermedia*. The study has integrated environmental considerations into the early stages of the development of a fungal food product, showing which scenario has the best environmental performance and highlighting trade-offs and the parts of the process that are hotspots and should, thereby, be in focus when optimizing the process. This approach is suggested to contribute to a sustainable way to handle otherwise wasted bread, consistent with a circular economy, and it provides a broader base for the developers of the technology to make sustainable decisions during process optimisation.

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This thesis presents work that was done within the Swedish Centre for Resource Recovery (SCRR). Research and education performed within SCRR identifies new and improved methods to convert residuals into value-added products. SCRR covers technical, environmental and social aspects of sustainable resource recovery.

The global food system is responsible for major environmental impacts, particularly those related to climate change, biodiversity loss, and depletion of freshwater resources. These problems are aggravated by a significant wastage of food throughout the supply chain, where retailers are responsible for large quantities of the waste. While food waste in retail has been studied throughout the past decade, bread is a product category that has remained relatively unexplored.

The prevention and valorisation of food waste have the potential to decrease the food system's pressure on the environment and on the natural resources, and to lead to positive outcomes in relation to food security, nutrition, and reduction in production costs. Through a variety of approaches, this PhD thesis has investigated the quantities, causes, and measures for the prevention, valorisation and management of surplus bread at the supplier–retailer interface. As bread is a product with a high potential for waste prevention, this thesis contributes to the design of effective waste prevention activities by suppliers and retailers through the identification of risk factors for the large amounts of bread waste and surplus. Furthermore, surplus bread presents an abundant, inexpensive and underutilized substrate that has the potential to be used for different valorisation pathways, such as the production of fungi based protein-enriched food, animal feed, ethanol and beer. These processes have been shown to have the potential to reduce the environmental impacts caused by the bread supply chain. By suggesting a variety of measures, the thesis supports the transition of bakeries and retailers towards circularity, contributing to a more sustainable future.

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